Supplementary Material

Modeling Soil–Water–Atmosphere Interactions for Irrigation Scheduling and Design

Derivations of the equations, the mechanisms behind the results, the data, and the origin of every table and figure

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S1. What is new in this article

Table S1: What this article adds, item by item: what was established before, what is new here, and the evidence in this article.

#	Established before	New in this article	Evidence here
1	Irrigation tools compute schedules and designs from standard procedures (Allen et al., 1998; Šimůnek et al., 2016; Steduto et al., 2009) but seldom show which relation produced a number or where it holds.	A *ledger* in which every numbered equation carries its origin (literature, fitted to data, engineered) and every number of the interface leads to one of them.	134 equations labelled: 38 literature, 18 fitted, 78 engineered.
2	Meta-models of process models speed up screening (Asher et al., 2015; Razavi et al., 2012).	A meta-model of a designed Daisy ensemble whose rank fidelity and absolute offset are both measured on held-out runs.	Spearman 0.64 to 0.97 on held-out simulations; off-axis nRMSE 6.2 % (ET) and 18.9 % (irrigation) after re-anchoring.
3	Surface, drip and sprinkler design procedures are textbook (Christiansen, 1942; Keller and Karmeli, 1974; Walker and Skogerboe, 1987).	The procedures implemented and checked against published worked examples, with three defects found and removed by that check.	Median error 0.08 %, maximum 1.81 %; a fixed advance exponent that under-predicted slow advance by up to 29 % removed; drip solver reliable to a head loss of 60 %.
4	A suction trigger is set from tables (Allen et al., 1998).	The coupled column shows how one trigger means different things in different soils.	A -300 cm trigger uses 24 to 84 % of the plant-available water; drainage ranges from 28 to 164 mm yr^{-1} under the same rain.
5	–	Two limitations of the system exposed by its own evaluation.	The leaf water potential is set by a conductivity floor; the drainage share of a mulch saving is understated in coarse soils.

#	Established before	New in this article	Evidence here
6	–	Scenario sweeps fast enough for interactive use while staying traceable.	10^5 scenarios in about 2 s on a laptop.

S2. How this supplement is organised

Each of the following sections gathers everything about one result of the article under one heading: the problem and the author's contribution with its pinpointed equation, the worked numbers that can be checked on a calculator or in the workbook, graphs that mark the exact points, the figures of the article with their numbered labels, the derivations, and why it happens. The terms, the constants, the tables, the path from the data to every result, the data themselves and the origin of every item follow at the end. Most derivations of this supplement are of established relations, given so that every equation of the article can be followed from first principles: the capillary law, the Kelvin relation, Green–Ampt infiltration, the volume balance of surface irrigation, Hazen–Williams losses with the Christiansen factor, emission uniformity and nozzle discharge (Christiansen, 1942; Green and Ampt, 1911; Keller and Karmeli, 1974; Walker and Skogerboe, 1987). New in this article are the inversion of the depletion fraction through the van Genuchten curve into a soil-specific trigger (S3.11), the smooth joining of the two evaporation stages (S5.9), and the choice of the interception store that makes the daily model reproduce the annual interception (S4.6); the contribution of the article is the system that joins these relations, its ledger and its verification (Table S1).

S3. Soil-specific triggers from one depletion

How one critical depletion becomes a different suction trigger in every soil.

S3.1 Contribution 1: one depletion, ten soil-specific triggers

The problem

Agronomic guidance gives a depletion fraction at which to irrigate (FAO-56), while sensors read suction. A single suction trigger is therefore used on every soil. What share of the plant-available water does that trigger really represent, and which suction corresponds to the same depletion on each soil?

How the author developed it

(a) The plant-available water lies between field capacity (100 cm) and the wilting point (15 849 cm). **(b)** Because $\theta - \theta_r \propto S_e$, the depletion at suction h needs only S_e: $f(h) = [S_e(100) - S_e(h)]/[S_e(100) - S_e(15\,849)]$. **(c)** Set $f = f^*$ and solve for S_e^*. **(d)** Invert the van Genuchten curve.

The result

How it is derived, line by line

1. **Depletion uses only S_e:**

$$f(h) = \frac{S_e(100) - S_e(h)}{S_e(100) - S_e(15\,849)}$$

2. **Set $f = f^*$ and solve for S_e:**

$$S_e^* = S_e(100) - f^*[S_e(100) - S_e(15\,849)]$$

3. **Invert van Genuchten:**

$$h^* = \alpha^{-1}\left(S_e^{*\,-1/m} - 1\right)^{1/n}$$

The result, with its parts marked

$$\underbrace{h^*}_{①} = \frac{1}{\alpha}\Big(\underbrace{\left[S_e(100) - f^*\{S_e(100) - S_e(15\,849)\}\right]}_{②}{}^{-1/m} - 1\Big)^{1/n}$$

$$\underbrace{f(300) = 0.24 \text{ to } 0.84}_{③}$$

developed by the author

Soil	h^* at $f^* = 0.45$ (cm)	water used at a 300 cm trigger
Sand	(4) 143	84 %
Loam	284	47 %
Silty clay loam	(5) 729	24 %

Pinpoints: what each circled mark shows and why

(1) The soil's own trigger: the suction at which this soil has used the chosen share of its available water.

(2) The effective saturation at the chosen depletion; the factor $\theta_s - \theta_r$ cancels, so only α and n are needed.

(3) The same 300 cm trigger means 24 % of the water used in silty clay loam but 84 % in sand: one trigger is not one decision.

(4) Sand must be irrigated at a much wetter suction because it releases most of its water at low suction.

(5) Fine soils release water over a wide range, so the same depletion is reached at a much drier suction.

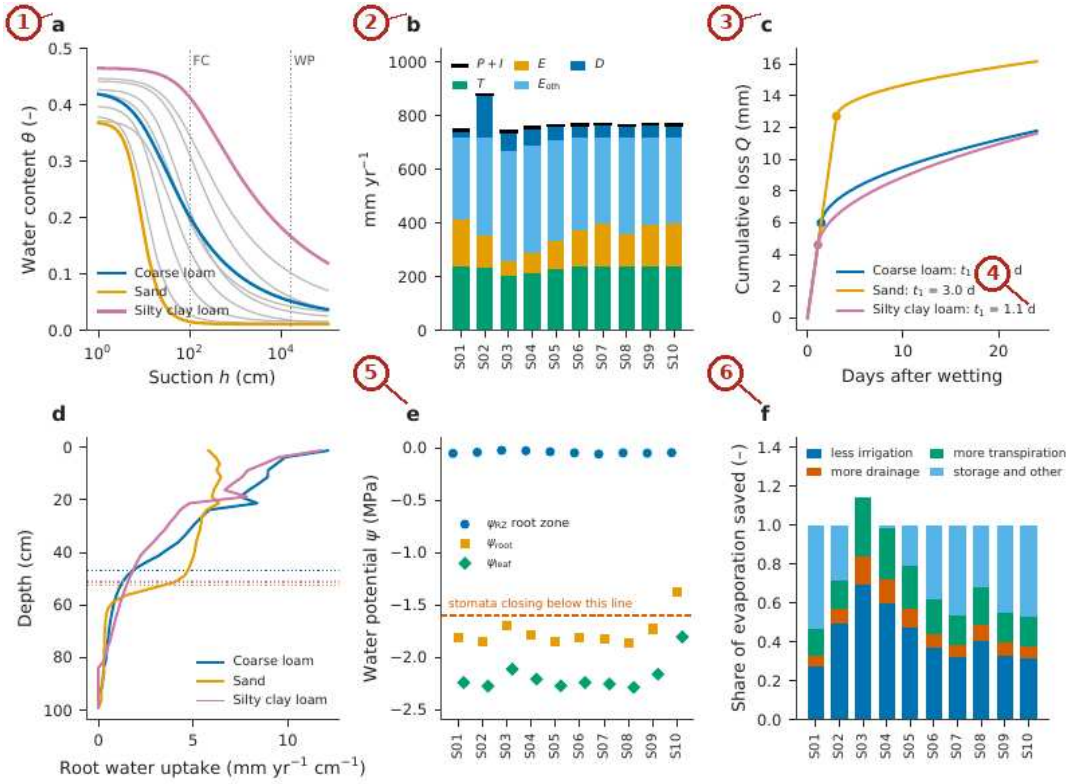
S3.2 Worked numbers, part 1: from a suction trigger to a share of the available water

The plant-available water lies between field capacity (100 cm) and the wilting point (15 849 cm). The depletion at a suction h is $f(h) = [S_e(100) - S_e(h)]/[S_e(100) - S_e(15\,849)]$. *Example (loam,* $\alpha = 0.036$, $n = 1.56$, $m = 0.359$*).* $S_e(100) = 0.466$, $S_e(15\,849) = 0.029$, $S_e(300) = 0.262$; $f(300) = (0.466 - 0.262)/(0.466 - 0.029) = 0.47$: at a 300 cm trigger the loam has used 47 % of its available water. In sand the same trigger means 84 %, in silty clay loam 24 %.

S3.3 Worked numbers, part 2: from a chosen depletion to a soil's own trigger

Choose $f^* = 0.45$. Then $S_e^* = S_e(100) - f^*[S_e(100) - S_e(15\,849)]$ and the trigger is $h^* = \alpha^{-1}(S_e^{*\,-1/m} - 1)^{1/n}$. *Example (loam).* $S_e^* = 0.466 - 0.45 \times 0.437 = 0.269$; $0.269^{-1/0.359} = 38.8$; $38.8 - 1 = 37.8$; $37.8^{1/1.56} = 10.2$; $10.2/0.036 = 284$ cm.

S3.4 Reading Figure 2 of the article



What the numbered labels point to.

(1) Panel (a): see the caption above. *Why:* Sands release their water at low suction, clays over a wide range, so the curves cross field capacity and wilting point at different contents.

(2) Panel (b): Annual balance: sinks stacked (T, E, E_{oth}, D) against the input $P + I$ (black marks). *Why:* The same rain leaves as drainage in coarse soils and as evaporation and transpiration in fine soils.

(3) Panel (c): Cumulative evaporation after a wetting at the season-mean demand (Eq. (11) of the article); dots mark the end of stage 1. *Why:* Fine soils keep the surface wet longer, so stage 1 lasts longer.

(4) Panel (d): Mean annual root water uptake by depth from Engine 2 (unscaled); dotted lines mark z_{90}. *Why:* Uptake follows root density, which falls with depth.

(5) Panel (e): Water potential of the root zone, the root surface and the leaf (Eq. (21) and Eq. (22) of the article), crop days of day-of-year 120–210; the dashed line is the stomatal-closure threshold of the view; the air is at -56 MPa (not shown). *Why:* Potential falls from root zone to leaf because water flows through resistances; dry soil adds a large soil resistance.

6 Panel (f): Fate of the soil evaporation saved by a straw mulch (Eq. (23) of the article); in sand the three positive shares sum to 1.14, so the remainder is zero. *Why:* A mulch saving goes to transpiration, less irrigation or drainage according to the soil.

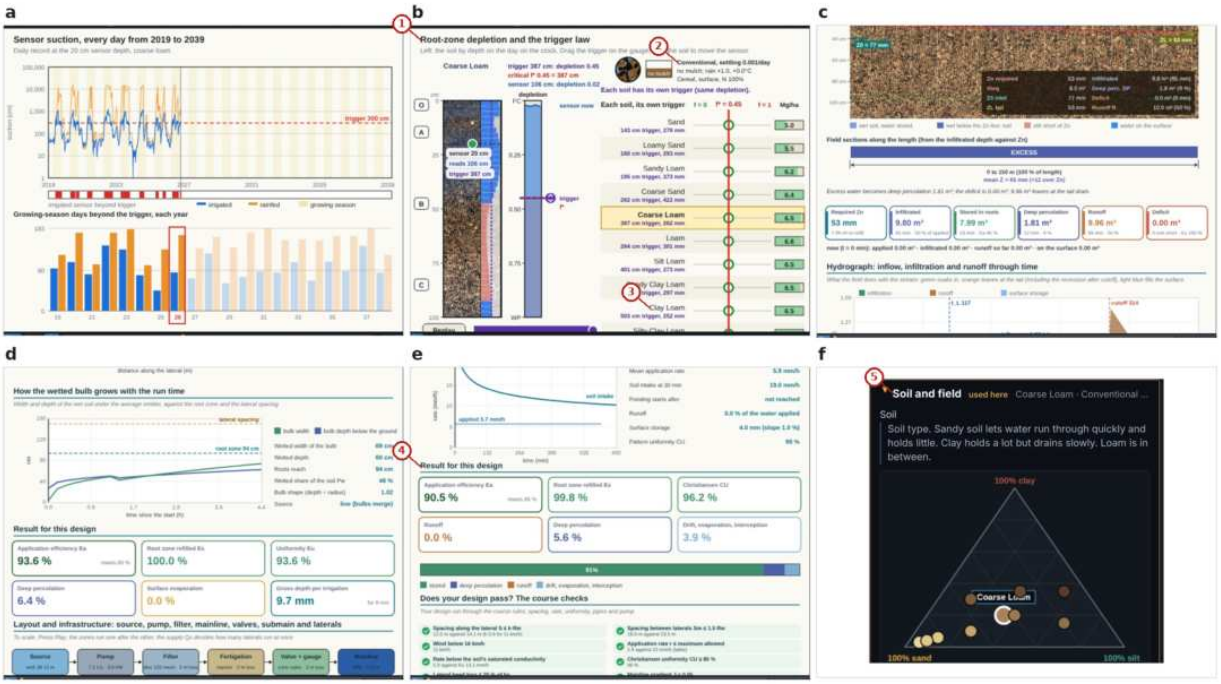
What it shows. Soil–water–atmosphere variables of the ten soils at the reference state. (a) Retention curves $\theta(h)$ (Eq. (27) and Eq. (2) of the article); sand, coarse loam and silty clay loam highlighted; dotted lines are field capacity (FC, 100 cm) and the wilting point (WP, 15 849 cm). (b) Annual balance: sinks stacked (T, E, E_{oth}, D) against the input $P + I$ (black marks). (c) Cumulative evaporation after a wetting at the season-mean demand (Eq. (11) of the article); dots mark the end of stage 1. (d) Mean annual root water uptake by depth from Engine 2 (unscaled); dotted lines mark z_{90}. (e) Water potential of the root zone, the root surface and the leaf (Eq. (21) and Eq. (22) of the article), crop days of day-of-year 120–210; the dashed line is the stomatal-closure threshold of the view; the air is at -56 MPa (not shown). (f) Fate of the soil evaporation saved by a straw mulch (Eq. (23) of the article); in sand the three positive shares sum to 1.14, so the remainder is zero.

How to read the labels, symbols and colours. (a) Retention curves with field capacity and wilting point dotted; (b) annual sinks stacked against the input (black marks); (c) cumulative evaporation after wetting, dots at the end of stage 1; (d) root uptake by depth; (e) water potential of root zone, root surface and leaf; (f) fate of a mulch saving.

What affects it, and why it happens. Soils differ in how much water they hold between field capacity and wilting point and how fast they pass it on; these differences propagate to evaporation, uptake, potential and the fate of a saving.

How it is made. Figure drawn by the script from the input data. Script: `Manuscripts/reproducibility/analysis/p3_swa_fig.py`. Input: `Data/daisy_ensemble.csv.gz`.

S3.5 Reading Figure 4 of the article



What the numbered labels point to.

1. "Root-zone depletion and the trigger law": the suction (or depletion) at which irrigation starts.
2. "Conventional": a heading or control of this screen; the caption above says which layer it belongs to.
3. "Clay Loam": the soil of the project, which every layer uses.
4. "Result for this design": a heading or control of this screen; the caption above says which layer it belongs to.
5. "Soil and field": the soil of the project, which every layer uses.

What it shows. Result views for irrigation timing and design. (a) Sensor and trigger; (b) irrigation rule collapsing ten soils onto one depletion curve; (c) surface irrigation; (d) drip irrigation; (e) sprinklers; (f) soil selection on the texture triangle, which drives every view.

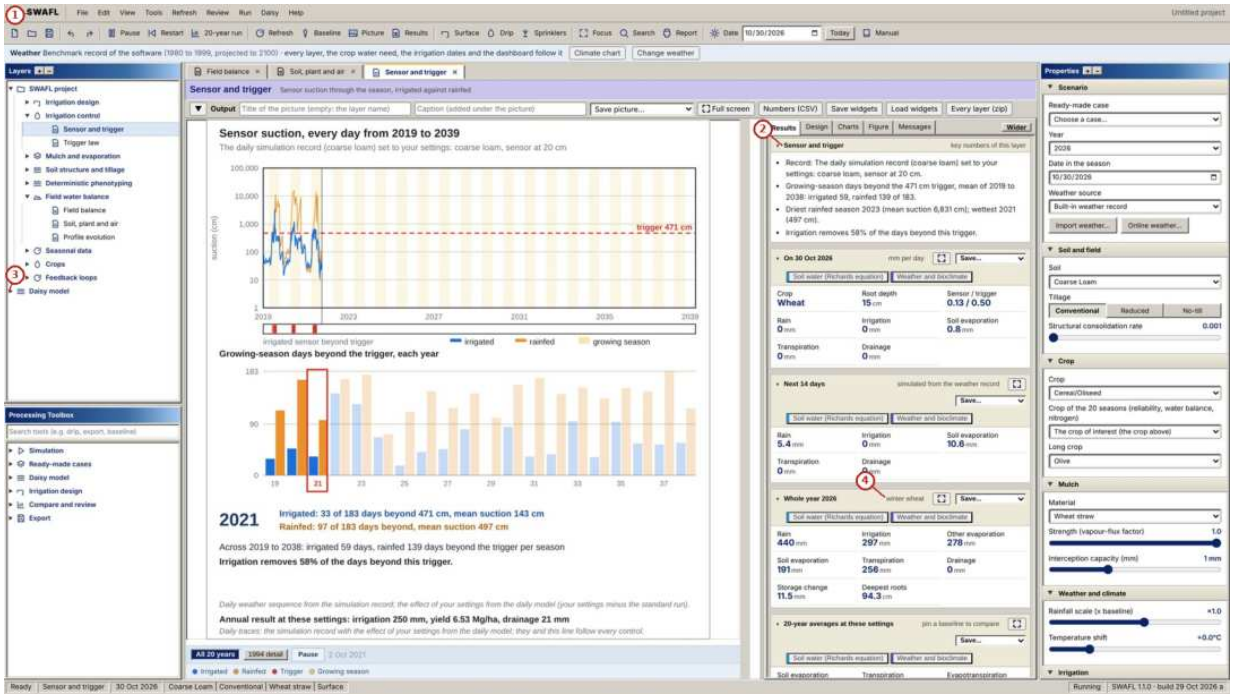
How to read the labels, symbols and colours. Views for timing and design (sensor and trigger, the trigger rule, surface, drip, sprinkler, soil triangle).

What affects it, and why it happens. Expressing the trigger as depletion collapses the soils onto one curve; the designs follow the soil intake and the water needed. The trigger rule collapses the soils onto one depletion curve because it uses depletion, not suction; the designs follow from

the soil's intake and the water needed.

How it is made. Screen capture of SWAFL Desktop 1.1.0 (not regenerated).

S3.6 Reading Figure 14 of the article



What the numbered labels point to (picture 1 of 2: Sensor and trigger: 20 years of daily suction at the sensor, irrigated and rainfed, against the trigger, and the days beyond the trigger per year (Eq. (8) of the article).).

1. "SWAFL View Run Daisy Help": the menu bar of the program (file, view, run, Daisy, help).
2. "Sensor and trigger": the suction (or depletion) at which irrigation starts.
3. "Daisy model": the Daisy model and its workbench.
4. "winter": a heading or control of this screen; the caption above says which layer it belongs to.



What the numbered labels point to (picture 2 of 2: Profile evolution: pressure-head profile through the season, irrigated and rainfed, and the share of the root zone beyond the trigger (Eq. (9) of the article).).

1. "SWAFL View Run Daisy Help": the menu bar of the program (file, view, run, Daisy, help).
2. "Focus Q Search": a heading or control of this screen; the caption above says which layer it belongs to.
3. "mm per day": a rate in millimetres per day.
4. "20-year averages at these settings": a heading or control of this screen; the caption above says which layer it belongs to.

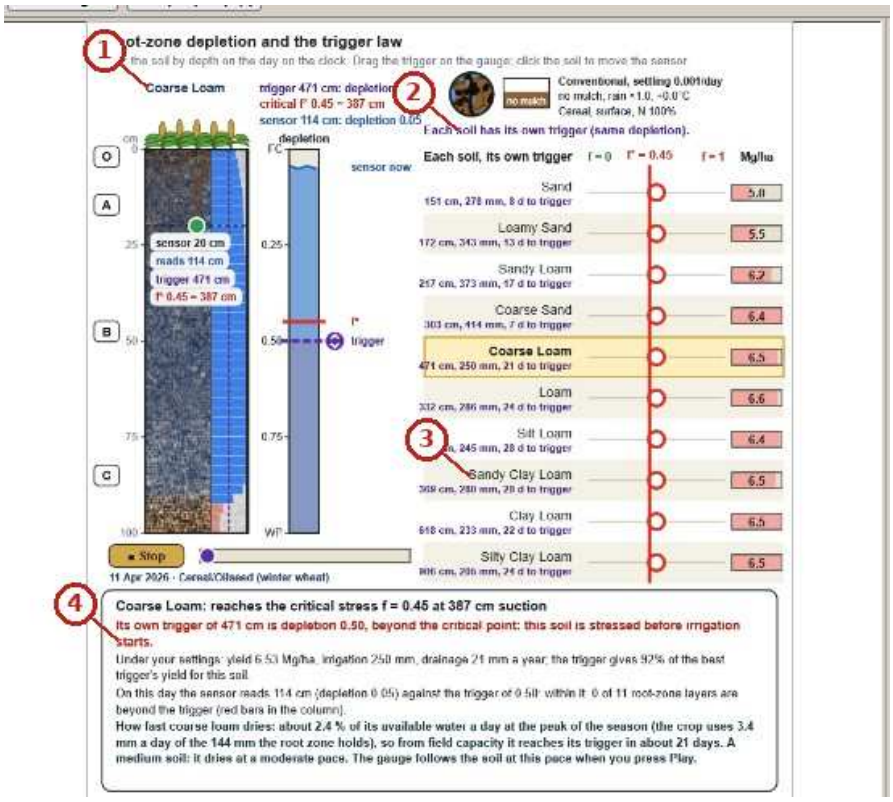
What it shows. SWAFL Desktop: soil water and irrigation timing. Panels: (a) Sensor and trigger: 20 years of daily suction at the sensor, irrigated and rainfed, against the trigger, and the days beyond the trigger per year (Eq. (8) of the article). (b) Profile evolution: pressure-head profile through the season, irrigated and rainfed, and the share of the root zone beyond the trigger (Eq. (9) of the article).

How to read the labels, symbols and colours. A screen of the software as it appears to the user; the labelled areas are the inputs, the figure and the numbers of the layer named in the caption.

What affects it, and why it happens. The suction at the sensor rises as the crop dries the soil and falls after each rain or irrigation; irrigation is triggered when it crosses the trigger line.

How it is made. Screen capture of SWAFL Desktop 1.1.0 (not regenerated).

S3.7 Reading Figure 15 of the article



What the numbered labels point to.

1. "Coarse Loam": the soil of the project, which every layer uses.
2. "soil has Its own trigger": the suction (or depletion) at which irrigation starts.
3. "Sandy": the soil of the project, which every layer uses.
4. "starts": a heading or control of this screen; the caption above says which layer it belongs to.

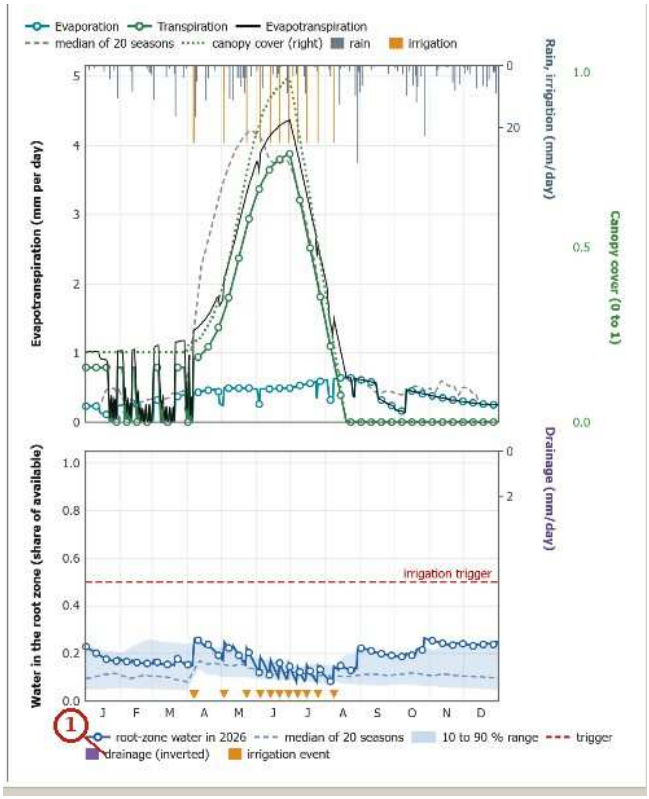
What it shows. SWAFL Desktop: trigger law. Panels: (a) Trigger law: root-zone depletion of ten soils on one axis with the critical depletion f^*, each soil's own trigger suction and yield, and the gauge of the sensor through the season (Eq. (29) and Eq. (25) and Eq. (26) of the article).

How to read the labels, symbols and colours. A screen of the software as it appears to the user; the labelled areas are the inputs, the figure and the numbers of the layer named in the caption.

What affects it, and why it happens. Each soil reaches the same depletion at a different suction because of the shape of its retention curve, so each soil has its own trigger.

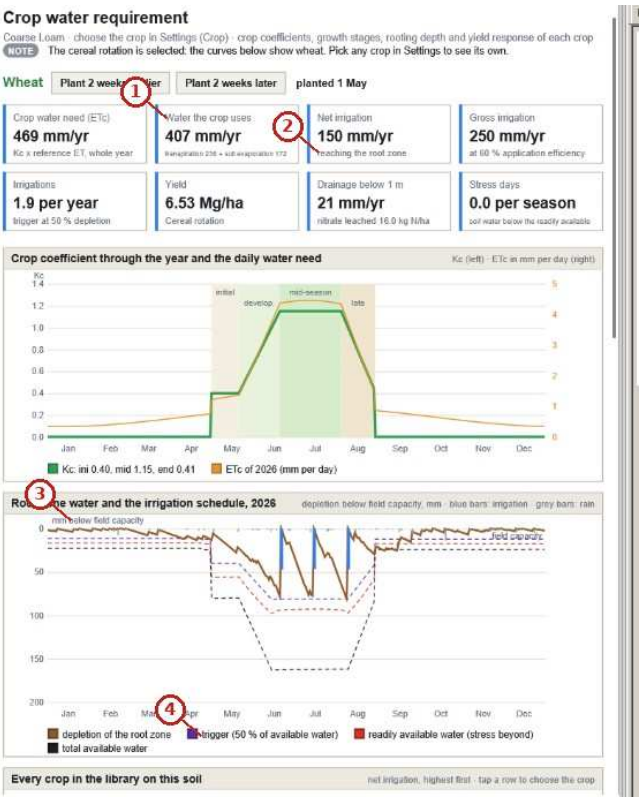
How it is made. Screen capture of SWAFL Desktop 1.1.0 (not regenerated).

S3.8 Reading Figure 30 of the article



What the numbered labels point to (picture 1 of 2: Field balance, charts: evaporation, transpiration and evapotranspiration of 2026 against the median of 20 seasons, with rain, irrigation and canopy cover, and the root-zone water against the trigger with the 10 to 90 % range and the irrigation events.).

(1) "drainage (inverted) event": rain, from the weather of the project.



What the numbered labels point to (picture 2 of 2: Crop water requirement: crop coefficient and daily need through the year, and the root-zone depletion with the trigger, the readily and the totally available water and the irrigations of 2026.).

1. "Water the crop uses": the crop of the project and its stage.
2. "reaching the root zone": the roots: depth, density and uptake.
3. "below capacity": a heading or control of this screen; the caption above says which layer it belongs to.
4. "trigger (50 % of available water)": the suction (or depletion) at which irrigation starts.

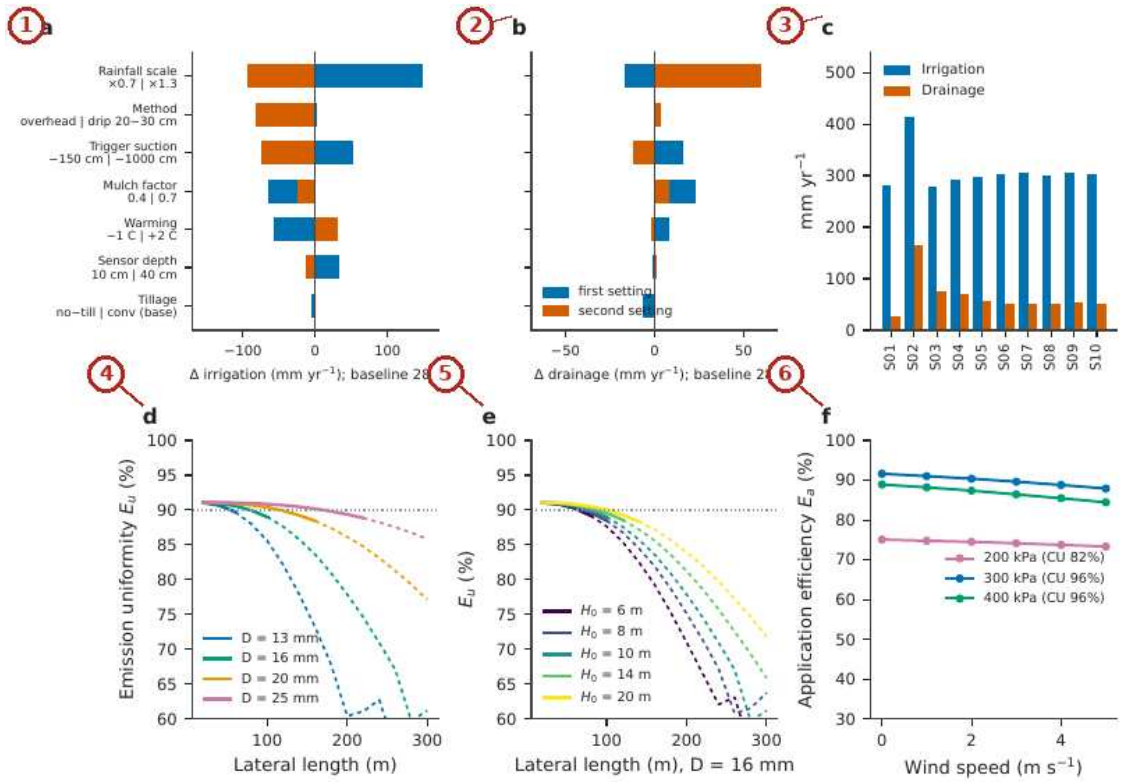
What it shows. SWAFL Desktop: the field water balance and the crop water requirement through the year (screen captures of release 1.1.0). Panels: (a) Field balance, charts: evaporation, transpiration and evapotranspiration of 2026 against the median of 20 seasons, with rain, irrigation and canopy cover, and the root-zone water against the trigger with the 10 to 90 % range and the irrigation events. (b) Crop water requirement: crop coefficient and daily need through the year, and the root-zone depletion with the trigger, the readily and the totally available water and the irrigations of 2026.

How to read the labels, symbols and colours. A screen of the software as it appears to the user; the labelled areas are the inputs, the figure and the numbers of the layer named in the caption.

What affects it, and why it happens. Evapotranspiration follows the crop coefficient through the season; root-zone water falls between irrigations and is refilled at the trigger.

How it is made. Screen capture of SWAFL Desktop 1.1.0 (not regenerated).

S3.9 Reading Figure 6 of the article



What the numbered labels point to.

1. Panel (a): Engine 1 annual irrigation and. *Why:* Controls that raise demand or lower supply raise irrigation.
2. Panel (b): drainage change for each control moved to the two settings listed under its name (blue: first setting; orange: second setting) from the reference state; nitrogen rate, which leaves both unchanged, is not shown. *Why:* Controls that keep the profile wet raise drainage.
3. Panel (c): Irrigation and drainage by soil. *Why:* Coarse soils need more irrigation and drain more.
4. Panel (d): Drip emission uniformity against lateral length by lateral diameter and. *Why:* Longer laterals carry more flow over more length, so friction lowers the head at the end and uniformity falls.
5. Panel (e): by inlet head; dashed where $h_f/H_0 > 0.2$. *Why:* A higher inlet head makes

friction a smaller share of the head, so uniformity rises.

6. Panel (f): Sprinkler application efficiency against wind speed at 200, 300 and 400 kPa (CU in the legend). *Why:* Wind distorts and blows away part of the spray, so efficiency falls; higher pressure makes smaller drops that drift more.

What it shows. One-at-a-time sensitivity of SWAFL Desktop outputs. (a) Engine 1 annual irrigation and (b) drainage change for each control moved to the two settings listed under its name (blue: first setting; orange: second setting) from the reference state; nitrogen rate, which leaves both unchanged, is not shown. (c) Irrigation and drainage by soil. (d) Drip emission uniformity against lateral length by lateral diameter and (e) by inlet head; dashed where $h_f/H_0 > 0.2$. (f) Sprinkler application efficiency against wind speed at 200, 300 and 400 kPa (CU in the legend).

How to read the labels, symbols and colours. (a, b) Change of irrigation and drainage when each control moves to two settings (blue, orange); (c) by soil; (d, e) drip uniformity against lateral length, diameter and inlet head; (f) sprinkler efficiency against wind.

What affects it, and why it happens. Controls that change water supply or demand move irrigation and drainage most; long or narrow laterals lose head and uniformity; wind distorts the sprinkler pattern.

How it is made. Figure drawn by the script from the input data. Script: `Manuscripts/reproducibility/analysis/p3_sens_fig.py`. Input: `Data/daisy_ensemble.csv.gz`.

S3.10 Derivation: Capillary radius

The pressure jump across a meniscus in a cylindrical pore of radius r is $\Delta p = 2\sigma\cos\gamma_c/r$ (Young–Laplace). Equating with the suction head $\rho_w g h$ gives $r_c = 2\sigma\cos\gamma_c/(\rho_w g h)$; with $\sigma = 0.0728\,\mathrm{N\,m^{-1}}$, $\gamma_c = 0$, $\rho_w = 1000\,\mathrm{kg\,m^{-3}}$ and $g = 9.81\,\mathrm{m\,s^{-2}}$, $r_c = 1.49\times10^{-5}\,\mathrm{m^2}/h = 0.149\,\mathrm{cm^2}/h$: the pores emptied at a suction h are those wider than r_c.

S3.11 Derivation: Depletion and the trigger law

The plant-available water lies between field capacity (100 cm) and the wilting point (15 849 cm). The depletion at suction h is the share of that water that has been used, $f(h) = [S_e(100) - S_e(h)]/[S_e(100) - S_e(15\,849)]$, because $\theta - \theta_r$ is proportional to S_e and the factor $\theta_s - \theta_r$ cancels. The suction that corresponds to a chosen depletion f^* is found by inverting: $S_e(h^*) = S_e(100) - f^*[S_e(100) - S_e(15\,849)]$ and $h^* = \alpha^{-1}(S_e^{-1/m} - 1)^{1/n}$. A soil-independent depletion therefore

gives a soil-specific suction, which is why one suction trigger uses different shares of the available water in different soils.

S3.12 Why it happens

Why the same trigger uses different shares of the available water. A suction trigger marks a point on the retention curve. In a sand most available water is released at low suctions, so a -300 cm trigger is reached late (much of the water used); in a clay the water is released over a wide range, so the same suction is reached early.

Why the daily model fails in sand at dry triggers. In sand dry triggers lie beyond the suction the crop can produce in the soil, so the full simulation seldom irrigates, while the daily bucket, which has no conductivity limit, still reaches the trigger.

S4. Rainfall and interception

The interception store of the daily model and the drainage that follows rain.

S4.1 Contribution 2: an interception store that matches the annual model

The problem

The daily model must lose to canopy interception the same water per year as the annual model, although its rain falls in random daily amounts. How large must the daily interception store be?

How the author developed it

Daily rain was taken as exponential with mean μ_0; the throughfall above a store I_c was integrated over that distribution; the result was set equal to the throughfall fraction of the annual model and solved for I_c.

The result

How it is derived, line by line

1. Daily rain exponential:

$$p(R) = \mu_0^{-1} e^{-R/\mu_0}$$

2. Expected throughfall above I_c:

$$\int_{I_c}^{\infty} (R - I_c) p(R)\, \mathrm{d}R = \mu_0 e^{-I_c/\mu_0}$$

3. Throughfall fraction:

$$r = e^{-I_c/\mu_0}$$

4. Solve for I_c:

$$I_c = \mu_0 \ln(1/r)$$

The result, with its parts marked

$$\underbrace{\mathrm{E}[(R - I_c)^+] = \mu_0 e^{-I_c/\mu_0}}_{(1)} \;\Rightarrow\; \underbrace{r = e^{-I_c/\mu_0}}_{(2)} \;\Rightarrow\; \underbrace{I_c = \mu_0 \ln \frac{1}{r}}_{(3)}$$

developed by the author

Pinpoints: what each circled mark shows and why

(1) Expected throughfall of an exponential daily rain of mean μ_0 above a store I_c, obtained by integrating the exponential density.

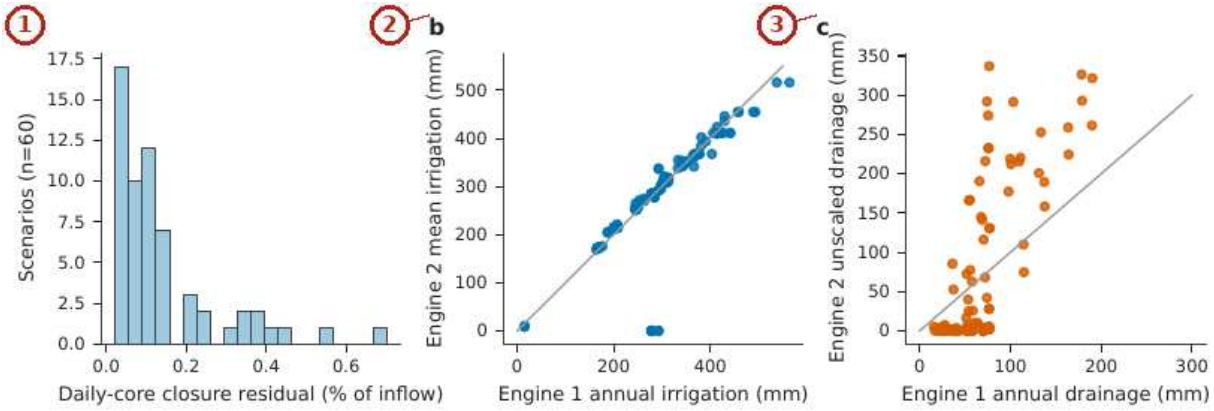
(2) The throughfall fraction; it is set equal to the throughfall of the annual model.

(3) The store follows in closed form: with $\mu_0 = 4.5$ mm and $r = 0.9$, $I_c = 0.47$ mm. The two engines therefore agree on interception by construction.

S4.2 Worked numbers, part 3: interception by the canopy

If daily rain is exponential with mean μ_0, a store I_c lets through the fraction $r = e^{-I_c/\mu_0}$, so $I_c = \mu_0 \ln(1/r)$. *Example:* $\mu_0 = 4.5$ mm and $r = 0.9$ give $I_c = 4.5 \times 0.105 = 0.47$ mm.

S4.3 Reading Figure 9 of the article



What the numbered labels point to.

(1) Panel (a): Closure residual of the unscaled daily core over 60 random states. *Why:* The daily model conserves water by construction, so the residual is near zero.

(2) Panel (b): Mean annual irrigation of Engine 2 against Engine 1 for the 100 scenarios of the realistic grid (10 soils × 5 triggers × 2 methods, cereal), including the failures at zero. *Why:* Irrigation of the two engines agrees except in sand at dry triggers, where the daily bucket reaches triggers that the full model cannot.

(3) Panel (c): Unscaled Engine 2 drainage against Engine 1 for the same scenarios. *Why:* Drainage agrees in order but the bucket drains in steps, so levels differ.

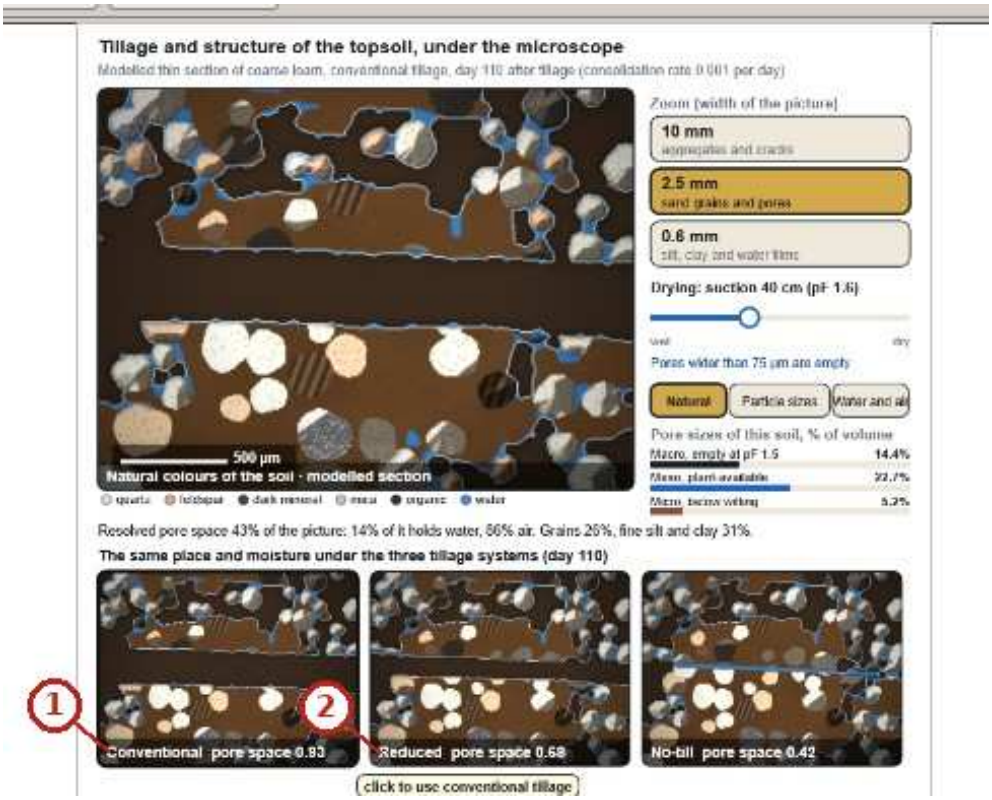
What it shows. Engine 2. (a) Closure residual of the unscaled daily core over 60 random states. (b) Mean annual irrigation of Engine 2 against Engine 1 for the 100 scenarios of the realistic grid (10 soils × 5 triggers × 2 methods, cereal), including the failures at zero. (c) Unscaled Engine 2 drainage against Engine 1 for the same scenarios.

How to read the labels, symbols and colours. (a) Closure residual over 60 states; (b) irrigation and (c) drainage of Engine 2 against Engine 1 for 100 scenarios.

What affects it, and why it happens. The residual is near zero because the daily model conserves water by construction; agreement with Engine 1 fails in sand at dry triggers where the daily bucket reaches triggers the full model cannot.

How it is made. Figure drawn by the script from the input data. Script: `Manuscripts/reproducibility/analysis/p3_figs.py`. Input: `Data/daisy_ensemble.csv.gz`.

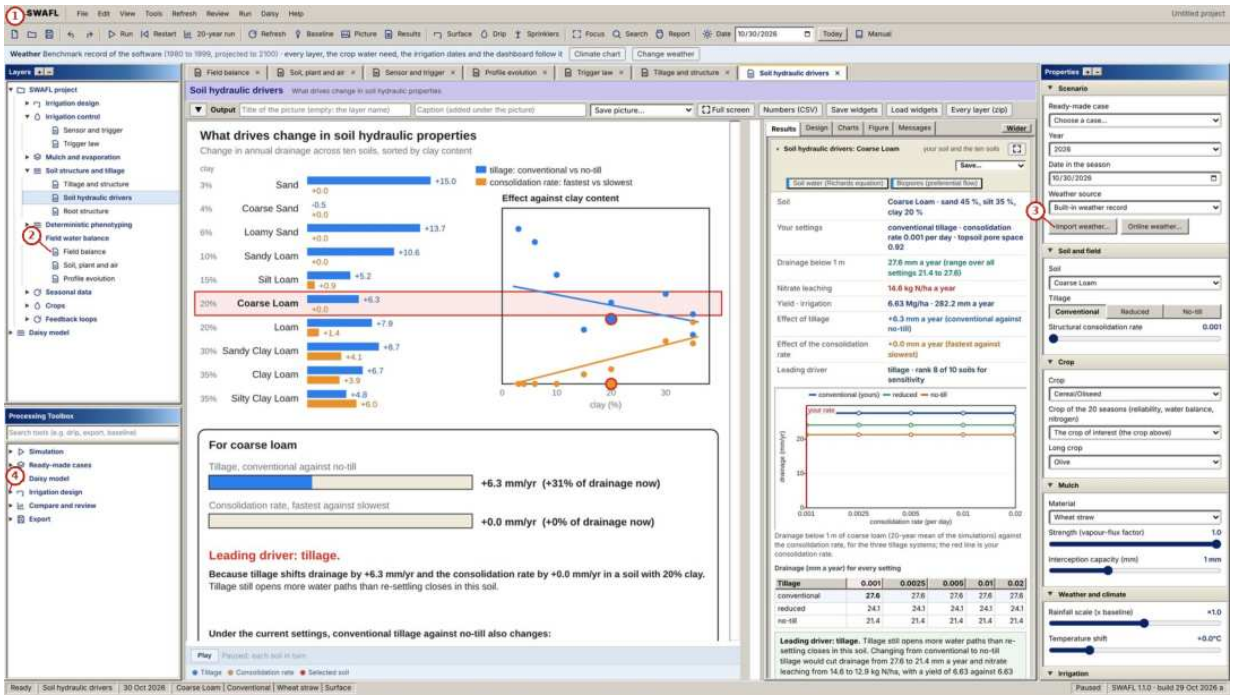
S4.4 Reading Figure 16 of the article



What the numbered labels point to (picture 1 of 2: Tillage and structure: modelled thin section of the topsoil with the water- and air-filled pores at the chosen suction, the pore sizes, and the same place under conventional, reduced and no tillage (Eq. (3) of the article)).

(1) "Conventional": a heading or control of this screen; the caption above says which layer it belongs to.

(2) "Reduced pore 0.68": a heading or control of this screen; the caption above says which layer it belongs to.



What the numbered labels point to (picture 2 of 2: Soil hydraulic drivers: tillage and consolidation effects on drainage for the ten soils and their regressions on clay content (Eq. (5) of the article).).

1. "SWAFL View Run Daisy Help": the menu bar of the program (file, view, run, Daisy, help).
2. "Field balance": the field water balance: inputs, outputs and storage of the root zone.
3. "Import weather... Online weather": the weather of the project: built-in record, a file of the user or online weather of any place.
4. "design": a heading or control of this screen; the caption above says which layer it belongs to.

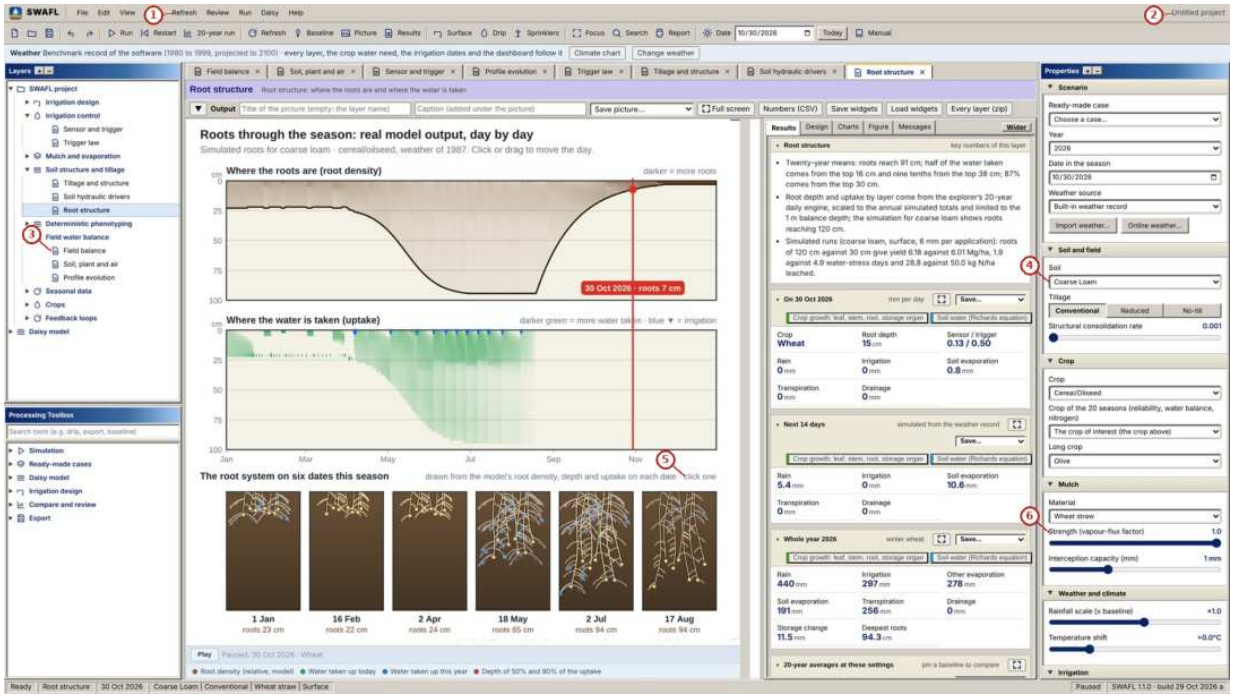
What it shows. SWAFL Desktop: soil structure and hydraulic drivers. Panels: (a) Tillage and structure: modelled thin section of the topsoil with the water- and air-filled pores at the chosen suction, the pore sizes, and the same place under conventional, reduced and no tillage (Eq. (3) of the article). (b) Soil hydraulic drivers: tillage and consolidation effects on drainage for the ten soils and their regressions on clay content (Eq. (5) of the article).

How to read the labels, symbols and colours. A screen of the software as it appears to the user; the labelled areas are the inputs, the figure and the numbers of the layer named in the caption.

What affects it, and why it happens. Tillage opens large pores that hold little water and drain fast; consolidation closes them over time, which changes drainage.

How it is made. Screen capture of SWAFL Desktop 1.1.0 (not regenerated).

S4.5 Reading Figure 17 of the article



What the numbered labels point to.

1. “Refresh Review Run Help”: a heading or control of this screen; the caption above says which layer it belongs to.
2. “Untitled project”: a heading or control of this screen; the caption above says which layer it belongs to.
3. “Field balance”: the field water balance: inputs, outputs and storage of the root zone.
4. “Coarse Loam v”: the soil of the project, which every layer uses.
5. “click one”: a heading or control of this screen; the caption above says which layer it belongs to.
6. “Strength (vapour-flux factor)”: a heading or control of this screen; the caption above says which layer it belongs to.

What it shows. SWAFL Desktop: roots. Panels: (a) Root structure: rooting depth and the depths holding 50 and 90 % of uptake in the ten soils (Eq. (17) and Eq. (18) and Eq. (19) and Eq. (20) of the article), with the E15 rooting-depth heat maps.

How to read the labels, symbols and colours. A screen of the software as it appears to

the user; the labelled areas are the inputs, the figure and the numbers of the layer named in the caption.

What affects it, and why it happens. Roots go deeper in soils that let them and where water is deeper; uptake follows root density.

How it is made. Screen capture of SWAFL Desktop 1.1.0 (not regenerated).

S4.6 Derivation: Rainfall and interception of the daily model

If daily rain depths are exponential with mean μ_0, the expected throughfall above an interception store I_c is $\mathrm{E}[(R - I_c)^+] = \int_{I_c}^{\infty}(R - I_c)\mu_0^{-1}e^{-R/\mu_0}\mathrm{d}R = \mu_0 e^{-I_c/\mu_0}$. The throughfall fraction is $r = e^{-I_c/\mu_0}$, hence $I_c = \mu_0 \ln(1/r)$: the store is chosen so that the expected interception equals the annual interception of the annual model.

S4.7 Why it happens

Why drainage differs under the same rain. Drainage is the water that exceeds what the profile holds and the crop uses; profiles that hold less (sands) and conduct more pass more water below 1 m.

S5. Evaporation and the soil–plant–atmosphere path

The two stages of soil evaporation and the water potential from the soil to the air.

S5.1 Contribution 3: two evaporation stages joined without a jump

How the author developed it

Stage 1 runs at the demand e_0 until the stage-1 limit Q_1; for stage 2 a square-root law of diffusion was taken, and its two constants were set so that the rate equals e_0 at the change of stage and the long-time behaviour is $\sqrt{t}$.

The result

How it is derived, line by line

1. **Stage 1 at demand:**

$$Q = q_0 + e_0 t,\ t \le t_1$$

2. **Stage 2, square-root form:**

$$Q = Q_1 + A(\sqrt{1 + (t - t_1)/\tau_E} - 1)$$

3. **Rate continuous at t_1:**

$$\mathrm{d}Q/\mathrm{d}t|_{t_1} = A/(2\tau_E) = e_0 \Rightarrow A = 2e_0\tau_E$$

The result, with its parts marked

$$Q(t) = Q_1 + \underbrace{2e_0\tau_E}_{(1)}\left(\sqrt{1 + \underbrace{(t - t_1)/\tau_E}_{(2)}} - 1\right)$$

$$\underbrace{\left.\frac{\mathrm{d}Q}{\mathrm{d}t}\right|_{t_1} = e_0}_{(3)}$$

$$\underbrace{Q \propto \sqrt{t}\ \ (t - t_1 \gg \tau_E)}_{(4)}$$

developed by the author

Pinpoints: what each circled mark shows and why

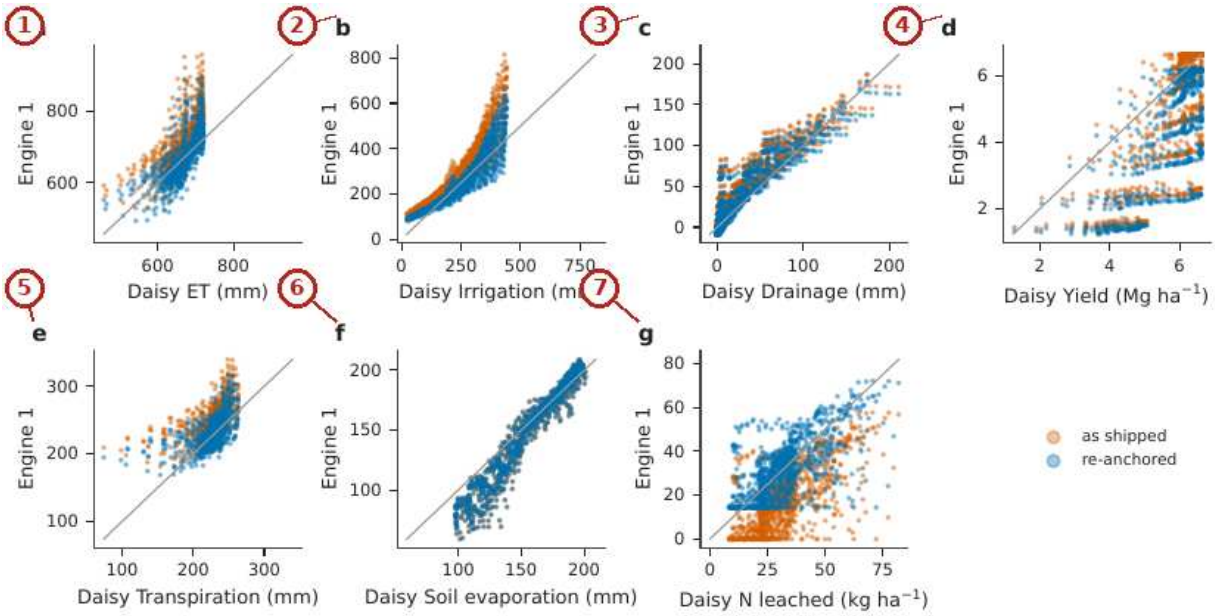
(1) The amplitude of stage 2, chosen so that the rate is continuous.

(2) Time since the end of stage 1, in units of the transition time τ_E.

(3) At the start of stage 2 the rate equals the stage-1 rate: no artificial jump in evaporation.

(4) Long after, cumulative evaporation grows as the square root of time, the diffusion-limited law of stage 2.

S5.2 Reading Figure 8 of the article



What the numbered labels point to.

1. Panel (a): ET. *Why:* Points follow the order of Daisy because the meta-model is built from Daisy runs; the offset before re-anchoring comes from anchoring to one reference run.
2. Panel (b): irrigation. *Why:* Points follow the order of Daisy because the meta-model is built from Daisy runs; the offset before re-anchoring comes from anchoring to one reference run.
3. Panel (c): drainage. *Why:* Points follow the order of Daisy because the meta-model is built from Daisy runs; the offset before re-anchoring comes from anchoring to one reference run.
4. Panel (d): yield. *Why:* Points follow the order of Daisy because the meta-model is built from Daisy runs; the offset before re-anchoring comes from anchoring to one reference run.
5. Panel (e): transpiration. *Why:* Points follow the order of Daisy because the meta-model is built from Daisy runs; the offset before re-anchoring comes from anchoring to one reference run.
6. Panel (f): soil evaporation. *Why:* Points follow the order of Daisy because the meta-model is built from Daisy runs; the offset before re-anchoring comes from anchoring to one reference run.
7. Panel (g): nitrogen leached. Grey lines are 1:1. *Why:* Points follow the order of Daisy because the meta-model is built from Daisy runs; the offset before re-anchoring comes from anchoring to one reference run.

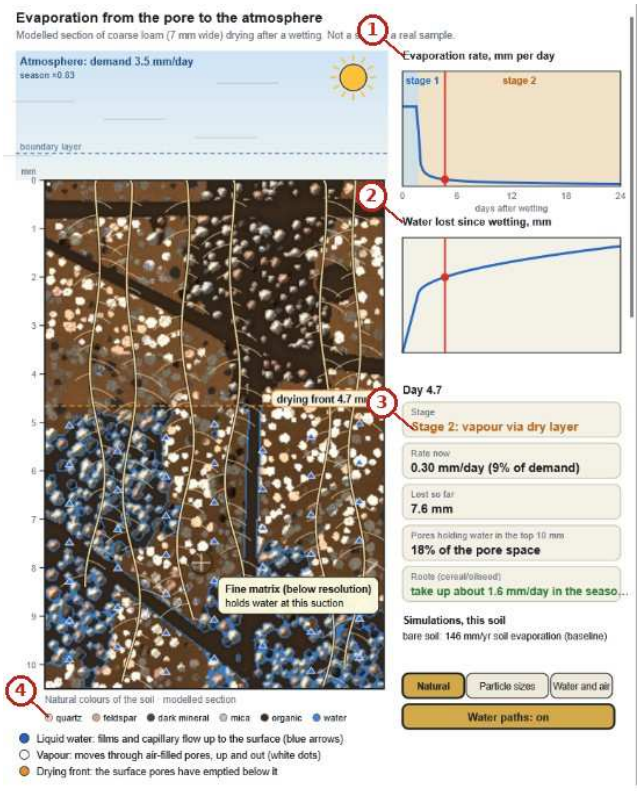
What it shows. Engine 1 versus Daisy for the 1 000 held-out E12 runs. Orange: as shipped; blue: re-anchored. (a) ET, (b) irrigation, (c) drainage, (d) yield, (e) transpiration, (f) soil evaporation, (g) nitrogen leached. Grey lines are 1:1.

How to read the labels, symbols and colours. Engine 1 against Daisy for 1 000 held-out runs, seven outputs; orange as shipped, blue re-anchored; grey 1:1.

What affects it, and why it happens. Points follow the order of Daisy (good ranking) but are offset as shipped because the meta-model is anchored to one reference run; re-anchoring removes most of the offset.

How it is made. Figure drawn by the script from the input data. Script: `Manuscripts/reproducibility/analysis/p3_figs.py`. Input: `Data/daisy_ensemble.csv.gz`.

S5.3 Reading Figure 18 of the article



What the numbered labels point to (picture 1 of 2: Evaporation from the pore to the atmosphere: a modelled section of coarse loam drying after a wetting, the drying front, stage-1 and stage-2 evaporation and the water lost since wetting (Eq. (11) and Eq. (12) of the article)).

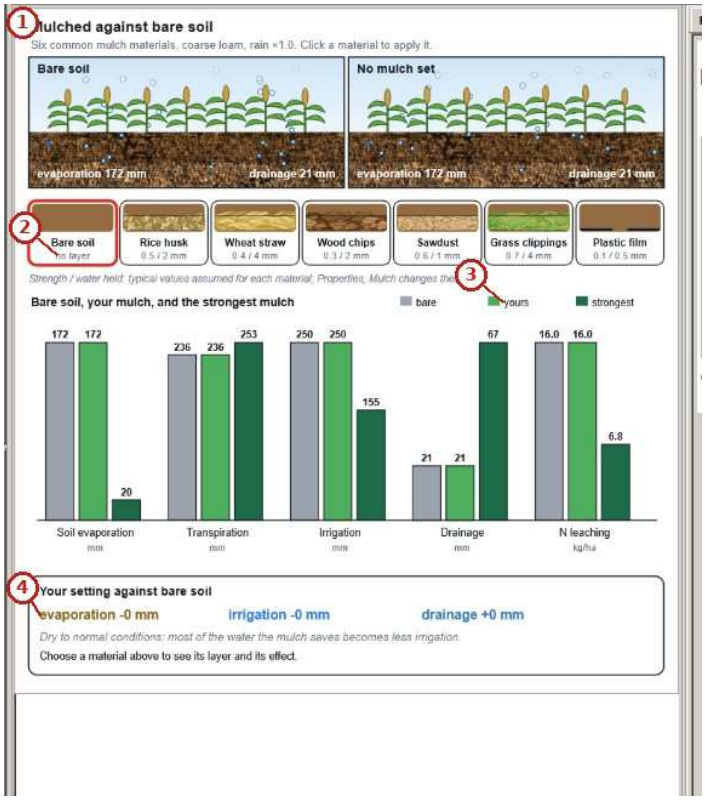
(1) "Evaporation rate, mm per day": evaporation from the soil (and, where stated, from mulch

or canopy).

(2) “Water lost since wetting”: a heading or control of this screen; the caption above says which layer it belongs to.

(3) “Stage 2: vapour via dry layer”: a heading or control of this screen; the caption above says which layer it belongs to.

(4) “O quartz mineral”: a heading or control of this screen; the caption above says which layer it belongs to.



What the numbered labels point to (picture 2 of 2: Mulch against bare soil: six mulch materials, and soil evaporation, transpiration, irrigation, drainage and nitrate leaching under bare soil, the chosen mulch and the strongest mulch (Eq. (13) of the article)).

(1) “Mulched against bare soil”: the surface cover and its effect on evaporation.

(2) “no layer”: a heading or control of this screen; the caption above says which layer it belongs to.

(3) “yours”: a heading or control of this screen; the caption above says which layer it belongs to.

(4) “evaporation -0 mm”: evaporation from the soil (and, where stated, from mulch or canopy).

What it shows. SWAFL Desktop: evaporation and mulch. Panels: (a) Evaporation from the pore to the atmosphere: a modelled section of coarse loam drying after a wetting, the drying

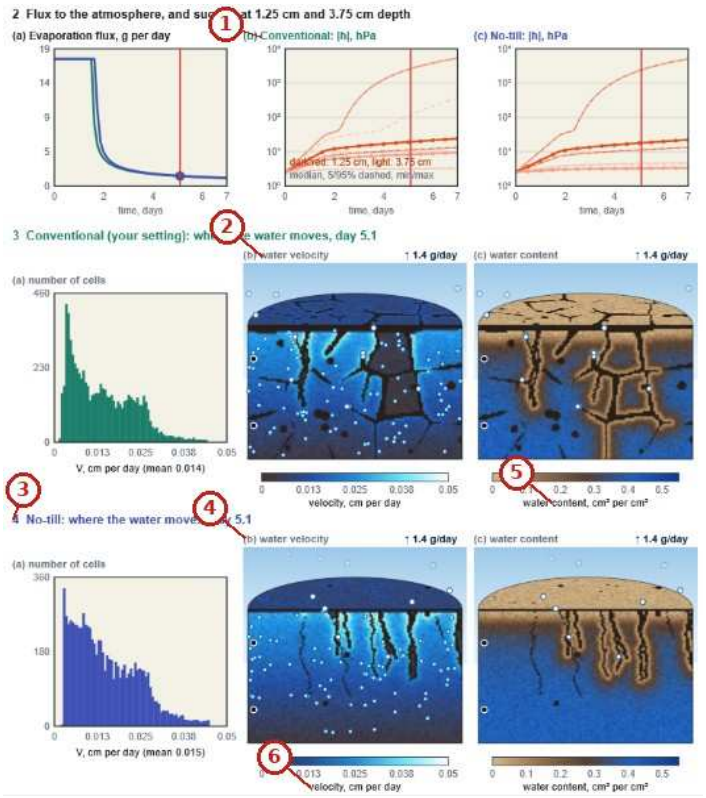
front, stage-1 and stage-2 evaporation and the water lost since wetting (Eq. (11) and Eq. (12) of the article). (b) Mulch against bare soil: six mulch materials, and soil evaporation, transpiration, irrigation, drainage and nitrate leaching under bare soil, the chosen mulch and the strongest mulch (Eq. (13) of the article).

How to read the labels, symbols and colours. A screen of the software as it appears to the user; the labelled areas are the inputs, the figure and the numbers of the layer named in the caption.

What affects it, and why it happens. Evaporation is fast while the surface is wet (stage 1) and slows when a dry layer forms (stage 2); a mulch reduces the vapour flux through it.

How it is made. Screen capture of SWAFL Desktop 1.1.0 (not regenerated).

S5.4 Reading Figure 31 of the article



What the numbered labels point to.

1. "Conventional": a heading or control of this screen; the caption above says which layer it belongs to.
2. "water velocity": a heading or control of this screen; the caption above says which layer it belongs to.

(3) "4 where the water moves": a heading or control of this screen; the caption above says which layer it belongs to.

(4) "(b) water velocity": a heading or control of this screen; the caption above says which layer it belongs to.

(5) "content, per cm?": a heading or control of this screen; the caption above says which layer it belongs to.

(6) "velocity, cm per day": a rate in millimetres per day.

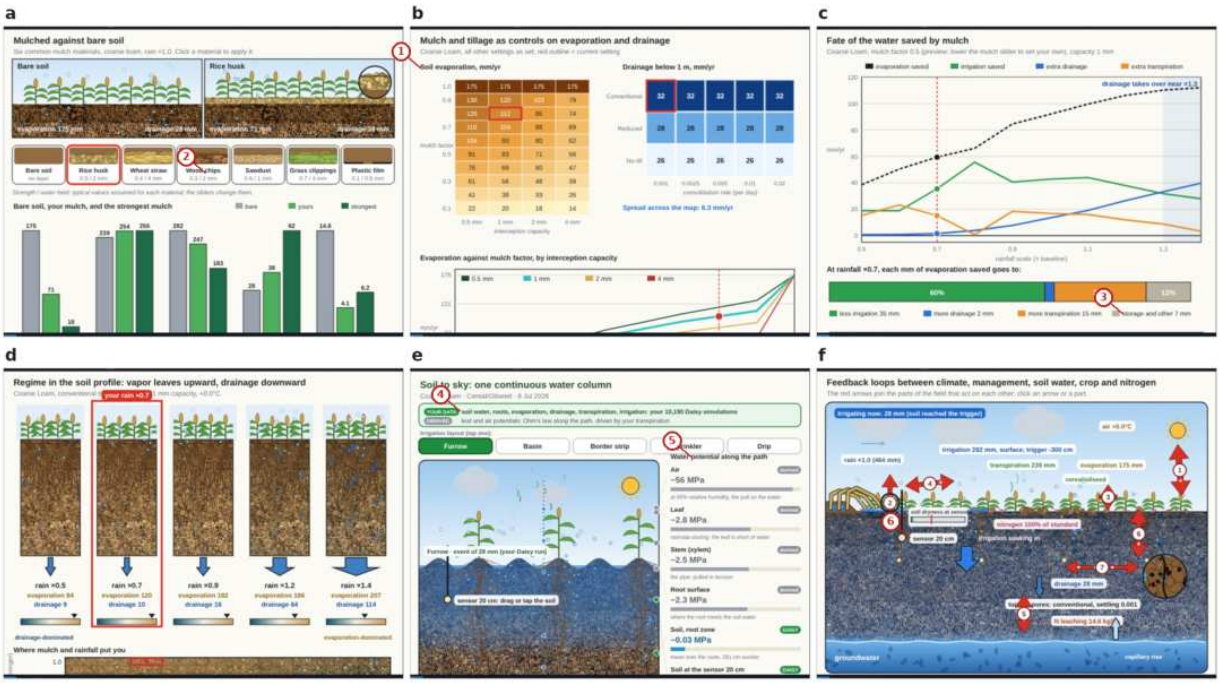
What it shows. SWAFL Desktop: pore-scale drying of the topsoil under conventional and no tillage: the evaporation flux and the suction at 1.25 and 3.75 cm, and the distribution, velocity and content of the water on day 5.1 (screen capture of release 1.1.0).

How to read the labels, symbols and colours. A screen of the software as it appears to the user; the labelled areas are the inputs, the figure and the numbers of the layer named in the caption.

What affects it, and why it happens. Under no-till the topsoil keeps more connected pores, so water moves to the surface differently and the suction rises more slowly.

How it is made. Screen capture of SWAFL Desktop 1.1.0 (not regenerated).

S5.5 Reading Figure 5 of the article



What the numbered labels point to.

1. "Soil evaporation": evaporation from the soil (and, where stated, from mulch or canopy).
2. "chips": a heading or control of this screen; the caption above says which layer it belongs to.
3. "storage and other 7 mm": a heading or control of this screen; the caption above says which layer it belongs to.
4. "soil water, roots, evaporation": evaporation from the soil (and, where stated, from mulch or canopy).
5. "potential along the path": a heading or control of this screen; the caption above says which layer it belongs to.
6. "sensor": the soil-water sensor: its depth and reading.

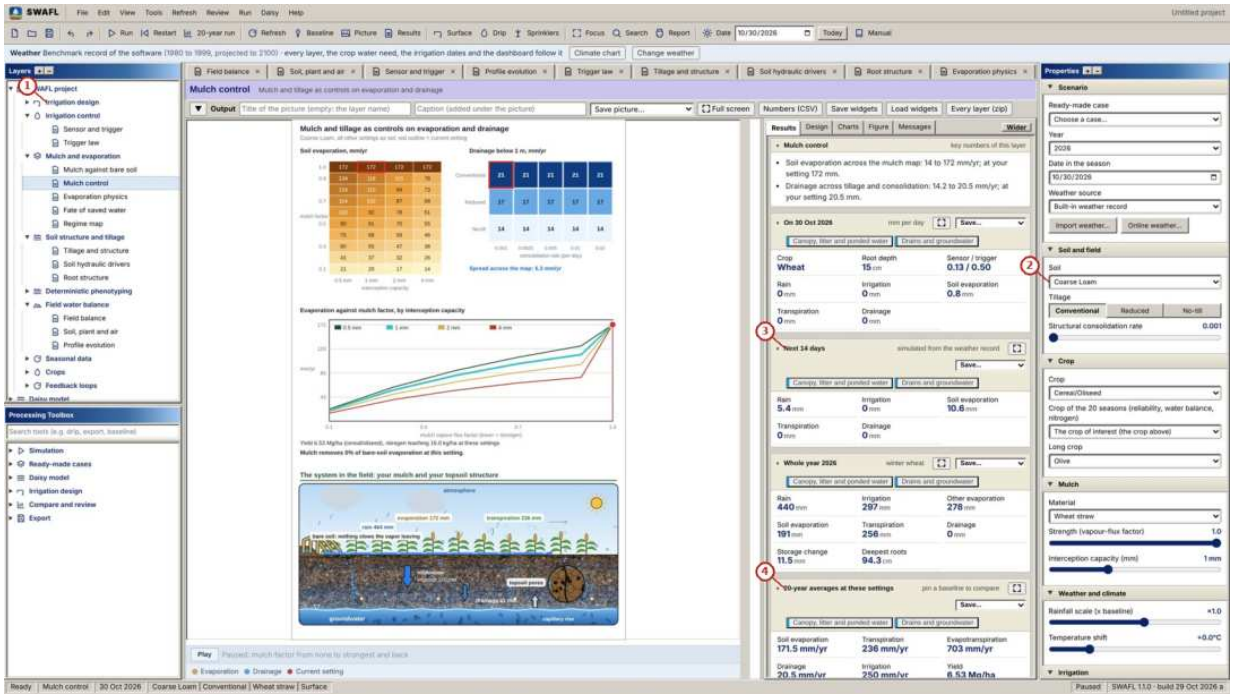
What it shows. Result views for mulch and the whole system. (a) Mulch against bare soil; (b) mulch strength; (c) fate of the saved water; (d) regime map of where mulch pays; (e) soil to sky; (f) how the effects connect.

How to read the labels, symbols and colours. Views for mulch and the whole system.

What affects it, and why it happens. A mulch lowers evaporation and catches rain; where the saving goes depends on how fast the soil drains. A mulch reduces evaporation, intercepts rain and warms the soil; where the saving goes depends on the soil's drainage.

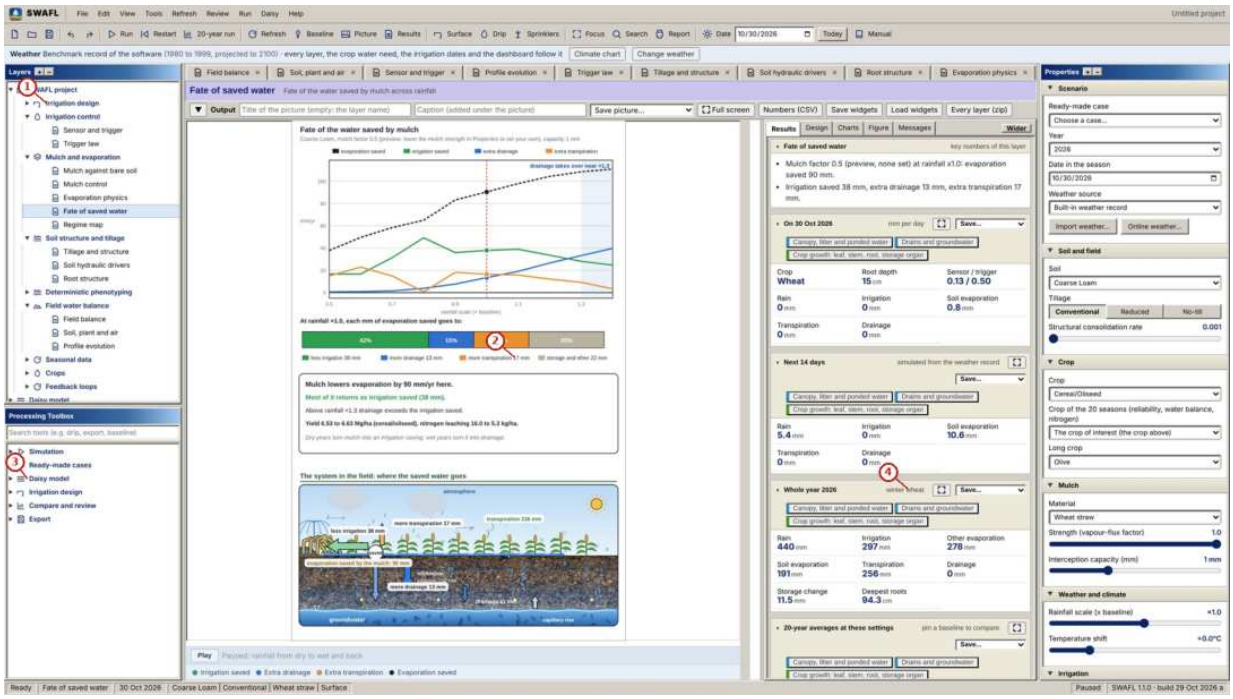
How it is made. Screen capture of SWAFL Desktop 1.1.0 (not regenerated).

S5.6 Reading Figure 19 of the article



What the numbered labels point to (picture 1 of 2: Mulch control: soil evaporation by mulch factor and interception capacity, and drainage by tillage and consolidation rate.).

1. "Irrigation design": the irrigation-design group of the Layers tree and its layers.
2. "coarse Loam": the soil of the project, which every layer uses.
3. "Next 14 days": the decision for the next 14 days, from the soil water of the day.
4. "20-year averages at these settings": a heading or control of this screen; the caption above says which layer it belongs to.



What the numbered labels point to (picture 2 of 2: Fate of saved water: evaporation saved, irrigation saved, extra drainage and extra transpiration against the rainfall scale (Eq. (23) of the article).).

1. "Irrigation design": the irrigation-design group of the Layers tree and its layers.
2. "17 mm storage and other 22": a heading or control of this screen; the caption above says which layer it belongs to.
3. "Daisy": the Daisy model and its workbench.
4. "wheat": the crop of the project and its stage.

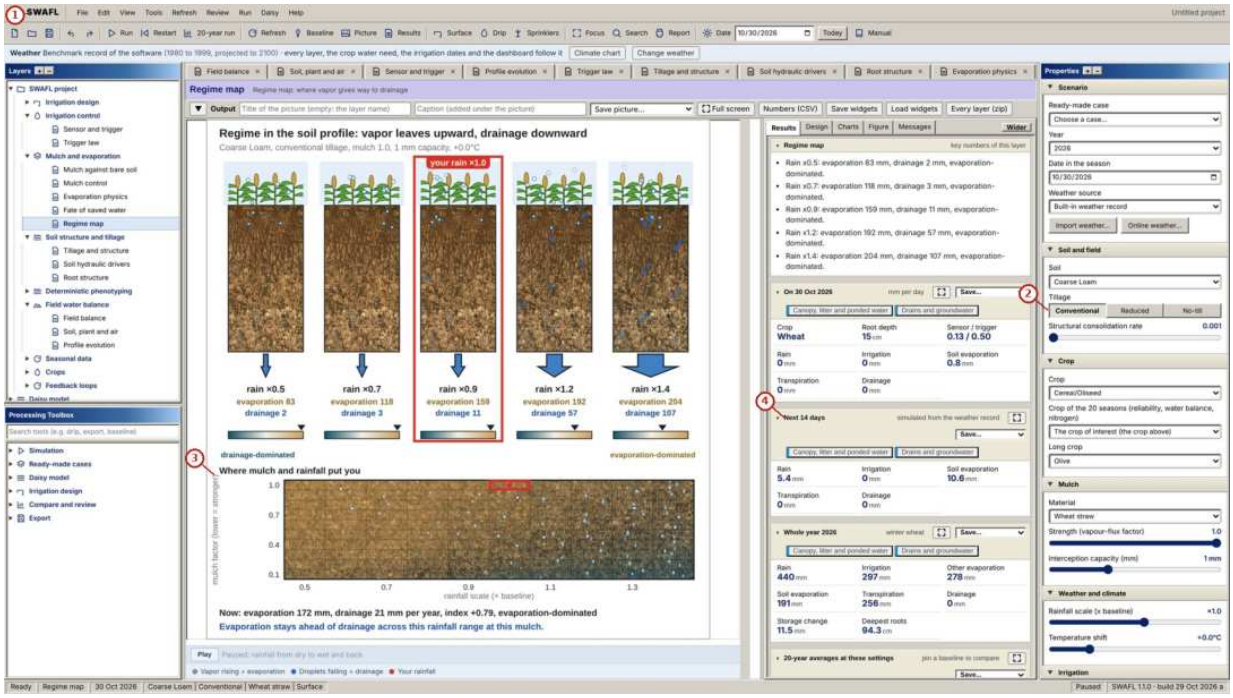
What it shows. SWAFL Desktop: mulch control and the fate of saved water. Panels: (a) Mulch control: soil evaporation by mulch factor and interception capacity, and drainage by tillage and consolidation rate. (b) Fate of saved water: evaporation saved, irrigation saved, extra drainage and extra transpiration against the rainfall scale (Eq. (23) of the article).

How to read the labels, symbols and colours. A screen of the software as it appears to the user; the labelled areas are the inputs, the figure and the numbers of the layer named in the caption.

What affects it, and why it happens. The water a mulch saves goes to transpiration, less irrigation or drainage, in shares set by the soil.

How it is made. Screen capture of SWAFL Desktop 1.1.0 (not regenerated).

S5.7 Reading Figure 20 of the article



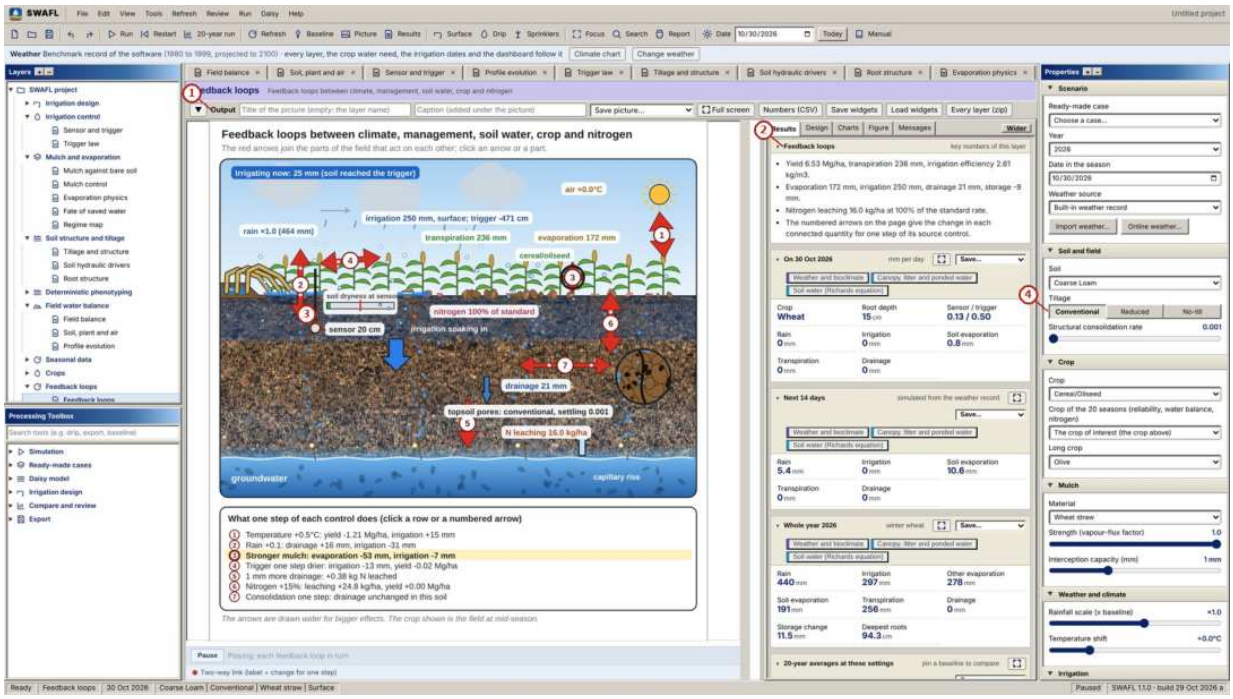
What the numbered labels point to (picture 1 of 2: Regime map: mulch factor against rainfall scale coloured by the dominant loss (Eq. (14) of the article).).

(1) "SWAFL View Run Daisy Help": the menu bar of the program (file, view, run, Daisy, help).

(2) "No-till": a heading or control of this screen; the caption above says which layer it belongs to.

(3) "Where mulch and rainfall pat you": rain, from the weather of the project.

(4) "Next 14 days": the decision for the next 14 days, from the soil water of the day.



What the numbered labels point to (picture 2 of 2: Feedback loops: the seven loops with arrow widths scaled by the effect of one control step (Table 4 of the article).).

1. "Output": the output bar of the layer: title and caption of the picture, save the picture, full screen, numbers as CSV.
2. "Feedback loops": the feedback loops between climate, management, soil water, crop and nitrogen.
3. "sensor 20 cm": the soil-water sensor: its depth and reading.
4. "No-till": a heading or control of this screen; the caption above says which layer it belongs to.

What it shows. SWAFL Desktop: regime map and feedback loops. Panels: (a) Regime map: mulch factor against rainfall scale coloured by the dominant loss (Eq. (14) of the article). (b) Feedback loops: the seven loops with arrow widths scaled by the effect of one control step (Table 4 of the article).

How to read the labels, symbols and colours. A screen of the software as it appears to the user; the labelled areas are the inputs, the figure and the numbers of the layer named in the caption.

What affects it, and why it happens. The regime map shows where the saved water drains (coarse soils, wet climate) and where it is used.

How it is made. Screen capture of SWAFL Desktop 1.1.0 (not regenerated).

S5.8 Derivation: Consolidation of the tilled layer

If the excess pore space decays at a rate proportional to itself, $\mathrm{d}p/\mathrm{d}t = -\tau_c(p - p_c)$, then $p(t) = p_c + (p_0 - p_c)e^{-\tau_c t}$, with p_0 set by the tillage and p_c the consolidated state.

S5.9 Derivation: Two-stage evaporation

In stage 1 the surface is wet and evaporation runs at the demand e_0, so $Q = q_0 + e_0 t$ until the cumulative stage-1 limit Q_1 is reached at $t_1 = (Q_1 - q_0)/e_0$. In stage 2 evaporation is limited by diffusion through the dry layer; for a desorptivity-controlled flux $e = S/(2\sqrt{t})$ the cumulative loss grows as $\sqrt{t}$. The form of the paper, $Q = Q_1 + 2e_0\tau_E(\sqrt{1 + (t - t_1)/\tau_E} - 1)$, starts stage 2 at the stage-1 rate (its derivative at $t = t_1$ is e_0) and tends to the $\sqrt{t}$ law for $t - t_1 \gg \tau_E$, which joins the two stages smoothly (Philip, 1957; Ritchie, 1972).

S5.10 Derivation: Water potential of the air

For water vapour in equilibrium with liquid water, the chemical potential difference per unit volume is $\psi = (\mathcal{R}T_K/V_w)\ln\mathrm{RH}$ (Kelvin). At 20 °C, $\mathcal{R}T_K/V_w = 8.314 \times 293.15/1.8 \times 10^{-5} = 1.353 \times 10^8$ Pa, that is 135.3 MPa, so $\psi_{\mathrm{air}} \simeq 135.3 \ln\mathrm{RH}$ MPa.

S5.11 Derivation: The soil–plant–atmosphere path

With steady flow q through soil, root, xylem and leaf in series, each segment drops the potential by qR_j (the van den Honert analogue of Ohm's law), so $\psi_{\mathrm{leaf}} = \psi_{\mathrm{RZ}} - q(R_{\mathrm{soil}} + R_{\mathrm{root}} + R_{\mathrm{xyl}} + R_{\mathrm{leaf}})$. The soil resistance grows as the conductivity falls; bounding K_r from below (the floor of the paper) bounds R_{soil} and therefore the leaf potential, which is why that potential is set by the floor rather than by the soil when the soil is dry.

S5.12 Why it happens

Why stage-1 evaporation lasts longer in fine soils. Fine soils hold more water near the surface and conduct it upward over longer distances, so the surface stays wet longer.

Why the leaf water potential is set by a conductivity floor. The series resistance of the soil grows without bound as the soil dries; the software bounds the relative conductivity from below to keep the calculation finite, and in dry soil that bound, not the soil, then decides the leaf potential.

Why the drainage share of a mulch saving is understated in coarse soils. The engines split the saving with ratios that do not contain the soil's drainage competition; in coarse soils, where most saved water drains, that split gives too little to drainage.

S6. The model, its equations and its views

The architecture, the ledger of equations and the result views.

S6.1 Contribution 4: a ledger of every equation

The result

Origin	Equations	Meaning
(1) Literature	38	taken from a cited source
(2) Fitted to data	18	coefficients fitted to the ensemble
(3) Engineered	78	form chosen for the software

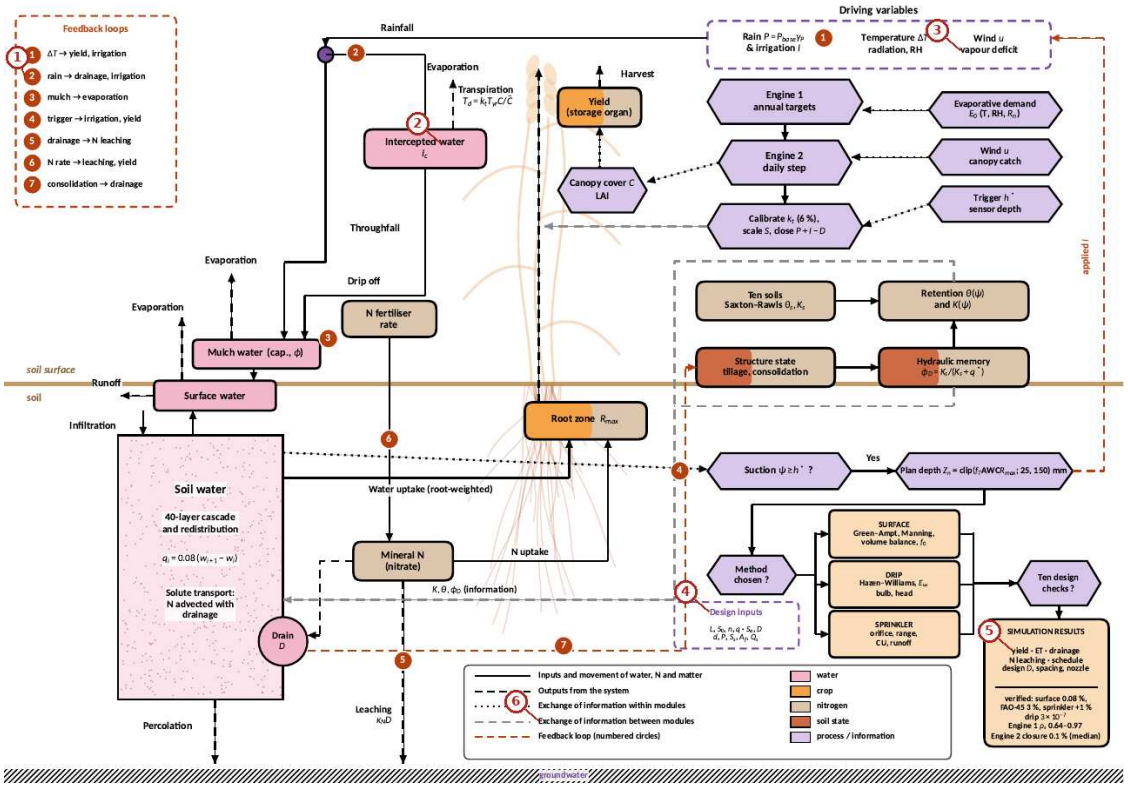
Pinpoints: what each circled mark shows and why

(1) These carry the authority of their source and need only to be implemented correctly.

(2) These hold only within the data they were fitted to; their range is stated.

(3) These are the largest class and the ones that most need testing; labelling them tells the reader where the evidence is weakest.

S6.2 Reading Figure 1 of the article



What the numbered labels point to.

1. "drainage": rain, from the weather of the project.
2. "water": a heading or control of this screen; the caption above says which layer it belongs to.
3. "vapour deficit": a heading or control of this screen; the caption above says which layer it belongs to.
4. "Design inputs": a heading or control of this screen; the caption above says which layer it belongs to.
5. "yield ET - drainage": rain, from the weather of the project.
6. "Exchange of information between modules": a heading or control of this screen; the caption above says which layer it belongs to.

What it shows. Schematic overview of the main processes in SWAFL Desktop. Boxes indicate storages and solid lines flows; dashed black lines are flows out of the system, dotted lines are flows of information and dashed grey lines exchange of information between modules. Dashed purple boxes with a triangle are the drop-down controls and driving variables of the Properties panel, and the drivers of soil hydraulic properties (SHP). Dashed red lines and numbered circles

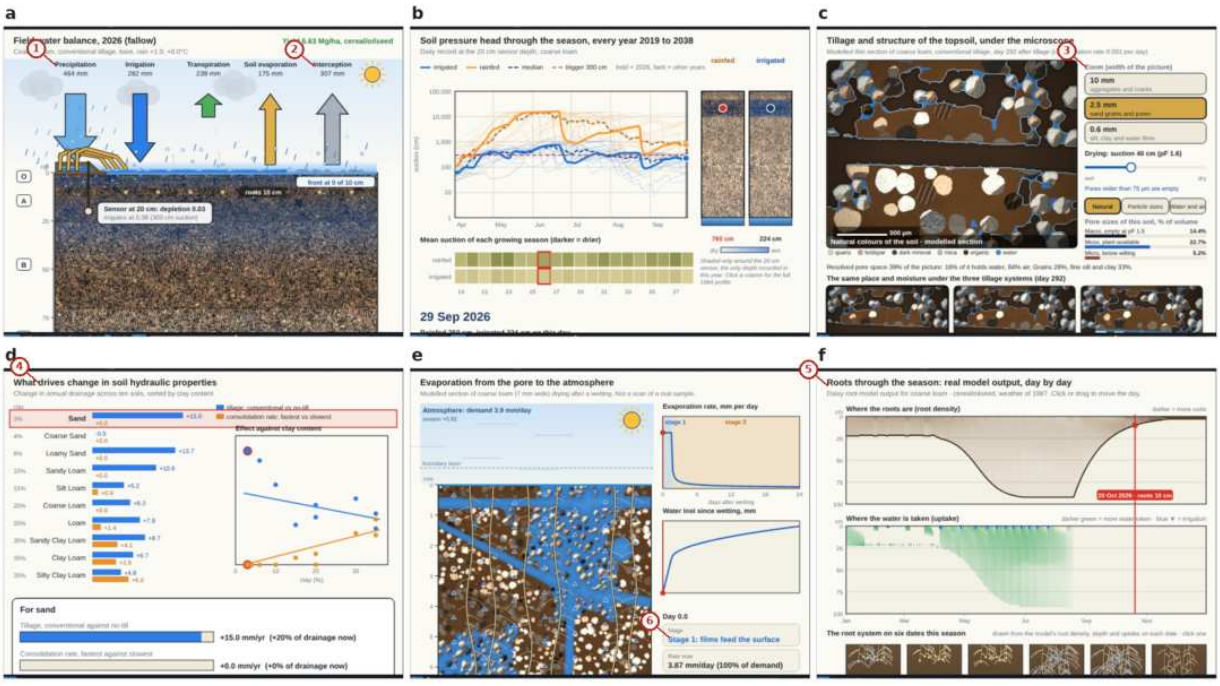
mark the seven feedback loops. Pink indicates water, orange the crop, light brown nitrogen and rust soil structure and hydraulic state; purple hexagons are processes and information (Engines 1 and 2, trigger, design layers) and peach boxes are design layers and results. The brown line is the soil surface, with groundwater hatched at the base.

How to read the labels, symbols and colours. Boxes are storages, solid lines flows, dotted lines information; dashed purple boxes the controls; red dashed lines with numbers the seven feedback loops; pink water, orange crop, brown nitrogen, rust structure.

What affects it, and why it happens. The diagram shows how one control (for example the trigger) reaches every output through the coupled flows and feedbacks.

How it is made. Figure drawn by the script from the input data. Script: `Manuscripts/reproducibility/make_a4.py`. Input: `Data/daisy_ensemble.csv.gz`.

S6.3 Reading Figure 3 of the article



What the numbered labels point to.

1. "Precipitation": rain, from the weather of the project.
2. "Interception": a heading or control of this screen; the caption above says which layer it belongs to.
3. "Zoom (width of the picture)": a heading or control of this screen; the caption above says

which layer it belongs to.

4 “drives change in soil properties”: the soil of the project, which every layer uses.

5 “Roots through the season: real model”: the roots: depth, density and uptake.

6 “stage 1: films feed the surface”: a heading or control of this screen; the caption above says which layer it belongs to.

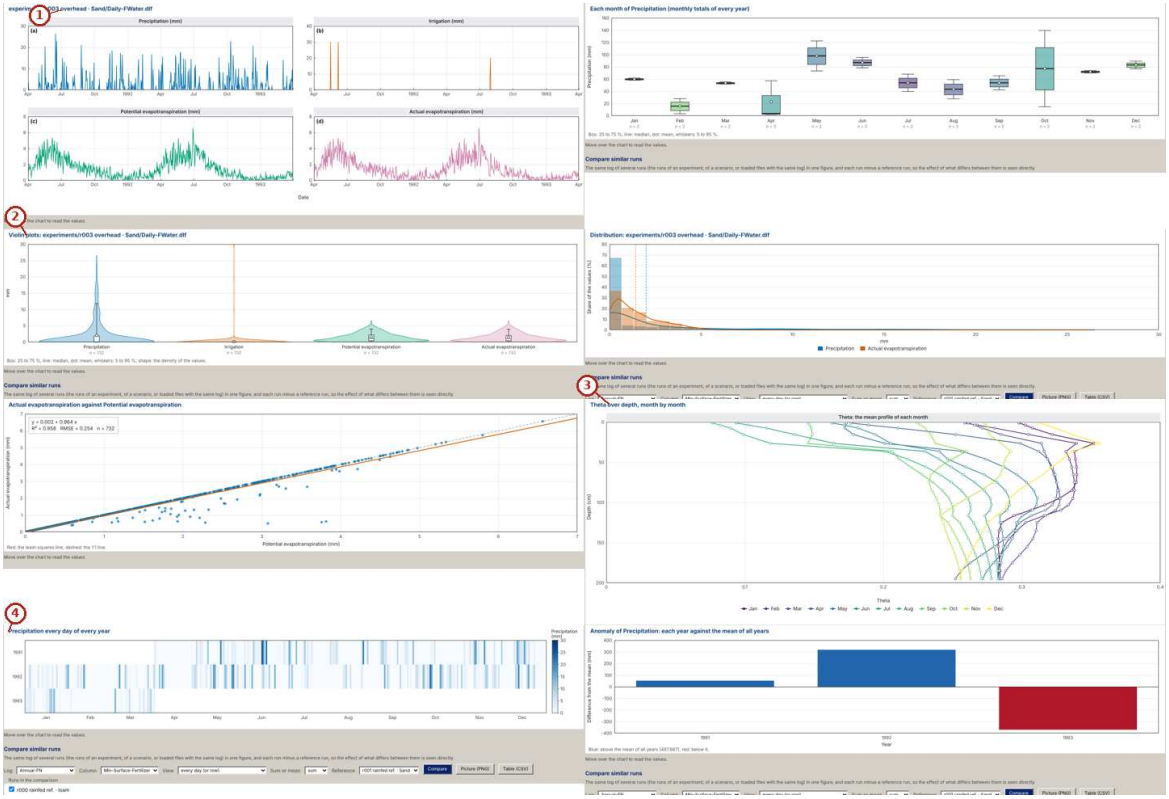
What it shows. Result views of the explorer for the soil and the field (coarse loam, conventional tillage, cereal rotation). (a) Field water balance of the root zone; (b) soil-water pressure-head profile; (c) pores and tillage; (d) soil hydraulic drivers by texture; (e) evaporation from the pore to the atmosphere; (f) root density and uptake. Governing relations are listed in Table 11 of the article.

How to read the labels, symbols and colours. Six result views of the program (field balance, pressure-head profile, pores and tillage, hydraulic drivers, evaporation, roots).

What affects it, and why it happens. Each view draws one relation of the engines for the soil chosen; a different soil changes every view at once. Each view draws one relation of the engines for the chosen soil; changing the soil or a control moves all of them together.

How it is made. Screen capture of SWAFL Desktop 1.1.0 (not regenerated).

S6.4 Reading Figure 12 of the article



What the numbered labels point to.

1. "overhead": a heading or control of this screen; the caption above says which layer it belongs to.
2. "plots: experiments/r003 overhead": a heading or control of this screen; the caption above says which layer it belongs to.
3. "over depth, month by month": a heading or control of this screen; the caption above says which layer it belongs to.
4. "Precipitation every day of every year": rain, from the weather of the project.

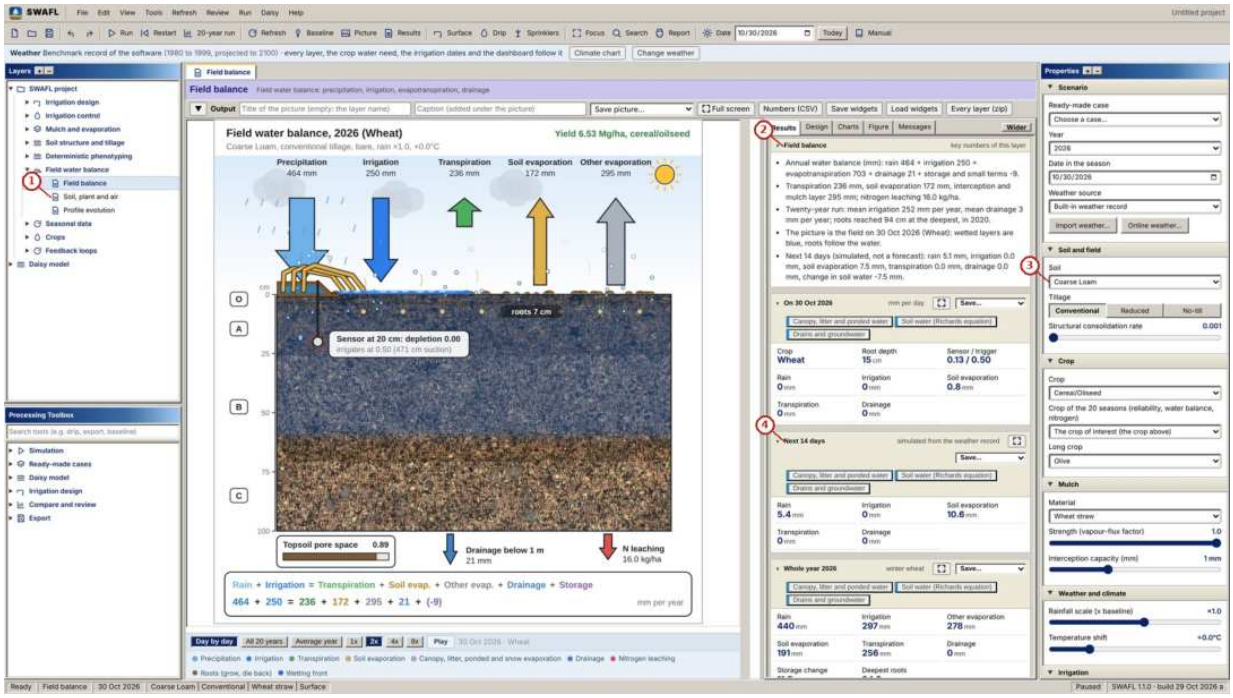
What it shows. Eight of the twenty kinds of figure of the Log plots of release 1.1.0, drawn from the daily logs of one Daisy run: small multiples, monthly box plots, violin plots, histogram with density, scatter with the least-squares line, R^2 and RMSE, the mean profile of each month, a calendar heat map and the anomaly of each year (screen captures).

How to read the labels, symbols and colours. Eight kinds of figure drawn from the daily logs of one run (small multiples, monthly boxes, violins, histogram, scatter with fit, monthly profile, calendar map, yearly anomaly).

What affects it, and why it happens. Each kind of plot shows the same daily log differently: through time, spread within months, distribution, relation to another quantity, departure of each year. Each shows the same log in a different way: course in time, spread within months, distribution, relation to another quantity and departure of each year from the mean.

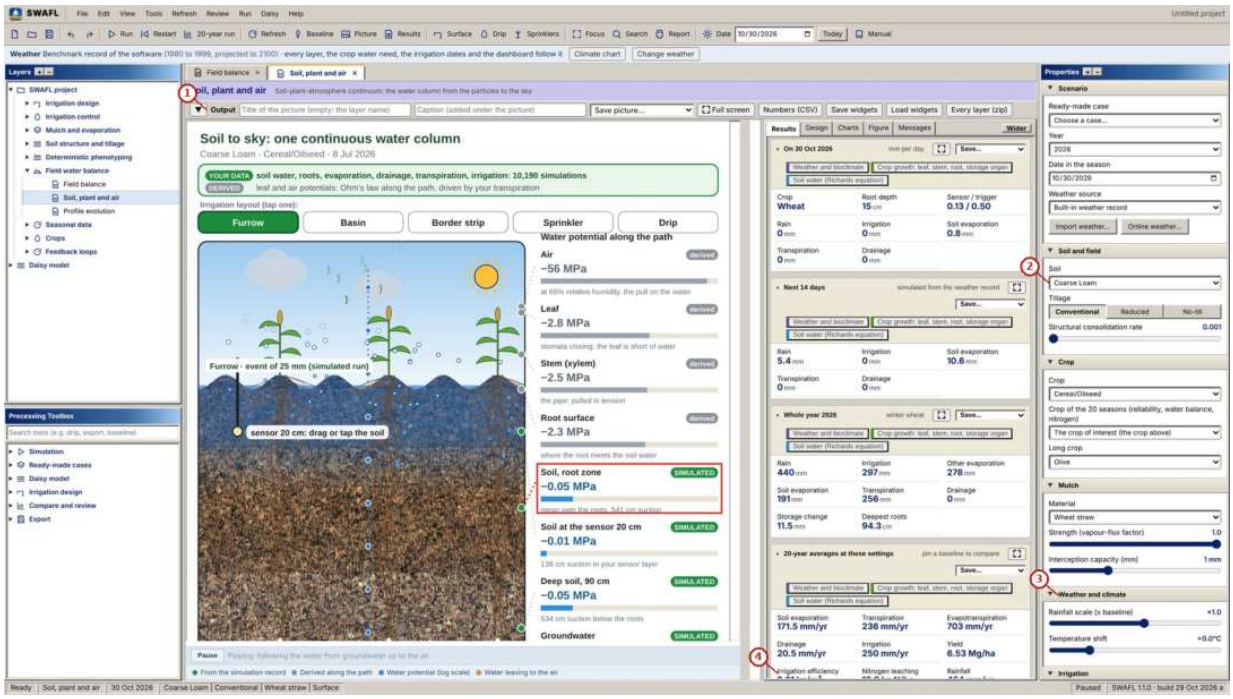
How it is made. Screen capture of SWAFL Desktop 1.1.0 (not regenerated).

S6.5 Reading Figure 13 of the article



What the numbered labels point to (picture 1 of 2: Field balance, the home view: daily fluxes on the field clock, the soil profile with sensor, roots and wetting front, and the closed budget (Eq. (1) of the article).).

1. "Soil, and air": the soil of the project, which every layer uses.
2. "Field balance": the field water balance: inputs, outputs and storage of the root zone.
3. "Coarse Loam v": the soil of the project, which every layer uses.
4. "Next 14 days": the decision for the next 14 days, from the soil water of the day.



What the numbered labels point to (picture 2 of 2: Soil, plant and air: layouts of the five irrigation methods and the water potential along the path from the deep soil to the air (Eq. (21) and Eq. (22) and Eq. (15) of the article).).

(1) "Output": the output bar of the layer: title and caption of the picture, save the picture, full screen, numbers as CSV.

(2) "Coarse Loam v": the soil of the project, which every layer uses.

(3) "Weather and climate": the weather of the project: built-in record, a file of the user or online weather of any place.

(4) "Irrigation efficiency": a heading or control of this screen; the caption above says which layer it belongs to.

What it shows. SWAFL Desktop: field and the soil–plant–air path. Panels: (a) Field balance, the home view: daily fluxes on the field clock, the soil profile with sensor, roots and wetting front, and the closed budget (Eq. (1) of the article). (b) Soil, plant and air: layouts of the five irrigation methods and the water potential along the path from the deep soil to the air (Eq. (21) and Eq. (22) and Eq. (15) of the article).

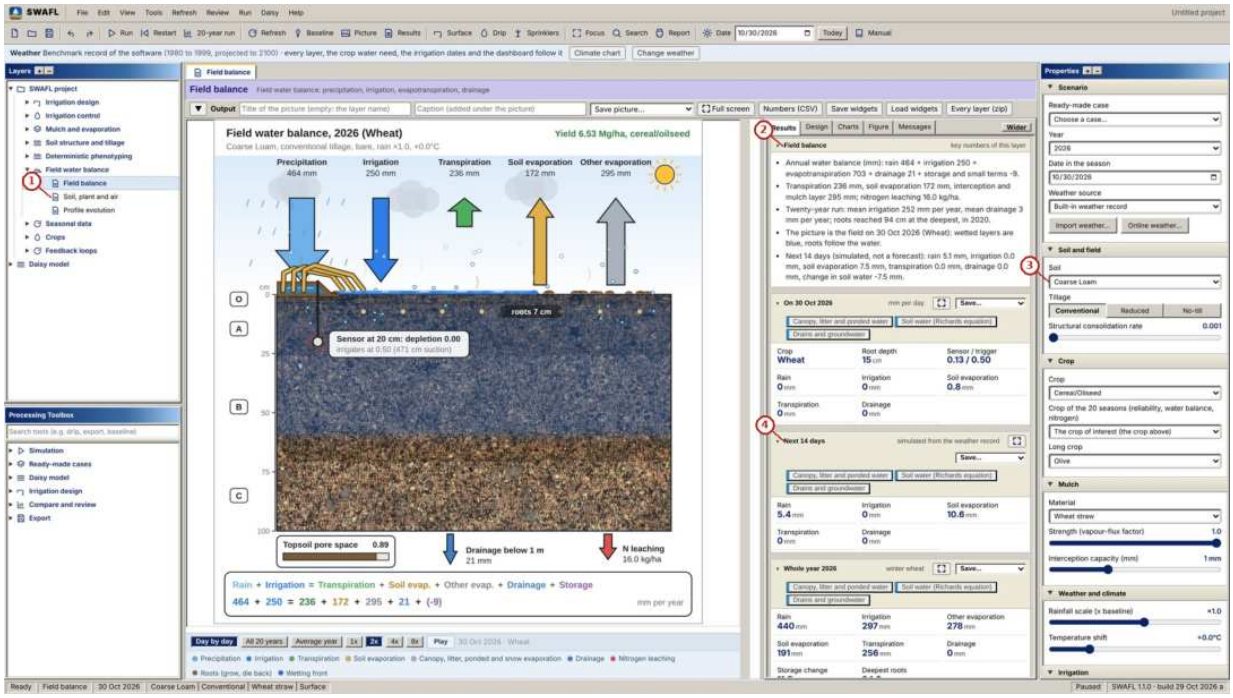
How to read the labels, symbols and colours. A screen of the software as it appears to the user; the labelled areas are the inputs, the figure and the numbers of the layer named in the caption.

What affects it, and why it happens. The field balance adds the day's rain and irrigation

and subtracts evaporation, transpiration and drainage; the path from soil to air shows the potential falling through each resistance, which is why the leaf is much drier than the soil.

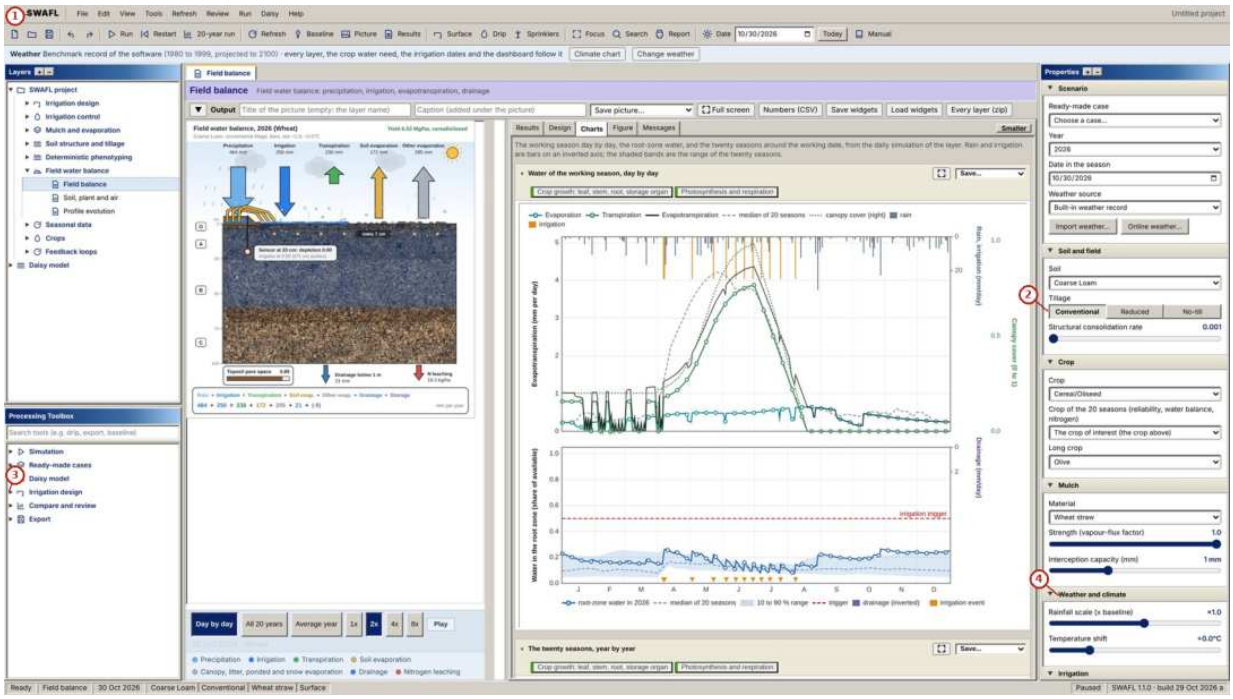
How it is made. Screen capture of SWAFL Desktop 1.1.0 (not regenerated).

S6.6 Reading Figure 28 of the article



What the numbered labels point to (picture 1 of 2: Dashboard, results tab: annual and daily flux tiles of the field water balance.).

1. "Soil, and air": the soil of the project, which every layer uses.
2. "Field balance": the field water balance: inputs, outputs and storage of the root zone.
3. "Coarse Loam v": the soil of the project, which every layer uses.
4. "Next 14 days": the decision for the next 14 days, from the soil water of the day.



What the numbered labels point to (picture 2 of 2: Dashboard, charts tab: seasonal rainfall, irrigation, drainage, evaporation and transpiration.).

1. "SWAFL View Run Daisy Help": the menu bar of the program (file, view, run, Daisy, help).
2. "No-till": a heading or control of this screen; the caption above says which layer it belongs to.
3. "Irrigation design": the irrigation-design group of the Layers tree and its layers.
4. "Weather and climate": the weather of the project: built-in record, a file of the user or online weather of any place.

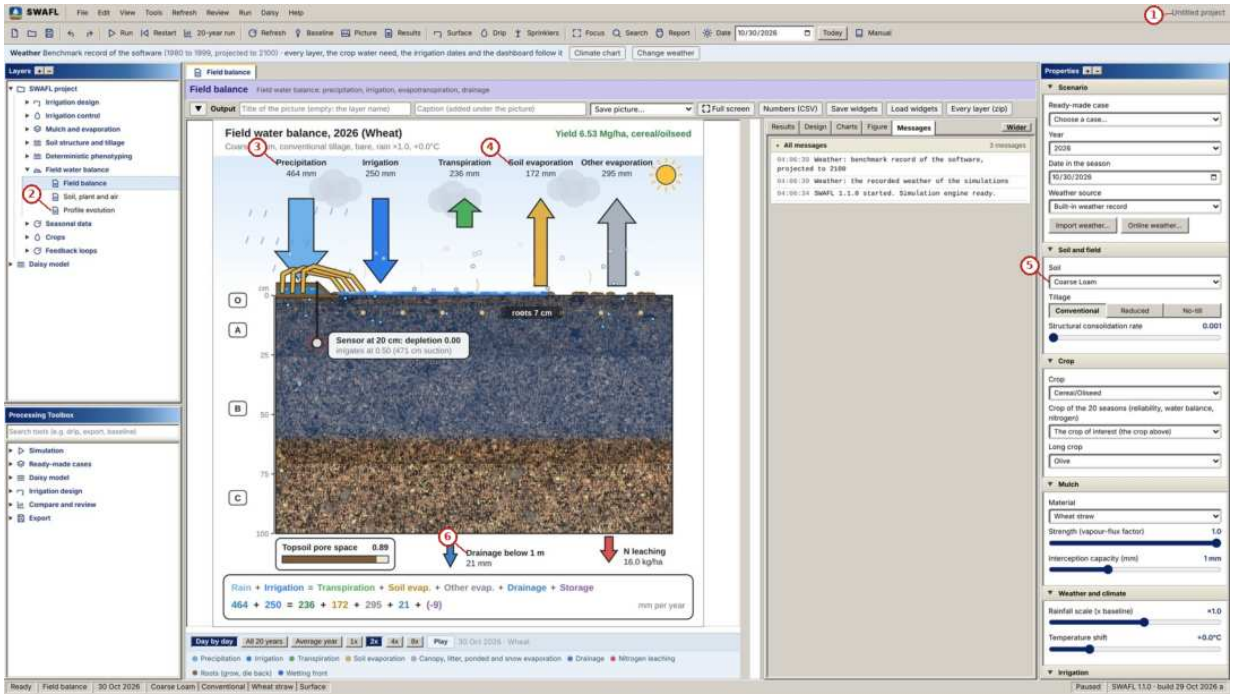
What it shows. SWAFL Desktop: dashboard. Panels: (a) Dashboard, results tab: annual and daily flux tiles of the field water balance. (b) Dashboard, charts tab: seasonal rainfall, irrigation, drainage, evaporation and transpiration.

How to read the labels, symbols and colours. A screen of the software as it appears to the user; the labelled areas are the inputs, the figure and the numbers of the layer named in the caption.

What affects it, and why it happens. The dashboard sums the fluxes of the season; each tile comes from one equation of the engines.

How it is made. Screen capture of SWAFL Desktop 1.1.0 (not regenerated).

S6.7 Reading Figure 29 of the article



What the numbered labels point to.

1. "Untitled project": a heading or control of this screen; the caption above says which layer it belongs to.
2. "Profile evolution": a heading or control of this screen; the caption above says which layer it belongs to.
3. "Precipitation": rain, from the weather of the project.
4. "Soil evaporation": evaporation from the soil (and, where stated, from mulch or canopy).
5. "Coarse Loam v": the soil of the project, which every layer uses.
6. "Drainage below 1m": rain, from the weather of the project.

What it shows. SWAFL Desktop: dashboard messages. Panels: (a) Dashboard, messages tab: the decision messages and warnings of the current state.

How to read the labels, symbols and colours. A screen of the software as it appears to the user; the labelled areas are the inputs, the figure and the numbers of the layer named in the caption.

What affects it, and why it happens. Messages fire when a number crosses a threshold (stress, drainage, efficiency), so the user sees what to change.

How it is made. Screen capture of SWAFL Desktop 1.1.0 (not regenerated).

S6.8 Derivation: Water balance of the column

Mass conservation of the column between the surface and 1 m gives $P+I = T+E+E_{\text{oth}}+D+\Delta S$ per unit area, with all terms in mm; every engine is constructed so that this balance closes, and its residual is the test of Engine 2.

S6.9 Why it happens

Why the meta-model ranks well but has an offset. It is built from ratios of simulated outcomes, which preserve the order of the scenarios, while its absolute level is anchored to one reference run, so an error in the anchor shifts all values.

S7. Infiltration and surface irrigation

Green–Ampt infiltration and the advance of water in a furrow.

S7.1 Contribution 5: a defect found and removed in surface irrigation

The problem

The surface layer under-predicted slow advance (up to $-29\,\%$ in a published example). Why?

The result

$$\underbrace{x(t)}_{(1)} = \frac{Qt}{\sigma_y A_0 + \underbrace{\sigma_z(r)}_{(2)}\, k t^A w}, \qquad \sigma_z = \frac{A + \underbrace{r}_{(3)}(1-A) + 1}{(1+A)(1+r)}$$

Pinpoints: what each circled mark shows and why

(1) The advance distance at time t.

(2) The subsurface shape factor depends on the advance exponent r.

(3) The first release fixed r; for slow advance the true r is smaller, σ_z too small and the advance too short. Iterating r with the computed advance removed the error, now within -0.2 to $+0.5\,\%$ of the published advance times.

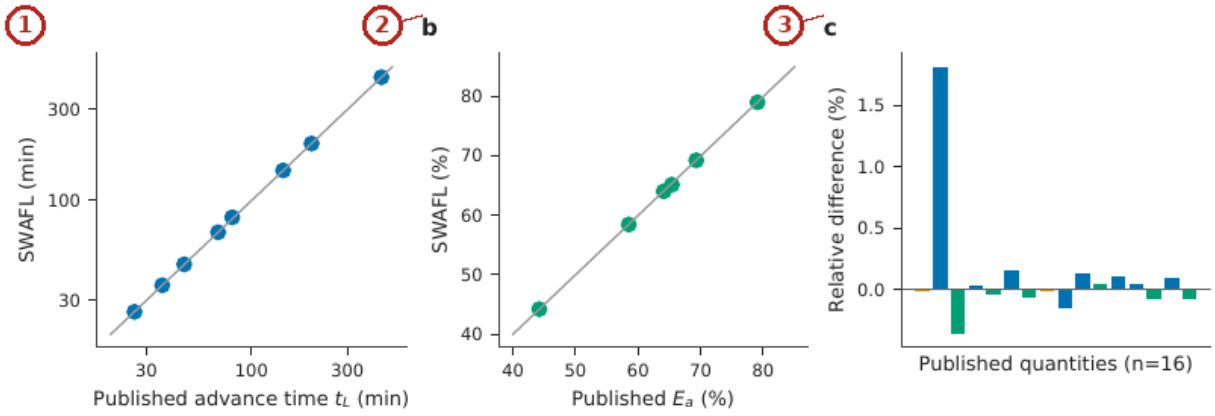
S7.2 Worked numbers, part 4: infiltration (Green–Ampt)

Water enters behind a sharp front. With K_s, front suction ψ_f and water-content jump $\Delta\theta$, the infiltrated depth F after time t solves $F - \psi_f \Delta\theta \ln(1 + F/(\psi_f \Delta\theta)) = K_s t$, found by trying values of F until both sides agree.

S7.3 Worked numbers, part 5: surface irrigation advance

The water admitted ($Q\,t$) is split between the surface and the soil; dividing by the volume per metre gives the advance $x(t) = Qt/(\sigma_y A_0 + \sigma_z k t^A w)$. The efficiencies follow from the stored, percolated and runoff volumes and add to one.

S7.4 Reading Figure 7 of the article



What the numbered labels point to.

1. Panel (a): Advance time t_L. *Why:* Points on the 1:1 line: the advance computed matches the published example.
2. Panel (b): application efficiency E_a (published versus SWAFL; grey line is 1:1). *Why:* Efficiencies match because they follow from the same volumes.
3. Panel (c): relative differences of the 16 quantities (blue: advance and required times; green: efficiencies). *Why:* All 16 quantities lie within a few per cent, so the procedure is reproduced.

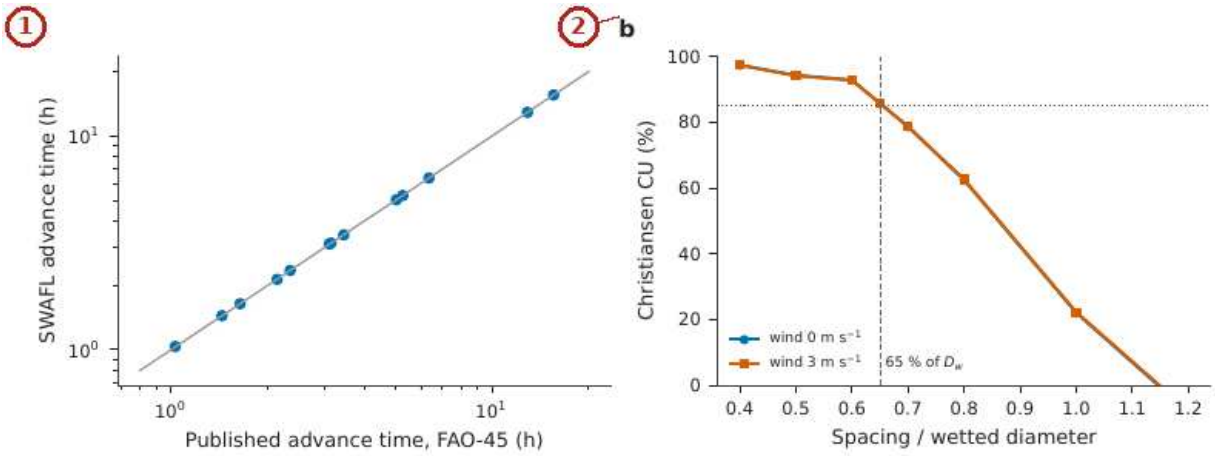
What it shows. Verification against Example 6.1. (a) Advance time t_L, (b) application efficiency E_a (published versus SWAFL; grey line is 1:1), (c) relative differences of the 16 quantities (blue: advance and required times; green: efficiencies).

How to read the labels, symbols and colours. (a) Advance time and (b) efficiency, published against SWAFL with the 1:1 line; (c) relative differences of 16 quantities.

What affects it, and why it happens. Points on the 1:1 line show the implementation reproduces the published worked example.

How it is made. Figure drawn by the script from the input data. Script: `Manuscripts/reproducibility/analysis/p3_figs.py`. Input: `Data/daisy_ensemble.csv.gz`.

S7.5 Reading Figure 11 of the article



What the numbered labels point to.

1. Panel (a): Thirteen advance times of the FAO Irrigation Paper 45 border examples (published against SWAFL; grey line 1:1). *Why:* Points on the 1:1 line: the shipped layer reproduces published advance times.
2. Panel (b): Christiansen coefficient of uniformity of the impact-sprinkler layer against spacing as a share of the wetted diameter, without wind and at 3 m s^{-1}; dashed: 65 % spacing rule, dotted: CU of 85 %. *Why:* Uniformity falls when sprinklers are spaced beyond about 65 % of the wetted diameter because the patterns overlap too little; wind makes it worse.

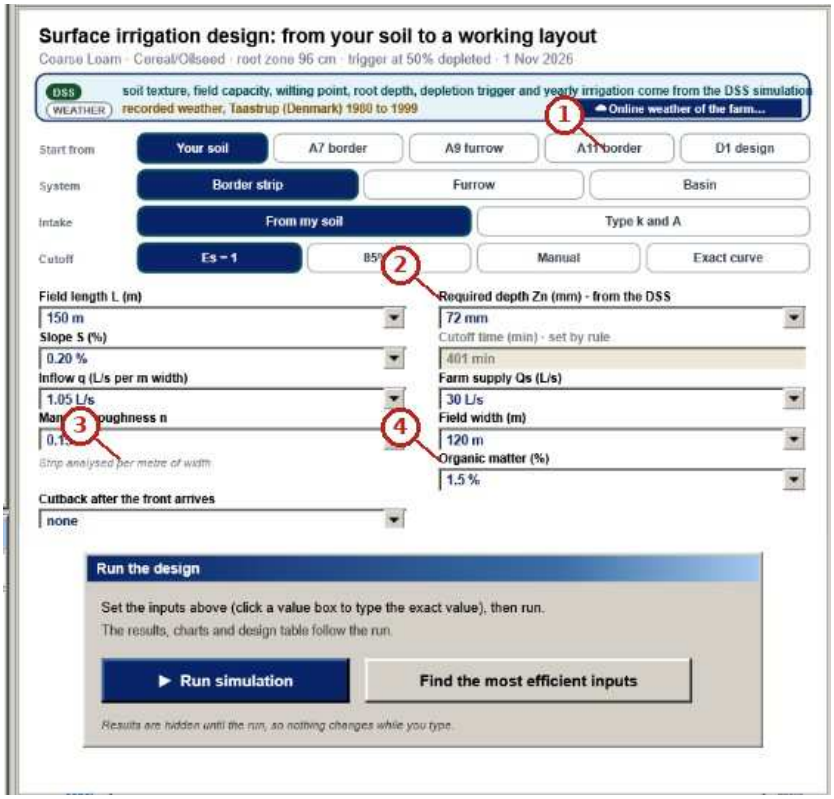
What it shows. Checks against published sources. (a) Thirteen advance times of the FAO Irrigation Paper 45 border examples (published against SWAFL; grey line 1:1). (b) Christiansen coefficient of uniformity of the impact-sprinkler layer against spacing as a share of the wetted diameter, without wind and at 3 m s^{-1}; dashed: 65 % spacing rule, dotted: CU of 85 %.

How to read the labels, symbols and colours. (a) Thirteen published advance times against SWAFL, 1:1 line; (b) uniformity against spacing, without and with wind.

What affects it, and why it happens. Uniformity falls as spacing exceeds about 65 % of the wetted diameter because the overlap of patterns becomes too small.

How it is made. Figure drawn by the script from the input data. Script: `Manuscripts/reproducibility/analysis/p3_lit_figs.py`. Input: `Data/daisy_ensemble.csv.gz`.

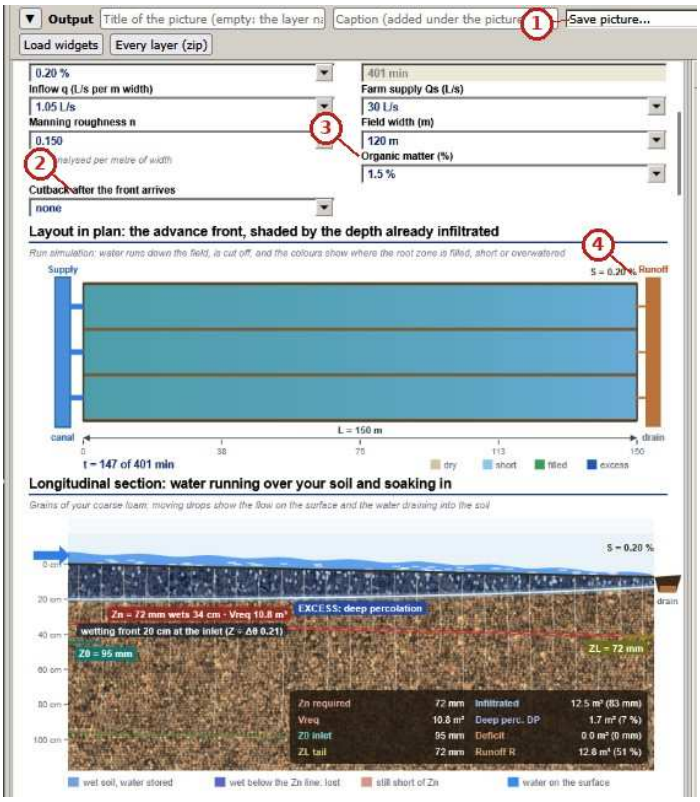
S7.6 Reading Figure 21 of the article



What the numbered labels point to (picture 1 of 2: Surface-irrigation layer: start from your soil or a textbook design (A7, A9, A11, D1), border strip, furrow or basin, intake from the soil or by k and A, the cutoff rule, and the inputs carried from the soil, root zone and trigger of the DSS (Section Section 4.4 of the article)).

1. "border": a heading or control of this screen; the caption above says which layer it belongs to.
2. "Required depth Zn from the DSS": a heading or control of this screen; the caption above says which layer it belongs to.
3. "per metre": a heading or control of this screen; the caption above says which layer it belongs to.
4. "Organic matter (%)": a heading or control of this screen; the caption above says which

layer it belongs to.



What the numbered labels point to (picture 2 of 2: Surface irrigation: layout in plan with the advance front shaded by the depth already infiltrated, and the longitudinal section with the wetting front, the required depth Z_n, deep percolation and runoff (Eq. (46) and Eq. (47) of the article)).

1. "Save picture": a heading or control of this screen; the caption above says which layer it belongs to.
2. "after the front arrives": a heading or control of this screen; the caption above says which layer it belongs to.
3. "Organic matter": a heading or control of this screen; the caption above says which layer it belongs to.
4. "Runoff": a heading or control of this screen; the caption above says which layer it belongs to.

What it shows. SWAFL Desktop: surface irrigation. Panels: (a) Surface-irrigation layer: start from your soil or a textbook design (A7, A9, A11, D1), border strip, furrow or basin, intake from the soil or by k and A, the cutoff rule, and the inputs carried from the soil, root zone and trigger of the DSS (Section Section 4.4 of the article). (b) Surface irrigation: layout in plan with

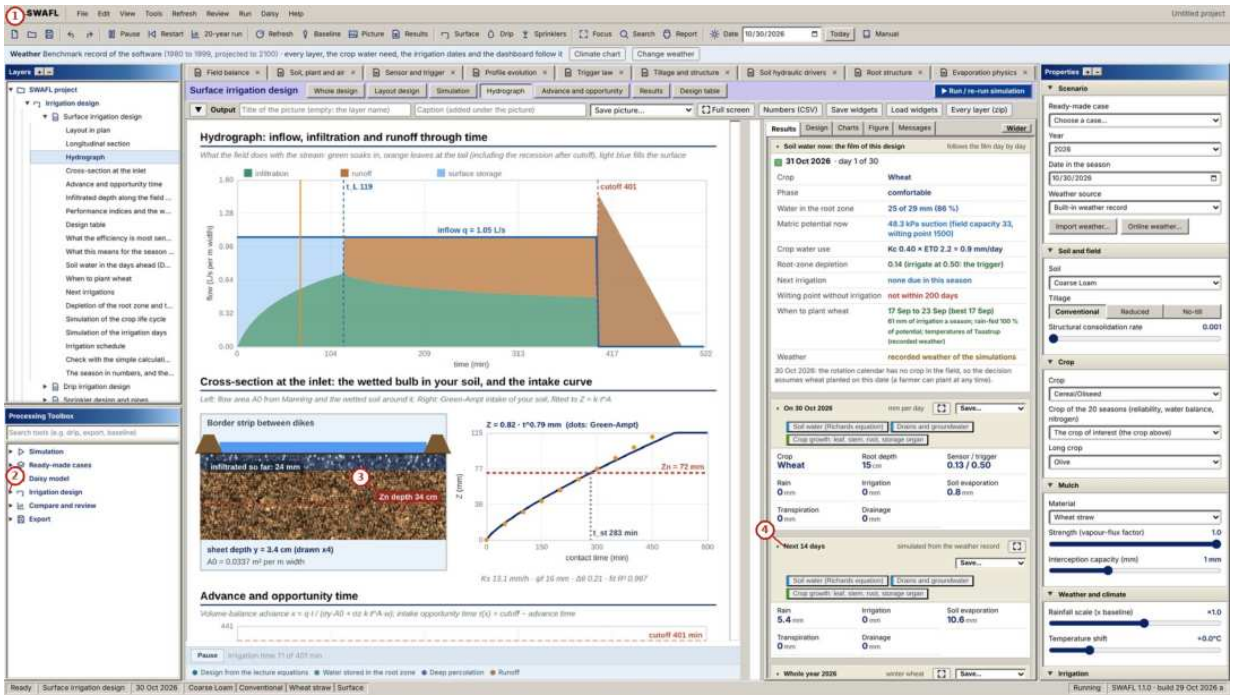
the advance front shaded by the depth already infiltrated, and the longitudinal section with the wetting front, the required depth Z_n, deep percolation and runoff (Eq. (46) and Eq. (47) of the article).

How to read the labels, symbols and colours. A screen of the software as it appears to the user; the labelled areas are the inputs, the figure and the numbers of the layer named in the caption.

What affects it, and why it happens. Water advances down the field while it infiltrates; the far end receives less, so the cutoff time sets the balance between shortfall and deep percolation.

How it is made. Screen capture of SWAFL Desktop 1.1.0 (not regenerated).

S7.7 Reading Figure 22 of the article



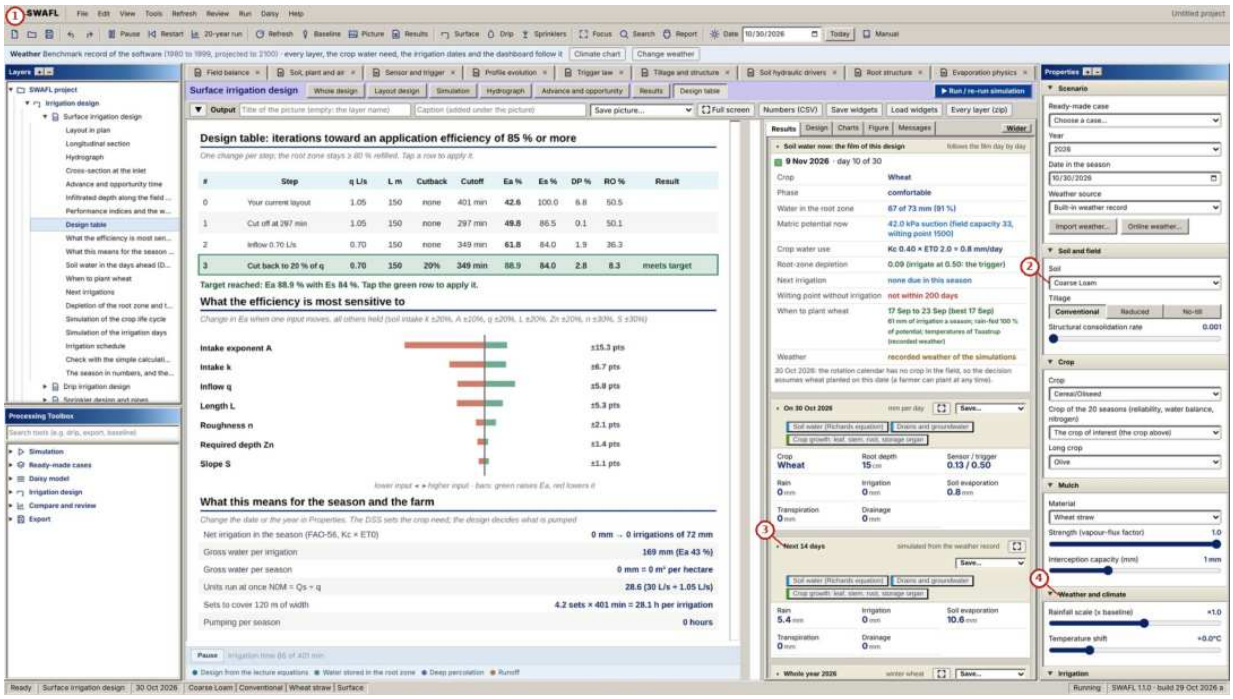
What the numbered labels point to (picture 1 of 2: Surface irrigation: hydrograph of inflow, infiltration and runoff through time.).

(1) "SWAFL View Run Daisy Help": the menu bar of the program (file, view, run, Daisy, help).

(2) "design": a heading or control of this screen; the caption above says which layer it belongs to.

(3) "Zn depth 34m": a heading or control of this screen; the caption above says which layer it belongs to.

④ “Next 14 days”: the decision for the next 14 days, from the soil water of the day.



What the numbered labels point to (picture 2 of 2: Surface irrigation: design table of options with application and storage efficiency, deep percolation and runoff (Eq. (48) of the article).).

① “SWAFL View Run Daisy Help”: the menu bar of the program (file, view, run, Daisy, help).

② “Coarse Loam”: the soil of the project, which every layer uses.

③ “Next 14 days”: the decision for the next 14 days, from the soil water of the day.

④ “Weather and climate”: the weather of the project: built-in record, a file of the user or online weather of any place.

What it shows. SWAFL Desktop: surface irrigation, schedule and design table. Panels: (a) Surface irrigation: hydrograph of inflow, infiltration and runoff through time. (b) Surface irrigation: design table of options with application and storage efficiency, deep percolation and runoff (Eq. (48) of the article).

How to read the labels, symbols and colours. A screen of the software as it appears to the user; the labelled areas are the inputs, the figure and the numbers of the layer named in the caption.

What affects it, and why it happens. The schedule repeats irrigation when the root zone is depleted to the trigger; the design table shows the volumes and efficiencies of each event.

How it is made. Screen capture of SWAFL Desktop 1.1.0 (not regenerated).

S7.8 Derivation: Green–Ampt infiltration

Behind a sharp wetting front at depth L_f with uniform water-content jump $\Delta\theta$ and front suction ψ_f, Darcy's law gives the infiltration rate $f = K_s(L_f + \psi_f)/L_f$ (ponding head neglected). With $F = L_f\Delta\theta$, $f = \mathrm{d}F/\mathrm{d}t = K_s(1 + \psi_f\Delta\theta/F)$. Separating variables and integrating from $F = 0$ at $t = 0$:

$$F - \psi_f\Delta\theta \ln\!\left(1 + \frac{F}{\psi_f\Delta\theta}\right) = K_s t. \tag{S1}$$

S7.9 Derivation: Surface irrigation: volume balance and advance

At time t after the start, the volume admitted, Qt, is stored on the surface (mean flow area $\sigma_y A_0$ over the advance length x) and in the soil (mean infiltrated depth $\sigma_z Z(t)$ over the wetted width w): $Qt = \sigma_y A_0 x + \sigma_z k t^A w x$, so $x(t) = Qt/(\sigma_y A_0 + \sigma_z k t^A w)$. If the advance follows a power law $x \propto t^r$ and the intake opportunity time at distance s is $t - t_x(s)$, the mean of Z along the advance is $Z(t)$ times

$$\sigma_z = \int_0^1 (1 - \xi^{1/r})^A \mathrm{d}\xi \;\simeq\; \frac{A + r(1 - A) + 1}{(1 + A)(1 + r)}, \tag{S2}$$

where the right-hand side is the closed approximation used in surface-irrigation practice (Walker and Skogerboe, 1987); it is exact for $A = 0$ and $A = 1$. The volumes applied, required, infiltrated, stored, percolated and run off satisfy $V_{\mathrm{app}} = V_{\mathrm{inf}} + V_{\mathrm{ro}}$ and $V_{\mathrm{inf}} = V_{\mathrm{st}} + V_{\mathrm{dp}}$, so the application efficiency, deep percolation and runoff fractions add to one.

S8. Drip and sprinkler irrigation

Friction in a lateral, uniformity and nozzle discharge.

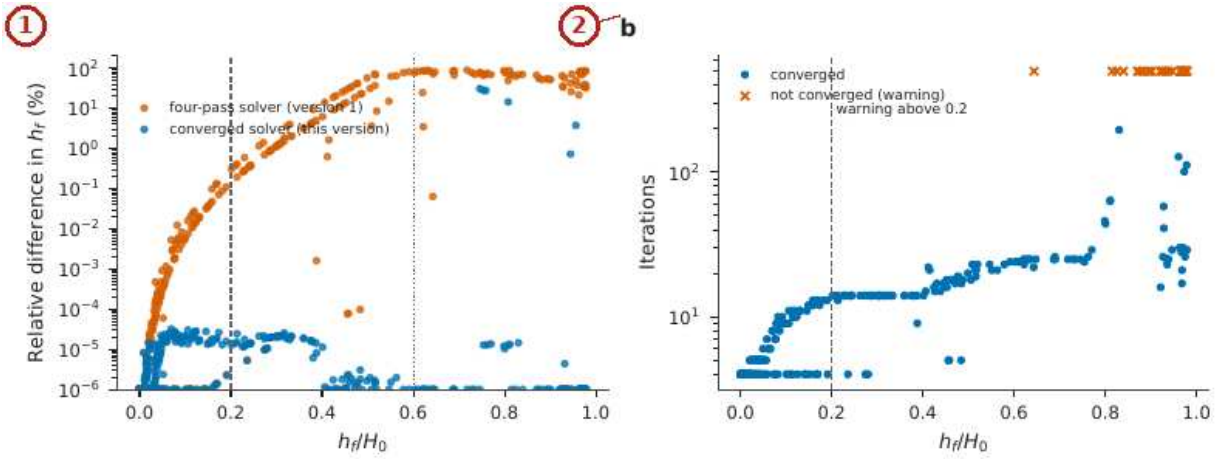
S8.1 Worked numbers, part 6: friction in a drip lateral

The Hazen–Williams loss of the full flow is reduced by the Christiansen factor because the flow falls along the lateral. *Example:* 50 outlets give $F_c = 1/2.852 + 1/100 + \sqrt{0.852}/15\,000 = 0.361$; a 100 m lateral of 16 mm with 0.5 L s^{-1} at the inlet loses about 17.6 m.

S8.2 Worked numbers, part 7: a sprinkler nozzle

$q = 0.95 A_n \sqrt{2 g h_n}$. *Example:* a 4 mm nozzle at 30 m head: $A_n = 1.26 \times 10^{-5}\,\mathrm{m}^2$, $\sqrt{2 \times 9.81 \times 30} = 24.3\,\mathrm{m\,s^{-1}}$, $q = 2.9 \times 10^{-4}\,\mathrm{m^3\,s^{-1}} = 17.4\,\mathrm{L\,min^{-1}}$.

S8.3 Reading Figure 10 of the article



What the numbered labels point to.

1. Panel (a): Relative difference in total head loss for the four-pass solver of version 1 and the converged solver, against the loss ratio h_f/H_0 (dashed: design limit 0.2; dotted: 0.6). *Why:* When friction is a large share of the head, emitter flows and heads are strongly coupled, so a few passes are not enough; the converged solver removes the error.
2. Panel (b): number of iterations of the converged solver; crosses mark configurations that did not converge. *Why:* More iterations are needed as the coupling grows.

What it shows. Drip-lateral solver versus an independent converged solver (400 random configurations). (a) Relative difference in total head loss for the four-pass solver of version 1 and the converged solver, against the loss ratio h_f/H_0 (dashed: design limit 0.2; dotted: 0.6); (b) number of iterations of the converged solver; crosses mark configurations that did not converge.

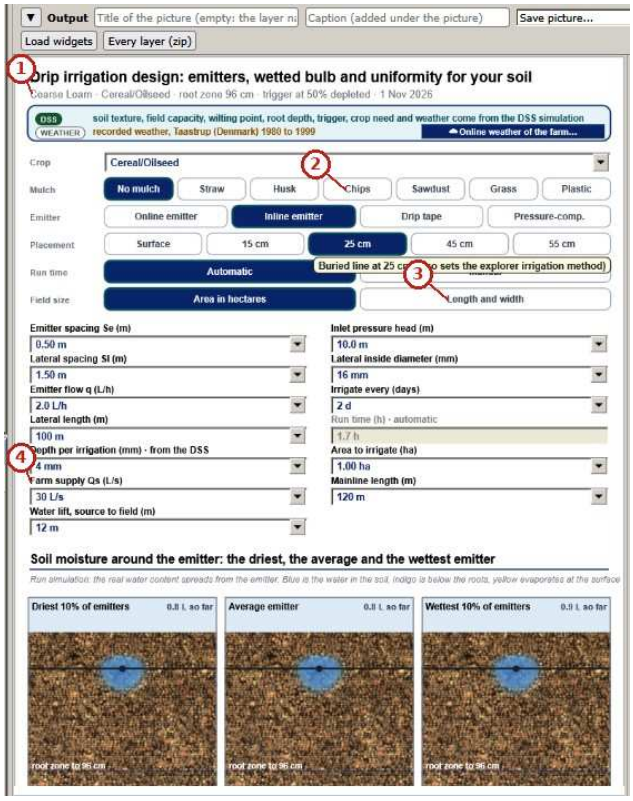
How to read the labels, symbols and colours. (a) Relative error of the lateral loss against the loss ratio, old and converged solver; dashed design limit 0.2; (b) iterations, crosses non-converged.

What affects it, and why it happens. Errors grow with the loss ratio because emitter flows and heads become strongly coupled; iterating to convergence removes them up to a large loss.

How it is made. Figure drawn by the script from the input data. Script: `Manuscripts/repr`

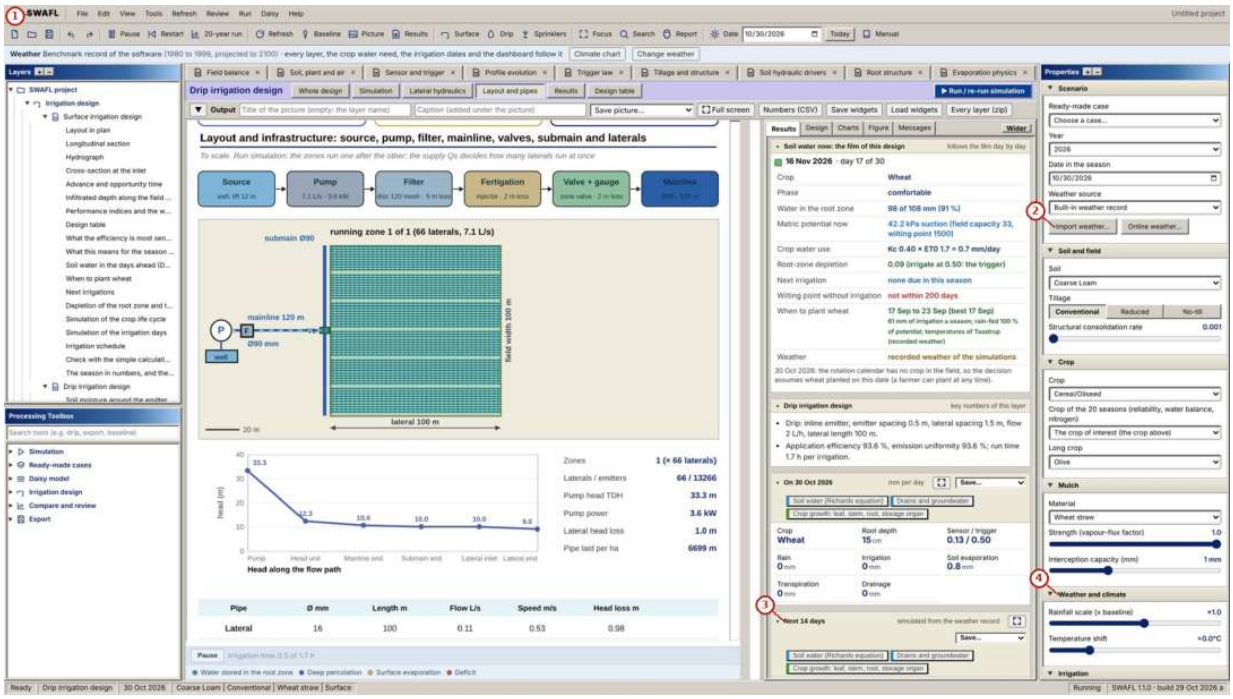
oducibility/analysis/p3_figs.py. Input: Data/daisy_ensemble.csv.gz.

S8.4 Reading Figure 23 of the article



What the numbered labels point to (picture 1 of 2: Drip-irrigation layer: crop, mulch, emitter type and placement, the design inputs, and the soil moisture around the driest, the average and the wettest emitter (Section Section 4.4 of the article)).

1. "Coarse Loam - Cereal": the soil of the project, which every layer uses.
2. "Chips": a heading or control of this screen; the caption above says which layer it belongs to.
3. "Length and width": a heading or control of this screen; the caption above says which layer it belongs to.
4. "Farm supply Qs": a heading or control of this screen; the caption above says which layer it belongs to.



What the numbered labels point to (picture 2 of 2: Drip irrigation: field layout and infrastructure.).

1. "SWAFL View Run Daisy Help": the menu bar of the program (file, view, run, Daisy, help).
2. "Import weather... Online weather": the weather of the project: built-in record, a file of the user or online weather of any place.
3. "Next 14 days": the decision for the next 14 days, from the soil water of the day.
4. "Weather and climate": the weather of the project: built-in record, a file of the user or online weather of any place.

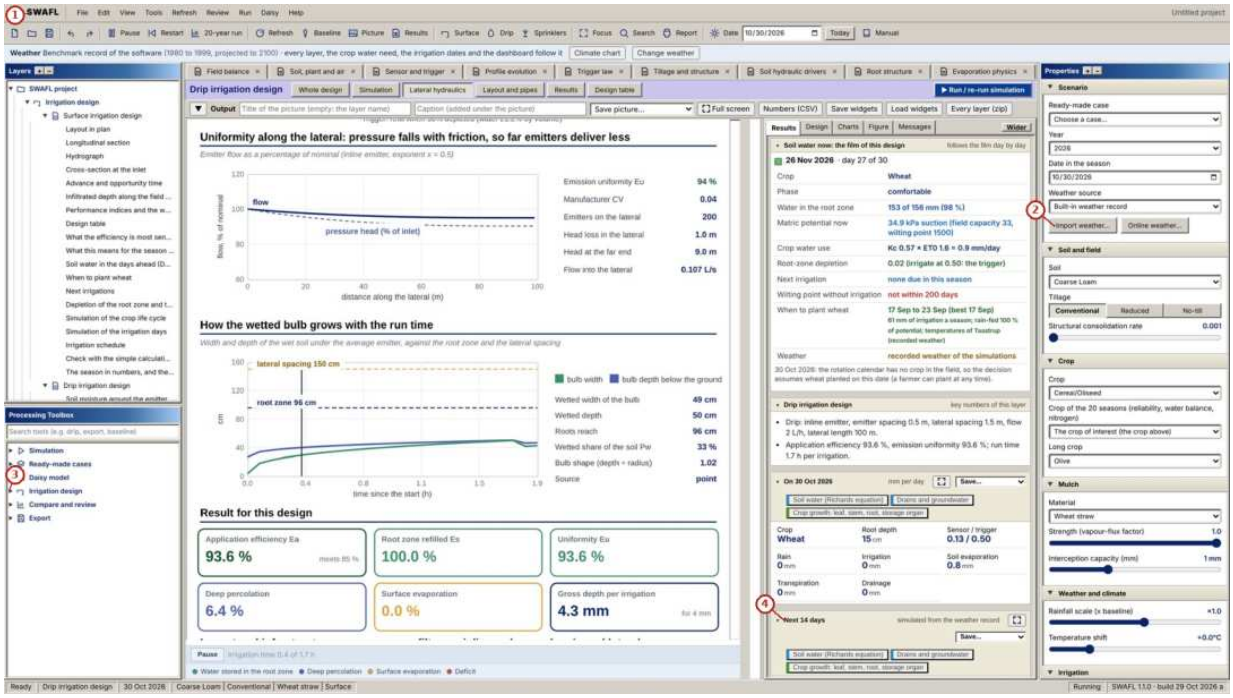
What it shows. SWAFL Desktop: drip irrigation. Panels: (a) Drip-irrigation layer: crop, mulch, emitter type and placement, the design inputs, and the soil moisture around the driest, the average and the wettest emitter (Section Section 4.4 of the article). (b) Drip irrigation: field layout and infrastructure.

How to read the labels, symbols and colours. A screen of the software as it appears to the user; the labelled areas are the inputs, the figure and the numbers of the layer named in the caption.

What affects it, and why it happens. Water spreads from each emitter as a bulb whose size depends on the soil; friction along the lateral lowers the flow of distant emitters.

How it is made. Screen capture of SWAFL Desktop 1.1.0 (not regenerated).

S8.5 Reading Figure 24 of the article



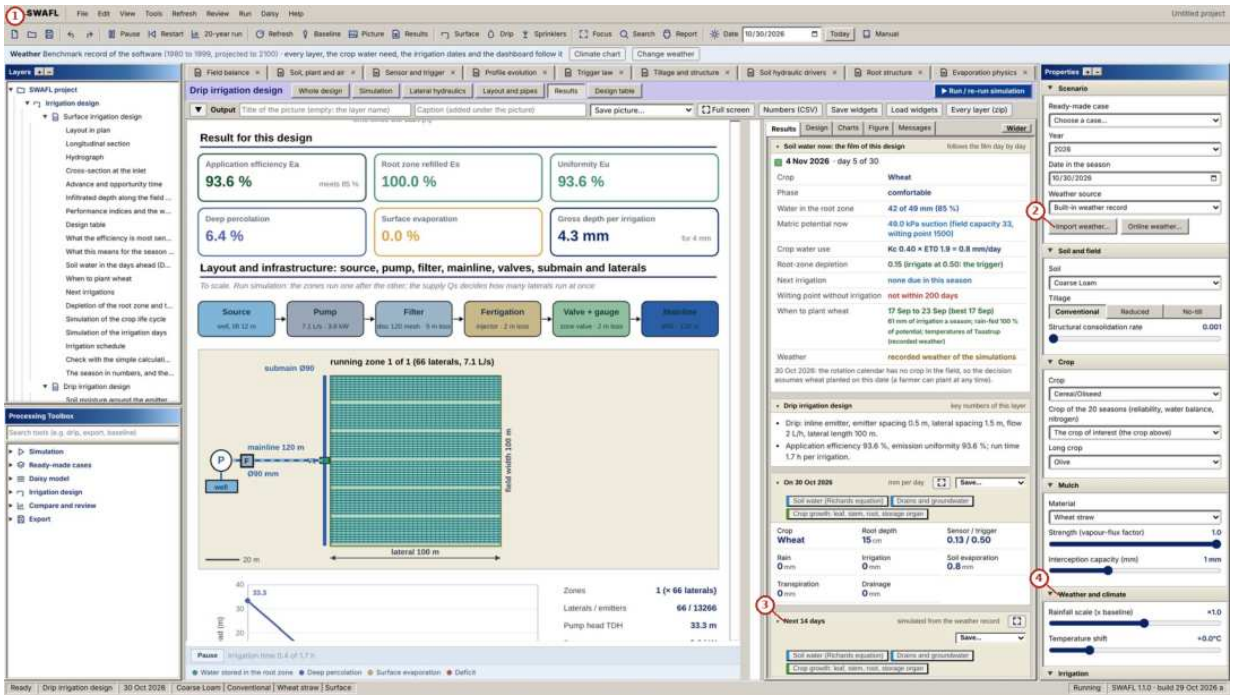
What the numbered labels point to (picture 1 of 2: Drip irrigation: head and emitter flow along the lateral (Eq. (49) of the article) with the 20 % loss warning.).

(1) "SWAFL View Run Daisy Help": the menu bar of the program (file, view, run, Daisy, help).

(2) "Import weather... Online weather": the weather of the project: built-in record, a file of the user or online weather of any place.

(3) "design": a heading or control of this screen; the caption above says which layer it belongs to.

(4) "Next 14 days": the decision for the next 14 days, from the soil water of the day.



What the numbered labels point to (picture 2 of 2: Drip irrigation: result summary with emission uniformity (Eq. (50) of the article) and efficiencies.).

1. "SWAFL View Run Daisy Help": the menu bar of the program (file, view, run, Daisy, help).
2. "Import weather... Online weather": the weather of the project: built-in record, a file of the user or online weather of any place.
3. "Next 14 days": the decision for the next 14 days, from the soil water of the day.
4. "Weather and climate": the weather of the project: built-in record, a file of the user or online weather of any place.

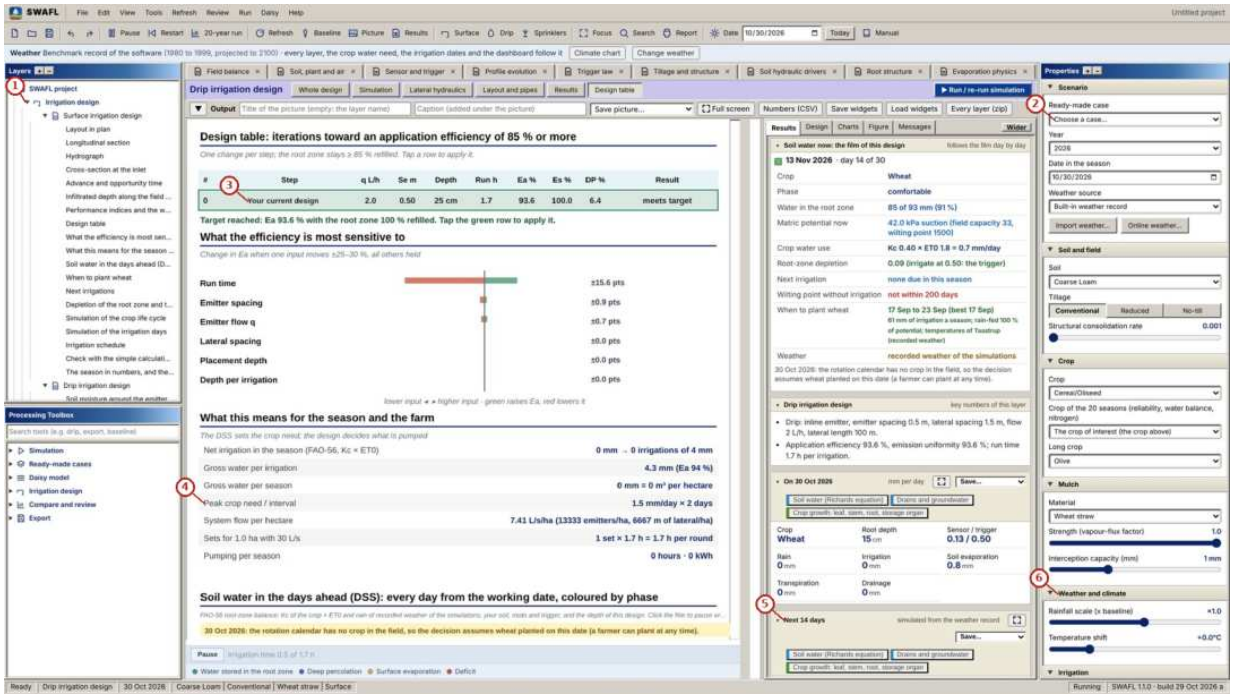
What it shows. SWAFL Desktop: drip irrigation, lateral and results. Panels: (a) Drip irrigation: head and emitter flow along the lateral (Eq. (49) of the article) with the 20 % loss warning. (b) Drip irrigation: result summary with emission uniformity (Eq. (50) of the article) and efficiencies.

How to read the labels, symbols and colours. A screen of the software as it appears to the user; the labelled areas are the inputs, the figure and the numbers of the layer named in the caption.

What affects it, and why it happens. Head falls along the lateral by friction, so emitter flow falls; the design keeps the loss within the limit.

How it is made. Screen capture of SWAFL Desktop 1.1.0 (not regenerated).

S8.6 Reading Figure 25 of the article



What the numbered labels point to.

1. "Irrigation design": the irrigation-design group of the Layers tree and its layers.
2. "Choose a case": the ready-made cases of the Properties panel.
3. "Your current design": a heading or control of this screen; the caption above says which layer it belongs to.
4. "Peak crop need / interval": the crop of the project and its stage.
5. "Next 14 days": the decision for the next 14 days, from the soil water of the day.
6. "Weather and climate": the weather of the project: built-in record, a file of the user or online weather of any place.

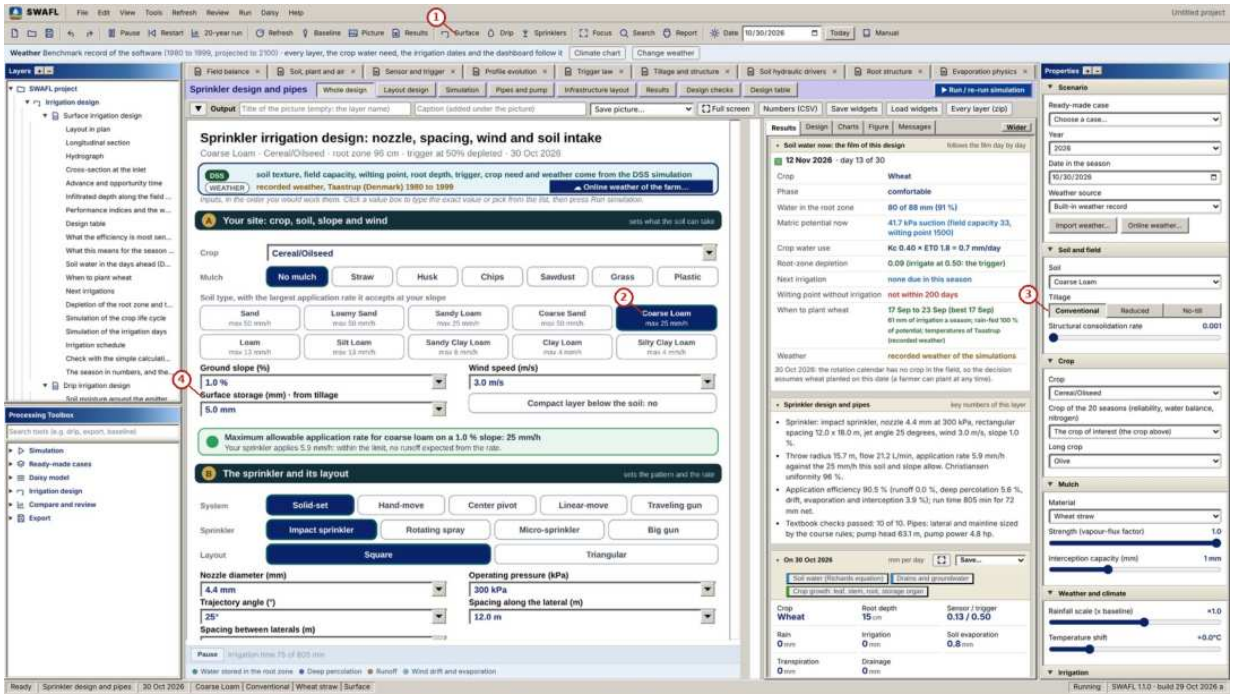
What it shows. SWAFL Desktop: drip irrigation, design table. Panels: (a) Drip irrigation: design table, iterations toward an application efficiency of 85 % or more.

How to read the labels, symbols and colours. A screen of the software as it appears to the user; the labelled areas are the inputs, the figure and the numbers of the layer named in the caption.

What affects it, and why it happens. The design iterates spacing and run time until the application efficiency and uniformity meet their targets.

How it is made. Screen capture of SWAFL Desktop 1.1.0 (not regenerated).

S8.7 Reading Figure 26 of the article



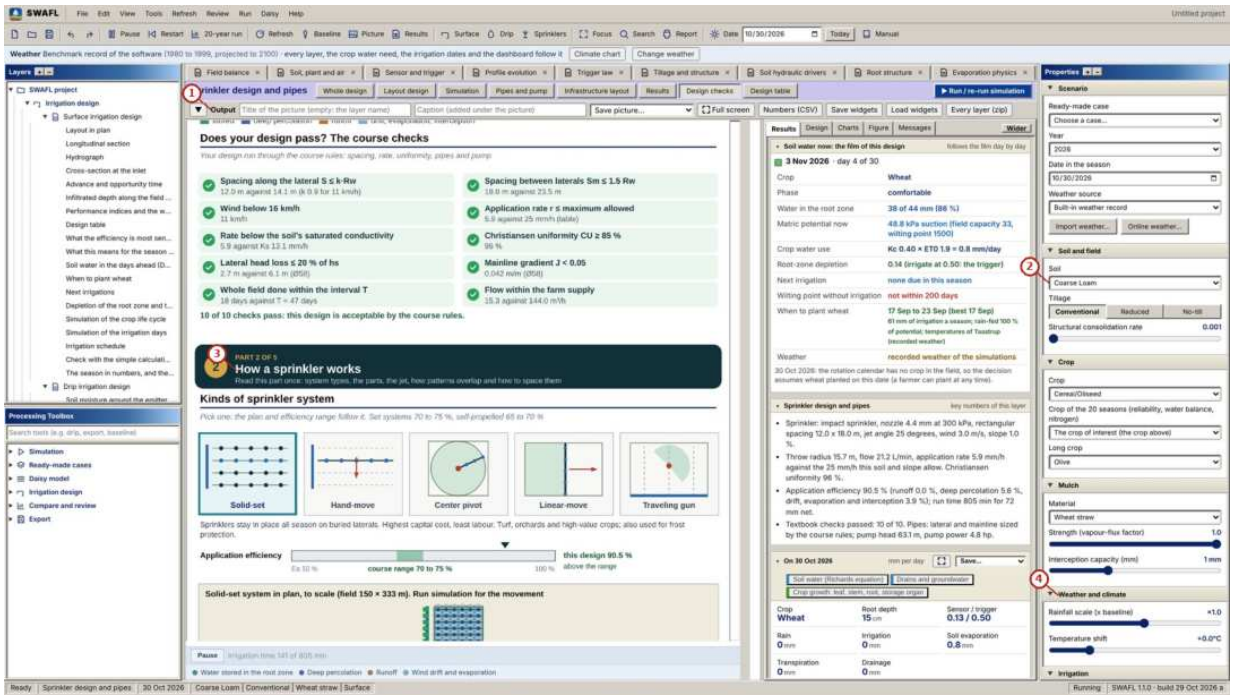
What the numbered labels point to (picture 1 of 2: Sprinkler layer: nozzle, wetted radius, pattern, overlap and uniformity (Eq. (51) of the article; Section Section 4.4 of the article).).

(1) "Surface © Sprinklers": the irrigation-design group of the Layers tree and its layers.

(2) "Coarse Lo": a heading or control of this screen; the caption above says which layer it belongs to.

(3) "Reduced No-till": a heading or control of this screen; the caption above says which layer it belongs to.

(4) "Surface storage (mm) - from tillage": a heading or control of this screen; the caption above says which layer it belongs to.



What the numbered labels point to (picture 2 of 2: Sprinkler design: textbook decision checklist (sprinkler class, spacing, application rate, sets per day).).

1. "Output": the output bar of the layer: title and caption of the picture, save the picture, full screen, numbers as CSV.
2. "Coarse Loam v": the soil of the project, which every layer uses.
3. "How a sprinkler works": a heading or control of this screen; the caption above says which layer it belongs to.
4. "Weather and climate": the weather of the project: built-in record, a file of the user or online weather of any place.

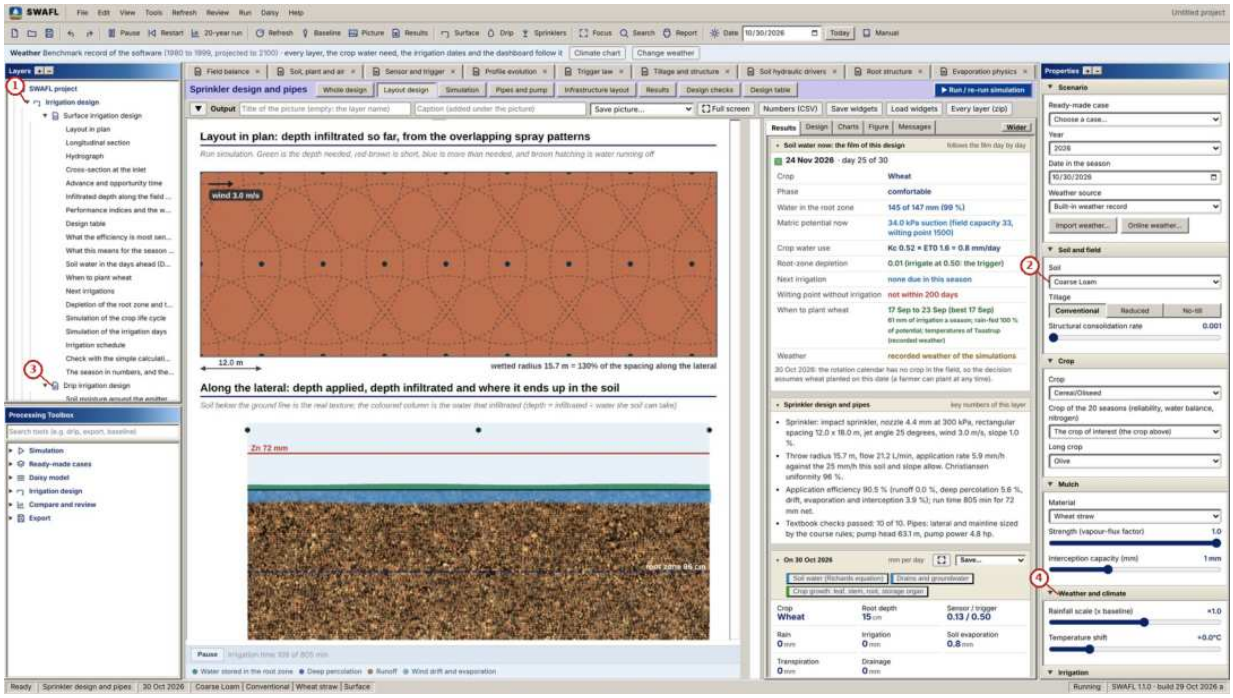
What it shows. SWAFL Desktop: sprinkler irrigation. Panels: (a) Sprinkler layer: nozzle, wetted radius, pattern, overlap and uniformity (Eq. (51) of the article; Section Section 4.4 of the article). (b) Sprinkler design: textbook decision checklist (sprinkler class, spacing, application rate, sets per day).

How to read the labels, symbols and colours. A screen of the software as it appears to the user; the labelled areas are the inputs, the figure and the numbers of the layer named in the caption.

What affects it, and why it happens. Each sprinkler wets a circle that is wettest near the nozzle; overlap of neighbouring circles evens out the depth.

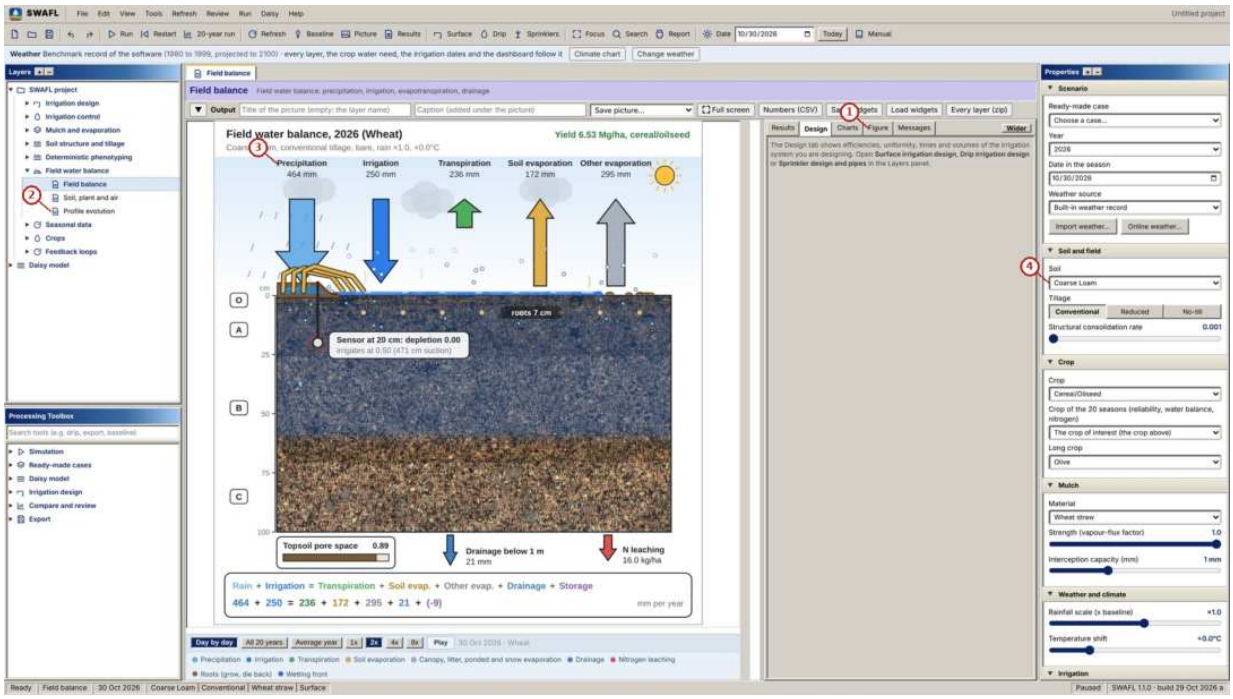
How it is made. Screen capture of SWAFL Desktop 1.1.0 (not regenerated).

S8.8 Reading Figure 27 of the article



What the numbered labels point to (picture 1 of 2: Sprinkler: layout in plan, with the depth infiltrated from the overlapping spray patterns.).

1. "Irrigation design": the irrigation-design group of the Layers tree and its layers.
2. "Coarse Loam v": the soil of the project, which every layer uses.
3. "Drip irrigation design": the irrigation-design group of the Layers tree and its layers.
4. "Weather and climate": the weather of the project: built-in record, a file of the user or online weather of any place.



What the numbered labels point to (picture 2 of 2: Dashboard with the design tab open beside the field-balance layer.).

1. “Figure”: a heading or control of this screen; the caption above says which layer it belongs to.
2. “Profile evolution”: a heading or control of this screen; the caption above says which layer it belongs to.
3. “Precipitation”: rain, from the weather of the project.
4. “Coarse Loam v”: the soil of the project, which every layer uses.

What it shows. SWAFL Desktop: sprinkler intake and the dashboard. Panels: (a) Sprinkler: layout in plan, with the depth infiltrated from the overlapping spray patterns. (b) Dashboard with the design tab open beside the field-balance layer.

How to read the labels, symbols and colours. A screen of the software as it appears to the user; the labelled areas are the inputs, the figure and the numbers of the layer named in the caption.

What affects it, and why it happens. Where the application rate exceeds the intake rate of the soil, water ponds and runs off.

How it is made. Screen capture of SWAFL Desktop 1.1.0 (not regenerated).

S8.9 Derivation: Drip laterals

The Hazen–Williams friction loss over a pipe segment of length Δx carrying Q_j is $h_{f,j} = 10.67 \Delta x\, Q_j^{1.852} / (C^{1.852} D^{4.871})$. Along a lateral with N equal outlets the flow falls linearly towards the end; summing the losses of the segments, $h_f = \frac{10.67 L Q_0^{1.852}}{C^{1.852} D^{4.871}} \frac{1}{N^{2.852}} \sum_{j=1}^{N} j^{1.852}$. The Euler–Maclaurin formula gives $\sum_{j=1}^{N} j^m \simeq \frac{N^{m+1}}{m+1} + \frac{N^m}{2} + \dots$ with $m = 1.852$, so the loss is the full-flow loss times $F_c = \frac{1}{m+1} + \frac{1}{2N} + \dots$; Christiansen (1942) approximated the remaining terms by $\sqrt{m-1}/(6N^2)$, which gives the factor of the paper, $F_c = \frac{1}{2.852} + \frac{1}{2N} + \frac{\sqrt{0.852}}{6N^2}$. Each emitter discharges $q_j = q(h_j/H_0)^{x_e}$, which couples the flows to the heads and is solved by iteration.

S8.10 Derivation: Emission uniformity

For emitters with manufacturing variation c_v and n_e emitters per plant, the lower quarter of plant flows lies about $1.27 c_v / \sqrt{n_e}$ below the mean (normal distribution), and the hydraulic variation adds the ratio $q_{\min}/\bar{q}$; their product is the design emission uniformity of Keller and Karmeli (1974).

S8.11 Derivation: Sprinkler discharge

Bernoulli's equation across a nozzle gives the jet speed $v = \sqrt{2 g h_n}$, and the discharge is $q = C_d A_n \sqrt{2 g h_n}$ with $C_d \approx 0.95$; the factor 6×10^4 converts $\mathrm{m^3\,s^{-1}}$ to $\mathrm{L\,min^{-1}}$. The Christiansen coefficient of uniformity is $CU = 1 - \sum |z_i - \bar{z}| / (n \bar{z})$.

S9. Every term of the author's equations

Soil-specific trigger, $h^* = \alpha^{-1}([S_e(100) - f^*\{S_e(100) - S_e(15\,849)\}]^{-1/m} - 1)^{1/n}$

Term	What it means	Unit	Where the value comes from	When it rises
f^*	chosen depletion of the plant-available water	–	agronomy (0.45)	h^* rises (drier trigger)
$S_e(100)$	effective saturation at field capacity	–	retention curve	h^* shifts
$S_e(15\,849)$	effective saturation at the wilting point	–	retention curve	h^* shifts

Term	What it means	Unit	Where the value comes from	When it rises
α, n, m	van Genuchten parameters	cm^{-1}, –, –	soil table	see the reach equations
h^*	suction at which the soil has used f^* of its water	cm	computed	–

Interception store, $I_c = \mu_0 \ln(1/r)$

Term	What it means	Unit	Where the value comes from	When it rises
μ_0	mean daily rain on rain days	mm	weather	I_c rises in proportion
r	throughfall fraction of the annual model	–	Engine 1	I_c falls

Two-stage evaporation, $Q(t) = Q_1 + 2e_0\tau_E(\sqrt{1 + (t - t_1)/\tau_E} - 1)$

Term	What it means	Unit	Where the value comes from	When it rises
e_0	evaporative demand (stage-1 rate)	$\mathrm{mm\,d}^{-1}$	weather	Q rises
Q_1	cumulative evaporation at the end of stage 1	mm	soil texture	stage 1 lasts longer
t_1	end of stage 1	d	$(Q_1 - q_0)/e_0$	–
τ_E	transition time of stage 2	d	soil	stage 2 decays more slowly

S10. How every constant is obtained, what it depends on, and how sensitive the results are

S10.1 How every constant of the article is obtained

Constant	Value	How it was obtained	What it depends on; sensitivity
f^*, critical depletion	0.45	from the simulations: the depletion beyond which the simulated yield falls by more than 5 %	the crop and the 5 % tolerance; a larger f^* gives drier triggers
Field capacity, wilting point	100 cm, 15 849 cm	FAO convention (Allen et al., 1998) (pF 2 and pF 4.2)	a different field capacity shifts every depletion
Rain-day fraction in $\mu_0 = P/(0.28 \cdot 365)$	0.28	engineered: about one day in four is a rain day in the benchmark weather	the climate; I_c scales with μ_0
Throughfall r	$\max(40, P - P_{\text{other}})/P$	chosen so that the daily model intercepts what the annual model intercepts	Engine 1
Stage-1 depth L_1	40 mm	engineered: the layer that dries in stage 1	a deeper layer lengthens stage 1
Surface-dry suction h_s	$4/\alpha$	engineered, scaled by α so it samples the steep part of every curve	the soil
Stage-2 time τ_E	$0.006 + 0.03(\text{clay} + \text{silt})/100$ d	engineered: finer soils supply the surface longer	the texture
Surface tension, contact angle	0.0728 N m^{-1}; 0	physical constants of water at 20 °C (Hillel, 1998)	temperature; repellent soils have a larger angle
Kelvin factor	135.3 MPa	$\mathcal{R}T_K/V_w$ at 20 °C	temperature
Nozzle coefficient	0.95	typical discharge coefficient of a sprinkler nozzle	the nozzle

Constant	Value	How it was obtained	What it depends on; sensitivity
Hazen–Williams exponent	1.852	from the Hazen–Williams formula	the pipe; C depends on the material
Keller–Karmeli factor	1.27	from the normal distribution of emitter flows (Keller and Karmeli, 1974)	the manufacturing variation

S11. How each table of the article is produced

Table 1 of the article. Symbols, abbreviations and units of the soil–water–atmosphere engines and of the irrigation design modules, grouped by subject.

How it is made. Table typed from the result file (4 of its numbers found there). Script: `Manuscripts/reproducibility/audit_numbers.py`. Input: `Data/daisy_ensemble.csv.gz`.

Table 2 of the article. State variables and fluxes of the modelled soil–water–atmosphere system, by compartment; the soil matrix and soil structure rows form the soil compartment, and the lower boundary (drainage) and the management decision are listed separately. Source: E1 = Engine 1 (annual), E2 = Engine 2 (daily), D = Daisy output stored in the software, R = relation evaluated by the layer. Provenance: L literature, D data-driven, E engineered.

How it is made. Table typed from the result file (10 of its numbers found there). Script: `Manuscripts/reproducibility/audit_numbers.py`. Input: `Data/daisy_ensemble.csv.gz`.

Table 3 of the article. The twelve controls of the framework, their values and defaults, and the variables that respond. Run set: the Daisy experiment from which the response is read.

How it is made. Table typed from the result file (5 of its numbers found there). Script: `Manuscripts/reproducibility/p3_js/run_drip.js`. Input: `Data/daisy_ensemble.csv.gz`.

Table 4 of the article. The seven feedback loops: control step, printed effect and arrow-width input μ.

How it is made. Table typed in the manuscript from the result files of the script. Script: `(seethereproducibilityappendixofthepaper)`. Input: `Data/daisy_ensemble.csv.gz`.

Table 5 of the article. Experiments used by Engine 1, the number of soils they cover and the equation family that consumes them.

How it is made. Descriptive table (definitions, design, parameters or tests); no computation.

Table 6 of the article. Output variables of each Daisy run (20-year means, 1980–1999) that form the evidence layer. The complete run table is in the Data folder of the repository and on Zenodo.

How it is made. Descriptive table (definitions, design, parameters or tests); no computation.

Table 7 of the article. Effective van Genuchten parameters used by Engine 1 and the available-water capacity (AWC, mm of water per cm of soil, an engineered texture-class constant), with the critical suction h^* at the soil-independent depletion $f^* = 0.45$.

How it is made. Table typed from the result file (11 of its numbers found there). Script: `Manuscripts/reproducibility/audit_numbers.py`. Input: `Data/daisy_ensemble.csv.gz`.

Table 8 of the article. Soil and surface variables of the ten soils. Texture as sand/silt/clay (%); θ_s, θ_r from Eq. (2) of the article; $S_e(100)$ effective saturation at field capacity; $f(300)$ depletion reached at a suction of 300 cm (Eq. (29) of the article); Q_1 stage-1 storage, τ_E stage-2 time scale, t_1 stage-1 duration and $Q(12)$ cumulative evaporation after 12 days at $e_0 = 4.2\,\mathrm{mm\,d^{-1}}$ (Eq. (11) of the article).

How it is made. Table typed from the result file (39 of its numbers found there). Script: `Manuscripts/reproducibility/p3_js/run_drip.js`. Input: `Data/daisy_ensemble.csv.gz`.

Table 9 of the article. Annual balance of the ten soils from Engine 1 ($\mathrm{mm\,yr^{-1}}$, 20-year means; Eq. (1) of the article): precipitation P, irrigation I, transpiration T, soil evaporation E, other evaporation E_{oth}, drainage D and storage change ΔS; yield Y ($\mathrm{Mg\,ha^{-1}}$), nitrate leaching N_L ($\mathrm{kg\,ha^{-1}\,yr^{-1}}$) and the regime index $\mathcal{I}$ (Eq. (14) of the article).

How it is made. Table typed from the result file (15 of its numbers found there). Script: `Manuscripts/reproducibility/audit_numbers.py`. Input: `Data/daisy_ensemble.csv.gz`.

Table 10 of the article. Crop, water-potential and mulch variables of the ten soils. Engine 2 (20 years): maximum rooting depth R_{max} (Eq. (17) of the article), mean of the yearly deepest roots $\bar{R}_d$, depths holding 50 and 90 % of uptake (Eq. (20) of the article), irrigation events per year, dose d (mm), median wetting depth of an irrigation (cm) and calibration factor k_t. Water-potential path (Eq. (21) and Eq. (22) of the article), means over crop days of day-of-year 120–210: ψ_{RZ}, ψ_{leaf} (MPa) and share of days with $\psi_{\mathrm{leaf}} < -1.6$ MPa. Straw mulch ($\varphi = 0.4$, $c = 4$ mm) against bare

soil: evaporation saved ΔE (mm yr^{-1}) and its fate shares (Eq. (23) of the article).

How it is made. Table typed from the result file (36 of its numbers found there). Script: `Manuscripts/reproducibility/p3_js/swa_vars.js`. Input: `Data/daisy_ensemble.csv.gz`.

Table 11 of the article. Result views, governing relations and verification. Equation numbers in brackets refer to the Manual; provenance labels are L (literature), D (data-driven), E (engineered).

How it is made. Table typed from the result file (11 of its numbers found there). Script: `Manuscripts/reproducibility/audit_numbers.py`. Input: `Data/daisy_ensemble.csv.gz`.

Table 12 of the article. Surface-irrigation Example 6.1 (Walker and Skogerboe, 1987): published and computed quantities. T_{req}: required infiltration time (min); t_L: advance time to the end of the furrow (min) at flow Q (m^3 min^{-1}); E_a: application efficiency (%). Error $= 100(SWAFL - book)/book$.

How it is made. Table typed from the result file (22 of its numbers found there). Script: `Manuscripts/reproducibility/audit_numbers.py`. Input: `Data/daisy_ensemble.csv.gz`.

Table 13 of the article. Border Example 6.4 (Walker and Skogerboe (1987)) and the sprinkler example: published and computed values.

How it is made. Table typed from the result file (11 of its numbers found there). Script: `Manuscripts/reproducibility/run_all.sh`. Input: `Data/daisy_ensemble.csv.gz`.

Table 14 of the article. Engine 1 against 1 000 held-out E12 runs (coarse loam), off-axis cells. Raw: as shipped; re-anchored: baseline aligned. nRMSE: RMSE as % of the observed mean; ρ: Spearman rank correlation.

How it is made. Table typed from the result file (15 of its numbers found there). Script: `Manuscripts/reproducibility/run_all.sh`. Input: `Data/daisy_ensemble.csv.gz`.

Table 15 of the article. Drip lateral solver of version 1 (four passes) and the converged solver against an independent solver, by loss ratio h_f/H_0 (400 random configurations). Maximum relative error in h_f and maximum absolute error in E_u.

How it is made. Table typed from the result file (4 of its numbers found there). Script: `Manuscripts/reproducibility/p3_js/run_drip.js`. Input: `Data/daisy_ensemble.csv.gz`.

Table 16 of the article. What this article adds, item by item: what was established before, what is new here, and the evidence in this article.

How it is made. Table typed from the result file (4 of its numbers found there). Script: `Manuscripts/reproducibility/audit_numbers.py`. Input: `Data/daisy_ensemble.csv.gz`.

S12. From the data to every result: experiments, runs and computations

Where every number starts. All simulated numbers come from one set of 10 190 Daisy runs, organised in seventeen experiments (E0 to E16), each of which varies a few factors on the same ten soils (S01 to S10) under the same weather series (Taastrup, Denmark). Every run writes 20-year mean annual fluxes (rain, irrigation, evapotranspiration and its parts, drainage at 1 m, yield, nitrogen); one row per run is stored in `Data/Comprehensive_Summary/00_summary_all_simulations.csv.gz` (sheet `All_runs_summary` of the workbook) and, for the ensemble analyses, in `Data/daisy_ensemble.csv.gz` (sheet `Ensemble_runs`). A run is used only when its status is `ok` (10 041 runs); the 149 runs whose solver failed are kept in the files but never enter a result. The workbook repeats the per-run algebra as live Excel formulas (the green columns at the end of `All_runs_summary`: other evaporation, balance residual, differences from the rainfed run of the same soil, water productivity), and the sheet `Column_Equations` lists the equation behind every calculated column. The rainfed run of each soil (experiment E0) is the reference against which every irrigated or treated run is compared (sheet `Baselines_E0`).

Result	Experiment and runs	How it is computed from the runs	Script, file, workbook
Critical depletion $f^* = 0.45$	E2_TriggerPressure (197 valid runs; soils with a maximum irrigation above 50 mm)	the trigger of every run is turned into a root-zone depletion f (two horizons, 0–25 and 25–100 cm, field capacity 100 cm, wilting point 15 849 cm); relative yield $Y/Y_{\max}$ of the soil; median of $Y/Y_{\max}$ in 15 bins of f; f^* is the first bin after the peak whose median falls below 0.95	`Results/08_Scripts/novel.py`; `D_N1_soil_specific_triggers`
Soil-specific triggers h^* (143–729 cm)	α, n of the soils and f^*	solve $f(h) = f^*$ for h (the closed form derived in this supplement)	live formulas in `D_N1_soil_specific_triggers`, `EQ_P2`; chart `CH_P2`
Share of the available water used at 300 cm (24–84 %)	same	$f(300)$ for each soil	column `f_at_300`

Result	Experiment and runs	How it is computed from the runs	Script, file, workbook
Soil–water–atmosphere variables of the ten soils	the soil parameters of the model	evaluated by the program's own functions (capillary radius, conductivity, evaporation stages, air potential)	`p3_js/swa_vars.js` → `p3_swa_vars.json`; sheet `R_p3_swa_vars`
Interception store matched to the annual model	daily rain as an exponential distribution with mean μ_0	throughfall above a store I_c integrated over the distribution and set equal to the annual throughfall fraction	live formula in `EQ_P2`
Sensitivity of the outputs to the controls	E1_Doses, E3, E4, E7 and E12	per-soil relative change of yield, drainage and irrigation per lever (`lev_pct`) and its rank correlation with the available water	`D_controllability_by_soil_lever`, `D_controllability_rho_vs_TAW`
Verification of every computing layer	textbook examples and the shipped engines	each layer recomputes a published example (surface irrigation, drip lateral, sprinkler, FAO-56) and the difference is reported	`p3_js/*.js` → `p3_surface*.json`, `drip_out.json`, `spr_out.json`, `fao_out.json`; sheets `R_*`

S13. The data

Table S7 lists the 17 experiments of the data folder and Fig. S1 shows their outputs. The tables that follow are the result files of the analyses of the paper, printed in full or up to 220 rows; each is also in the workbook `Data/Manuscript_Data/SWAFL_Manuscripts_Data.xlsx`, and every run is in `Data/Comprehensive_Summary`.

Table S7: The data of this study: the 17 Daisy experiments of `Data/Comprehensive_Summary` (10190 runs, 10041 valid; 20 years, 1980–1999, of Taastrup weather). Means of the valid runs; Y yield (Mg ha^{-1}), I irrigation, D drainage at 1 m and ET (mm yr^{-1}), N_L nitrate leached (kg N ha^{-1} yr^{-1}), t run time of Daisy (s). Every row is one file of the folder, linked.

Exp.	File	Runs	Valid	Factors varied	Y	I	D	ET	N_L	t
E0	`Baselines`	10	10	soil	2.19	0	55	418	158	231
E1	`Doses`	200	189	soil, rate, dose	5.13	198	98	564	79	195
E2	`TriggerPressure`	200	197	soil, trigger	4.70	169	80	559	90	139
E3	`SensorDepth`	200	200	soil, sensor depth	4.94	199	86	583	86	146
E4	`Methods`	100	96	soil, method	5.01	164	88	542	83	147
E5	`SHP Dynamics`	300	300	soil, method, tillage	4.13	152	43	579	84	189
E6	`Mediterranean Pollution`	210	193	soil, method, trigger, N rate	3.64	84	239	620	62	118
E7	`Mulching`	200	200	soil, mulch	5.29	183	96	555	74	161
E8	`Rain Intensity Moisture`	1000	1000	initial suction, rain scale	0.68	0	-1	282	-0	6
E9	`Soil Moisture Rain`	1000	987	soil, initial suction, rain scale	0.75	0	19	247	33	7
E10	`Amount Timing Soil`	1000	954	soil, rate, dose, window	4.42	139	76	532	97	146
E11	`Temp Evap Rain`	1000	1000	rain scale, warming, demand	3.81	324	18	754	66	151
E12	`Sensor Trigger Rain`	1000	1000	trigger, sensor depth, rain scale	5.88	260	35	670	32	155
E13	`N Water Soil`	1000	975	soil, method, rate, dose, trigger, N rate	4.00	162	83	547	79	207
E14	`Mulch Rain`	1000	1000	rain scale, mulch	6.15	235	43	634	25	173
E15	`Method Depth Amount`	1000	1000	method, rate, dose, root depth	6.02	217	61	619	37	157
E16	`Edge Cases`	770	740	soil, method, rate, dose, trigger, sensor depth, rain scale, warming, N rate, demand	2.17	344	185	600	93	224

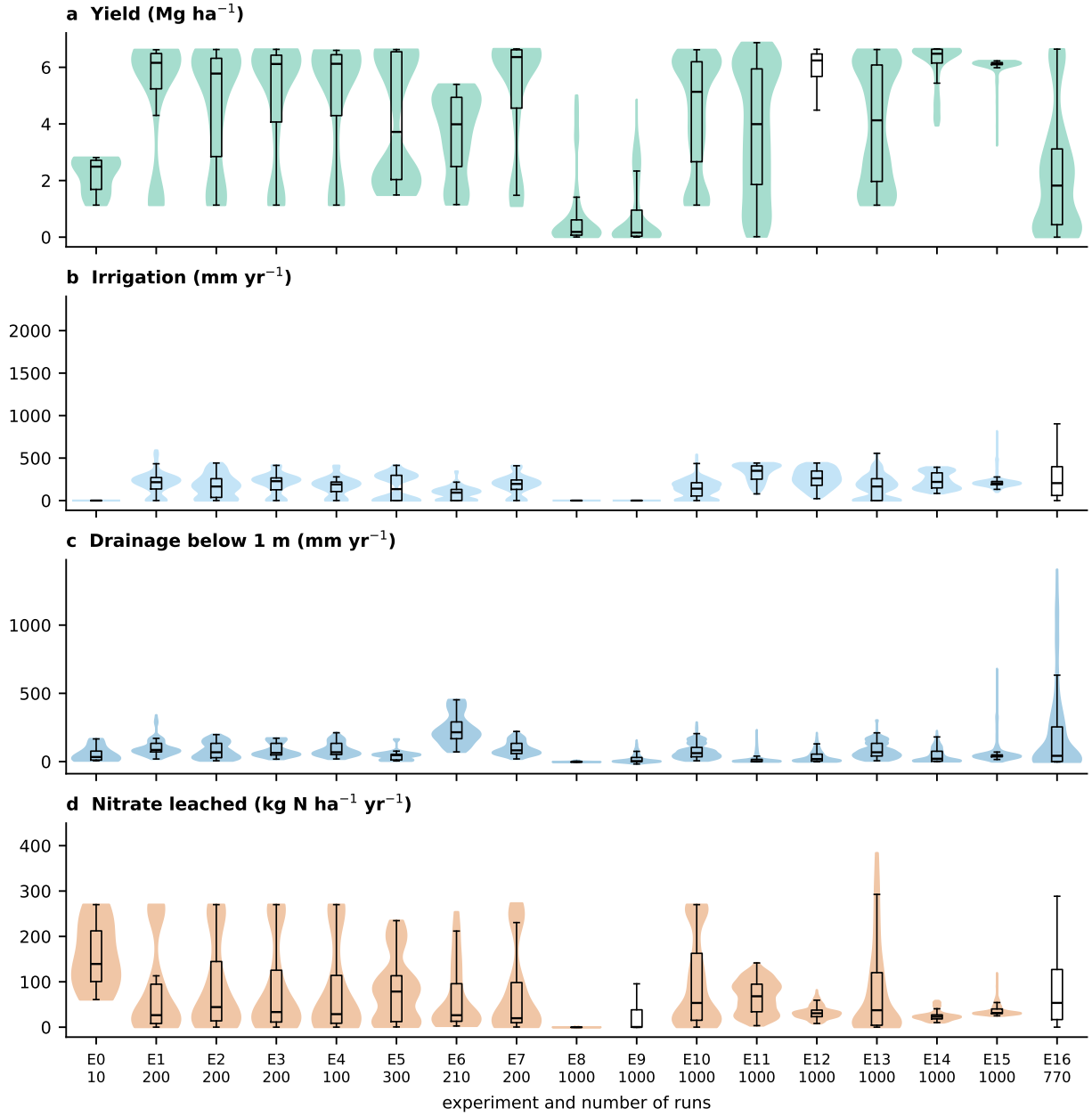


Figure S1: Yield, irrigation, drainage and nitrate leaching of the valid runs, by experiment.

p3_swa_vars.json

Source: `Manuscripts/results/p3_swa_vars.json`; 650 values, first 220 shown. Workbook sheet `R_p3_swa_vars`.

quantity	value
[0].soil	S01_CoarseLoam
[0].label	Coarse Loam
[0].hyd.alpha	0.09613
[0].hyd.n	1.353
[0].hyd.sand	45

quantity	value
[0].hyd.silt	35
[0].hyd.clay	20
[0].hyd.thetaS	0.4235
[0].hyd.thetaR	0.022
[0].hyd.AWC	1.5
[0].hyd.f300	0.3795
[0].hyd.hStar	387.3
[0].hyd.Se100	0.4443
[0].e1.evap	174.6
[0].e1.transp	239
[0].e1.et	720.3
[0].e1.irr	282.2
[0].e1.yield	6.627
[0].e1.drain	27.62
[0].e1.nl	14.56
[0].e1.rain	464
[0].e1.other	306.7
[0].e1.store	-1.67
[0].e1.ie	2.348
[0].e2.Rmax	93.58
[0].e2.meanRootMax	89.94
[0].e2.z50	19.25
[0].e2.z90	47
[0].e2.kt	1.641
[0].e2.dose	28.22
[0].e2.fTrig	0.3795
[0].e2.irrMean	283.7
[0].e2.eventsPerYear	10.05
[0].e2.uptakeProfile	30.27, 24.66, 23.52, 22.37, 22.36, 21.16, 19.86, 18.68, 20.95, 14.61, 13.69, 12.82 ... (40 values)
[0].e2.sProfile	0.4658, 0.5419, 0.6077, 0.6541, 0.7093, 0.7365, 0.7559, 0.7728, 0.7867, 0.7988, 0.81, 0.821 ... (40 values)
[0].e2.wetFrontIrr	45
[0].e2.drainUnscaled	7.334
[0].e2.interceptionUnscaled	312.7
[0].e2.evapUnscaled	117
[0].e2.transpUnscaled	335.7
[0].spac.psiRZ	-0.0494
[0].spac.psiRoot	-1.814
[0].spac.psiLeaf	-2.243
[0].spac.Kr	8.391e-06
[0].spac.closedShare	0.7405
[0].spac.psiAir	-56.3
[0].spac.days	1715
[0].evap.e0	4.2
[0].evap.Q1	5.991
[0].evap.tau	0.0225
[0].evap.t1	1.426

quantity	value
[0].evap.lost12	9.903
[0].evap.dtheta	0.3514
[0].evap.hi	2.601
[0].evap.hs	41.61
[0].mulch.dE	127.4
[0].mulch.dI	34.86
[0].mulch.dD	7.26
[0].mulch.dT	17.84
[0].mulch.fI	0.2736
[0].mulch.fD	0.05697
[0].mulch.fT	0.14
[0].mulch.fRest	0.5294
[0].regime	0.7268
[0].kappaN	0.3783
[1].soil	S02_CoarseSand
[1].label	Coarse Sand
[1].hyd.alpha	0.05358
[1].hyd.n	1.6
[1].hyd.sand	88
[1].hyd.silt	8
[1].hyd.clay	4
[1].hyd.thetaS	0.3736
[1].hyd.thetaR	0.0124
[1].hyd.AWC	0.5
[1].hyd.f300	0.4964
[1].hyd.hStar	261.9
[1].hyd.Se100	0.3563
[1].e1.evap	117.8
[1].e1.transp	236.7
[1].e1.et	716.3
[1].e1.irr	413.8
[1].e1.yield	6.449
[1].e1.drain	163.6
[1].e1.nl	62.55
[1].e1.rain	464
[1].e1.other	361.8
[1].e1.store	-2.18
[1].e1.ie	1.559
[1].e2.Rmax	100
[1].e2.meanRootMax	96.15
[1].e2.z50	24.5
[1].e2.z90	64.75
[1].e2.kt	1.12
[1].e2.dose	35
[1].e2.fTrig	0.4964
[1].e2.irrMean	423.5
[1].e2.eventsPerYear	12.1

quantity	value
[1].e2.uptakeProfile	8.878, 10.93, 12.22, 12.53, 13.49, 12.93, 12.01, 11.32, 11.73, 9.564, 8.149, 7.661 ... (40 values)
[1].e2.sProfile	0.2935, 0.4015, 0.4911, 0.552, 0.6278, 0.6603, 0.6845, 0.7038, 0.7202, 0.7356, 0.7505, 0.7642 ... (40 values)
[1].e2.wetFrontIrr	100
[1].e2.drainUnscaled	259
[1].e2.interceptionUnscaled	369.4
[1].e2.evapUnscaled	61.06
[1].e2.transpUnscaled	219.3
[1].spac.psiRZ	-0.03899
[1].spac.psiRoot	-1.848
[1].spac.psiLeaf	-2.275
[1].spac.Kr	5.581e-05
[1].spac.closedShare	0.7504
[1].spac.psiAir	-56.3
[1].spac.days	1715
[1].evap.e0	4.2
[1].evap.Q1	7.849
[1].evap.tau	0.0096
[1].evap.t1	1.869
[1].evap.lost12	10.39
[1].evap.dtheta	0.3392
[1].evap.hi	4.666
[1].evap.hs	74.66
[1].mulch.dE	102.5
[1].mulch.dI	51.11
[1].mulch.dD	7.26
[1].mulch.dT	14.76
[1].mulch.fI	0.4988
[1].mulch.fD	0.07086
[1].mulch.fT	0.1441
[1].mulch.fRest	0.2862
[1].regime	-0.1628
[1].kappaN	0.3783
[2].soil	S03_Sand
[2].label	Sand
[2].hyd.alpha	0.145
[2].hyd.n	2.68
[2].hyd.sand	92
[2].hyd.silt	5
[2].hyd.clay	3
[2].hyd.thetaS	0.3693
[2].hyd.thetaR	0.0118
[2].hyd.AWC	0.6
[2].hyd.f300	0.8422
[2].hyd.hStar	142.8
[2].hyd.Se100	0.01119
[2].e1.evap	58.2

quantity	value
[2].e1.transp	202.3
[2].e1.et	669
[2].e1.irr	278.2
[2].e1.yield	4.984
[2].e1.drain	76.52
[2].e1.nl	85.16
[2].e1.rain	464
[2].e1.other	408.5
[2].e1.store	-3.32
[2].e1.ie	1.792
[2].e2.Rmax	100
[2].e2.meanRootMax	95.84
[2].e2.z50	27
[2].e2.z90	52.38
[2].e2.kt	2.045
[2].e2.dose	27.82
[2].e2.fTrig	0.8422
[2].e2.irrMean	285.1
[2].e2.eventsPerYear	10.25
[2].e2.uptakeProfile	14.55, 15.46, 16.02, 15.59, 16.18, 15.72, 15.22, 14.97, 15.88, 14.28, 13.66, 13.55 ... (40 values)
[2].e2.sProfile	0.2917, 0.338, 0.3738, 0.396, 0.4246, 0.4392, 0.4515, 0.4625, 0.4765, 0.4922, 0.5086, 0.5228 ... (40 values)
[2].e2.wetFrontIrr	55
[2].e2.drainUnscaled	5.15
[2].e2.interceptionUnscaled	424.5
[2].e2.evapUnscaled	32.31
[2].e2.transpUnscaled	314.8
[2].spac.psiRZ	-0.02021
[2].spac.psiRoot	-1.694
[2].spac.psiLeaf	-2.114
[2].spac.Kr	2.075e-09
[2].spac.closedShare	0.712
[2].spac.psiAir	-56.3
[2].spac.days	1715
[2].evap.e0	4.2
[2].evap.Q1	12.71
[2].evap.tau	0.0084
[2].evap.t1	3.027
[2].evap.lost12	14.95
[2].evap.dtheta	0.3521
[2].evap.hi	1.724
[2].evap.hs	27.59
[2].mulch.dE	49.62
[2].mulch.dI	34.36
[2].mulch.dD	7.26
[2].mulch.dT	15.1
[2].mulch.fI	0.6924

quantity	value
[2].mulch.fD	0.1463
[2].mulch.fT	0.3043
[2].mulch.fRest	0
[2].regime	-0.136
[2].kappaN	0.3783
[3].soil	S04_LoamySand
[3].label	Loamy Sand
[3].hyd.alpha	0.124
[3].hyd.n	2.28
[3].hyd.sand	82
[3].hyd.silt	12
[3].hyd.clay	6
[3].hyd.thetaS	0.3804
[3].hyd.thetaR	0.0136
[3].hyd.AWC	0.9
[3].hyd.f300	0.7557
[3].hyd.hStar	159.5
[3].hyd.Se100	0.03978
[3].e1.evap	73.95
[3].e1.transp	215
[3].e1.et	691.2
[3].e1.irr	293
[3].e1.yield	5.456
[3].e1.drain	70.14
[3].e1.nl	61.13
[3].e1.rain	464
[3].e1.other	402.2
[3].e1.store	-4.31
[3].e1.ie	1.862
[3].e2.Rmax	100

p3_e1_sens.json

Source: `Manuscripts/results/p3_e1_sens.json`; 223 values, first 220 shown. Workbook sheet `R_p3_e1_sens`.

quantity	value
base.irr	282.2
base.et	720.3
base.drain	27.62
base.yield	6.627
base.nl	14.56
base.evap	174.6
base.transp	239
F[0].name	Rainfall scale

quantity	value
F[0].labLo	x0.7
F[0].labHi	x1.3
F[0].lo.irr	430.2
F[0].lo.et	725.9
F[0].lo.drain	10.64
F[0].lo.yield	3.655
F[0].lo.nl	10.5
F[0].lo.evap	124.6
F[0].lo.transp	249
F[0].hi.irr	188.7
F[0].hi.et	719.2
F[0].hi.drain	87.7
F[0].hi.yield	6.633
F[0].hi.nl	17.31
F[0].hi.evap	199.9
F[0].hi.transp	235.4
F[1].name	Warming
F[1].labLo	-1 C
F[1].labHi	+2 C
F[1].lo.irr	224.1
F[1].lo.et	643.4
F[1].lo.drain	35.81
F[1].lo.yield	6.633
F[1].lo.nl	6.34
F[1].lo.evap	178.2
F[1].lo.transp	218.3
F[1].hi.irr	314.3
F[1].hi.et	759.8
F[1].hi.drain	26.06
F[1].hi.yield	2.084
F[1].hi.nl	100.5
F[1].hi.evap	178.8
F[1].hi.transp	171.9
F[2].name	Trigger suction
F[2].labLo	-150 cm
F[2].labHi	-1000 cm
F[2].lo.irr	334.6
F[2].lo.et	747.1
F[2].lo.drain	43.59
F[2].lo.yield	6.633
F[2].lo.nl	13.14
F[2].lo.evap	174.8
F[2].lo.transp	244.4
F[2].hi.irr	208.4
F[2].hi.et	671.7
F[2].hi.drain	15.74
F[2].hi.yield	6.379
F[2].hi.nl	18.33

quantity	value
F[2].hi.evap	164.7
F[2].hi.transp	230.8
F[3].name	Sensor depth
F[3].labLo	10 cm
F[3].labHi	40 cm
F[3].lo.irr	315.1
F[3].lo.et	751.1
F[3].lo.drain	26.24
F[3].lo.yield	6.633
F[3].lo.nl	1.89
F[3].lo.evap	172.1
F[3].lo.transp	255.8
F[3].hi.irr	269.6
F[3].hi.et	709.6
F[3].hi.drain	28.63
F[3].hi.yield	6.583
F[3].hi.nl	18.36
F[3].hi.evap	173.3
F[3].hi.transp	234.2
F[4].name	Mulch vapour-flux factor
F[4].labLo	0.4
F[4].labHi	0.7
F[4].lo.irr	217.7
F[4].lo.et	639.3
F[4].lo.drain	50.51
F[4].lo.yield	6.633
F[4].lo.nl	4.66
F[4].lo.evap	76.19
F[4].lo.transp	257.2
F[4].hi.irr	257.8
F[4].hi.et	690
F[4].hi.drain	36.01
F[4].hi.yield	6.633
F[4].hi.nl	8.87
F[4].hi.evap	115.6
F[4].hi.transp	247.7
F[5].name	Irrigation method
F[5].labLo	overhead
F[5].labHi	drip 20-30 cm
F[5].lo.irr	281.9
F[5].lo.et	718.4
F[5].lo.drain	28.73
F[5].lo.yield	6.633
F[5].lo.nl	14.14
F[5].lo.evap	186.9
F[5].lo.transp	140.9
F[5].hi.irr	200.7
F[5].hi.et	645

quantity	value
F[5].hi.drain	31.14
F[5].hi.yield	6.564
F[5].hi.nl	15.87
F[5].hi.evap	183.3
F[5].hi.transp	258
F[6].name	Tillage
F[6].labLo	no-till
F[6].labHi	conv (base)
F[6].lo.irr	276
F[6].lo.et	720.4
F[6].lo.drain	21.35
F[6].lo.yield	6.626
F[6].lo.nl	12.95
F[6].lo.evap	178.8
F[6].lo.transp	239
F[6].hi.irr	282.2
F[6].hi.et	720.3
F[6].hi.drain	27.62
F[6].hi.yield	6.627
F[6].hi.nl	14.56
F[6].hi.evap	174.6
F[6].hi.transp	239
F[7].name	Nitrogen rate
F[7].labLo	55%
F[7].labHi	130%
F[7].lo.irr	282.2
F[7].lo.et	720.3
F[7].lo.drain	27.62
F[7].lo.yield	6.372
F[7].lo.nl	0
F[7].lo.evap	174.6
F[7].lo.transp	239
F[7].hi.irr	282.2
F[7].hi.et	720.3
F[7].hi.drain	27.62
F[7].hi.yield	6.627
F[7].hi.nl	70.09
F[7].hi.evap	174.6
F[7].hi.transp	239
SO[0].soil	S01_CoarseLoam
SO[0].irr	282.2
SO[0].et	720.3
SO[0].drain	27.62
SO[0].yield	6.627
SO[0].nl	14.56
SO[0].evap	174.6
SO[0].transp	239
SO[1].soil	S02_CoarseSand

quantity	value
SO[1].irr	413.8
SO[1].et	716.3
SO[1].drain	163.6
SO[1].yield	6.449
SO[1].nl	62.55
SO[1].evap	117.8
SO[1].transp	236.7
SO[2].soil	S03_Sand
SO[2].irr	278.2
SO[2].et	669
SO[2].drain	76.52
SO[2].yield	4.984
SO[2].nl	85.16
SO[2].evap	58.2
SO[2].transp	202.3
SO[3].soil	S04_LoamySand
SO[3].irr	293
SO[3].et	691.2
SO[3].drain	70.14
SO[3].yield	5.456
SO[3].nl	61.13
SO[3].evap	73.95
SO[3].transp	215
SO[4].soil	S05_SandyLoam
SO[4].irr	298.3
SO[4].et	710
SO[4].drain	57.78
SO[4].yield	6.198
SO[4].nl	30.79
SO[4].evap	102.5
SO[4].transp	229.8
SO[5].soil	S06_Loam
SO[5].irr	301.6
SO[5].et	719.8
SO[5].drain	50.36
SO[5].yield	6.585
SO[5].nl	10.12
SO[5].evap	137.8
SO[5].transp	238.4
SO[6].soil	S07_SiltLoam
SO[6].irr	305.4
SO[6].et	720.5
SO[6].drain	52.15
SO[6].yield	6.619
SO[6].nl	7.85
SO[6].evap	159.5
SO[6].transp	239.2
SO[7].soil	S08_SandyClayLoam

quantity	value
SO[7].irr	298.9
SO[7].et	718.6
SO[7].drain	49.98
SO[7].yield	6.539
SO[7].nl	9.03
SO[7].evap	123.9
SO[7].transp	237.2
SO[8].soil	S09_ClayLoam
SO[8].irr	305.6
SO[8].et	720.5
SO[8].drain	54.02
SO[8].yield	6.623
SO[8].nl	4.42
SO[8].evap	155.3
SO[8].transp	239.2
SO[9].soil	S10_SiltyClayLoam
SO[9].irr	303.2
SO[9].et	720.5
SO[9].drain	51.68
SO[9].yield	6.624

p3_l_sens.json

Source: `Manuscripts/results/p3_l_sens.json`; 649 values, first 220 shown. Workbook sheet `R_p3_l_sens`.

quantity	value
sprKeys	T, arr, ar, gross, I, ro, st, ex, lq, Ea, Es, DU ... (32 values)
spr[0].P	200
spr[0].u	0
spr[0].CU	0.8197
spr[0].Ea	0.7509
spr[0].drift	None
spr[0].keys	None
spr[1].P	200
spr[1].u	1
spr[1].CU	0.82
spr[1].Ea	0.7478
spr[1].drift	None
spr[1].keys	None
spr[2].P	200
spr[2].u	2
spr[2].CU	0.8211
spr[2].Ea	0.7451
spr[2].drift	None

quantity	value
spr[2].keys	None
spr[3].P	200
spr[3].u	3
spr[3].CU	0.822
spr[3].Ea	0.7416
spr[3].drift	None
spr[3].keys	None
spr[4].P	200
spr[4].u	4
spr[4].CU	0.823
spr[4].Ea	0.7375
spr[4].drift	None
spr[4].keys	None
spr[5].P	200
spr[5].u	5
spr[5].CU	0.8246
spr[5].Ea	0.7333
spr[5].drift	None
spr[5].keys	None
spr[6].P	300
spr[6].u	0
spr[6].CU	0.962
spr[6].Ea	0.9163
spr[6].drift	None
spr[6].keys	None
spr[7].P	300
spr[7].u	1
spr[7].CU	0.962
spr[7].Ea	0.9103
spr[7].drift	None
spr[7].keys	None
spr[8].P	300
spr[8].u	2
spr[8].CU	0.9622
spr[8].Ea	0.9039
spr[8].drift	None
spr[8].keys	None
spr[9].P	300
spr[9].u	3
spr[9].CU	0.9624
spr[9].Ea	0.8964
spr[9].drift	None
spr[9].keys	None
spr[10].P	300
spr[10].u	4
spr[10].CU	0.9628
spr[10].Ea	0.8881
spr[10].drift	None

quantity	value
spr[10].keys	None
spr[11].P	300
spr[11].u	5
spr[11].CU	0.9634
spr[11].Ea	0.8791
spr[11].drift	None
spr[11].keys	None
spr[12].P	400
spr[12].u	0
spr[12].CU	0.957
spr[12].Ea	0.8892
spr[12].drift	None
spr[12].keys	None
spr[13].P	400
spr[13].u	1
spr[13].CU	0.9569
spr[13].Ea	0.8821
spr[13].drift	None
spr[13].keys	None
spr[14].P	400
spr[14].u	2
spr[14].CU	0.9568
spr[14].Ea	0.8738
spr[14].drift	None
spr[14].keys	None
spr[15].P	400
spr[15].u	3
spr[15].CU	0.9567
spr[15].Ea	0.8644
spr[15].drift	None
spr[15].keys	None
spr[16].P	400
spr[16].u	4
spr[16].CU	0.9567
spr[16].Ea	0.8547
spr[16].drift	None
spr[16].keys	None
spr[17].P	400
spr[17].u	5
spr[17].CU	0.9569
spr[17].Ea	0.8446
spr[17].drift	None
spr[17].keys	None
drip[0].D	13
drip[0].LL	20
drip[0].Eu	0.9104
drip[0].hf	0.05446
drip[1].D	13

quantity	value
drip[1].LL	40
drip[1].Eu	0.9067
drip[1].hf	0.3546
drip[2].D	13
drip[2].LL	60
drip[2].Eu	0.8972
drip[2].hf	1.069
drip[3].D	13
drip[3].LL	80
drip[3].Eu	0.8802
drip[3].hf	2.193
drip[4].D	13
drip[4].LL	100
drip[4].Eu	0.8556
drip[4].hf	3.519
drip[5].D	13
drip[5].LL	120
drip[5].Eu	0.8212
drip[5].hf	4.934
drip[6].D	13
drip[6].LL	140
drip[6].Eu	0.7786
drip[6].hf	6.208
drip[7].D	13
drip[7].LL	160
drip[7].Eu	0.7313
drip[7].hf	7.218
drip[8].D	13
drip[8].LL	180
drip[8].Eu	0.6773
drip[8].hf	8.028
drip[9].D	13
drip[9].LL	200
drip[9].Eu	0.604
drip[9].hf	9.5
drip[10].D	13
drip[10].LL	220
drip[10].Eu	0.611
drip[10].hf	9.5
drip[11].D	13
drip[11].LL	240
drip[11].Eu	0.6272
drip[11].hf	8.392
drip[12].D	13
drip[12].LL	260
drip[12].Eu	0.5585
drip[12].hf	9.082
drip[13].D	13

quantity	value
drip[13].LL	280
drip[13].Eu	0.511
drip[13].hf	9.5
drip[14].D	13
drip[14].LL	300
drip[14].Eu	0.5431
drip[14].hf	9.5
drip[15].D	16
drip[15].LL	20
drip[15].Eu	0.9109
drip[15].hf	0.01986
drip[16].D	16
drip[16].LL	40
drip[16].Eu	0.9095
drip[16].hf	0.1314
drip[17].D	16
drip[17].LL	60
drip[17].Eu	0.906
drip[17].hf	0.4117
drip[18].D	16
drip[18].LL	80
drip[18].Eu	0.8996
drip[18].hf	0.9007
drip[19].D	16
drip[19].LL	100
drip[19].Eu	0.8901
drip[19].hf	1.572
drip[20].D	16
drip[20].LL	120
drip[20].Eu	0.8762
drip[20].hf	2.44
drip[21].D	16
drip[21].LL	140
drip[21].Eu	0.8579
drip[21].hf	3.414
drip[22].D	16
drip[22].LL	160
drip[22].Eu	0.8361
drip[22].hf	4.379
drip[23].D	16
drip[23].LL	180
drip[23].Eu	0.8093
drip[23].hf	5.341
drip[24].D	16
drip[24].LL	200
drip[24].Eu	0.7787
drip[24].hf	6.214
drip[25].D	16

quantity	value
drip[25].LL	220
drip[25].Eu	0.746
drip[25].hf	6.946
drip[26].D	16
drip[26].LL	240
drip[26].Eu	0.7094
drip[26].hf	7.586
drip[27].D	16
drip[27].LL	260
drip[27].Eu	0.6708

p3_e2_closure.json

Source: `Manuscripts/results/p3_e2_closure.json`; 900 values, first 220 shown. Workbook sheet `R_p3_e2_closure`.

quantity	value
[0].soil	S03_Sand
[0].method	M02_Surface
[0].crop	vegetable
[0].meanIrrTarget	447
[0].meanIrrSim	0
[0].meanDrainTarget	59.19
[0].meanDrainSim	12.07
[0].resid	5.373
[0].inflow	954.3
[0].relres	0.00563
[0].Esim	4.676
[0].Etarget	5.381
[0].Tsim	29.88
[0].Ttarget	125
[0].kt	1
[1].soil	S10_SiltyClayLoam
[1].method	M04_Drip10-20cm
[1].crop	legume
[1].meanIrrTarget	48.81
[1].meanIrrSim	0
[1].meanDrainTarget	127.3
[1].meanDrainSim	196.8
[1].resid	8.853
[1].inflow	9428
[1].relres	0.0009389
[1].Esim	69.42
[1].Etarget	78.57
[1].Tsim	202.5

quantity	value
[1].Ttarget	252.7
[1].kt	1
[2].soil	S05_SandyLoam
[2].method	M06_Drip20-30cm
[2].crop	vegetable
[2].meanIrrTarget	135.2
[2].meanIrrSim	0
[2].meanDrainTarget	245
[2].meanDrainSim	197.4
[2].resid	7.123
[2].inflow	6486
[2].relres	0.001098
[2].Esim	24.49
[2].Etarget	27.05
[2].Tsim	100.4
[2].Ttarget	176.6
[2].kt	1
[3].soil	S03_Sand
[3].method	M10_Drip50-60cm
[3].crop	vegetable
[3].meanIrrTarget	262.9
[3].meanIrrSim	0
[3].meanDrainTarget	67.23
[3].meanDrainSim	10.26
[3].resid	1.062
[3].inflow	985.7
[3].relres	0.001078
[3].Esim	6.975
[3].Etarget	9.251
[3].Tsim	31.21
[3].Ttarget	156.1
[3].kt	1
[4].soil	S01_CoarseLoam
[4].method	M06_Drip20-30cm
[4].crop	vegetable
[4].meanIrrTarget	197.6
[4].meanIrrSim	0
[4].meanDrainTarget	68.64
[4].meanDrainSim	84.22
[4].resid	8.305
[4].inflow	5360
[4].relres	0.001549
[4].Esim	82.59
[4].Etarget	107.7
[4].Tsim	98.73
[4].Ttarget	281.3
[4].kt	1
[5].soil	S02_CoarseSand

quantity	value
[5].method	M02_Surface
[5].crop	legume
[5].meanIrrTarget	262.6
[5].meanIrrSim	206.1
[5].meanDrainTarget	78.92
[5].meanDrainSim	45.57
[5].resid	2.823
[5].inflow	5097
[5].relres	0.0005539
[5].Esim	9.416
[5].Etarget	13.19
[5].Tsim	199.3
[5].Ttarget	190.5
[5].kt	2.45
[6].soil	S09_ClayLoam
[6].method	M06_Drip20-30cm
[6].crop	vegetable
[6].meanIrrTarget	156.8
[6].meanIrrSim	0
[6].meanDrainTarget	150.8
[6].meanDrainSim	172.8
[6].resid	7.893
[6].inflow	6577
[6].relres	0.0012
[6].Esim	43.68
[6].Etarget	49.53
[6].Tsim	109.7
[6].Ttarget	268.1
[6].kt	1
[7].soil	S07_SiltLoam
[7].method	M10_Drip50-60cm
[7].crop	cereal
[7].meanIrrTarget	325.8
[7].meanIrrSim	327.5
[7].meanDrainTarget	59.89
[7].meanDrainSim	191
[7].resid	7.322
[7].inflow	7526
[7].relres	0.000973
[7].Esim	16.34
[7].Etarget	24.11
[7].Tsim	169.2
[7].Ttarget	86.34
[7].kt	2.681
[8].soil	S05_SandyLoam
[8].method	M02_Surface
[8].crop	vegetable
[8].meanIrrTarget	252.9

quantity	value
[8].meanIrrSim	3.793
[8].meanDrainTarget	44.3
[8].meanDrainSim	0.0631
[8].resid	3.09
[8].inflow	1053
[8].relres	0.002935
[8].Esim	25.18
[8].Etarget	39.34
[8].Tsim	28.81
[8].Ttarget	71.66
[8].kt	1
[9].soil	S09_ClayLoam
[9].method	M06_Drip20-30cm
[9].crop	cereal
[9].meanIrrTarget	188.4
[9].meanIrrSim	182.3
[9].meanDrainTarget	23.79
[9].meanDrainSim	0.08001
[9].resid	2.201
[9].inflow	4761
[9].relres	0.0004624
[9].Esim	28.69
[9].Etarget	83.31
[9].Tsim	208.9
[9].Ttarget	105.8
[9].kt	3.837
[10].soil	S02_CoarseSand
[10].method	M09_Drip40-50cm
[10].crop	legume
[10].meanIrrTarget	258.5
[10].meanIrrSim	9.694
[10].meanDrainTarget	58.34
[10].meanDrainSim	2.344
[10].resid	3.93
[10].inflow	1678
[10].relres	0.002342
[10].Esim	34.03
[10].Etarget	104.1
[10].Tsim	48.23
[10].Ttarget	174.5
[10].kt	6.15
[11].soil	S01_CoarseLoam
[11].method	M10_Drip50-60cm
[11].crop	legume
[11].meanIrrTarget	253.9
[11].meanIrrSim	243.8
[11].meanDrainTarget	57.29
[11].meanDrainSim	138.9

quantity	value
[11].resid	4.042
[11].inflow	7026
[11].relres	0.0005752
[11].Esim	57.37
[11].Etarget	92.84
[11].Tsim	155.4
[11].Ttarget	210.5
[11].kt	0.918
[12].soil	S04_LoamySand
[12].method	M04_Drip10-20cm
[12].crop	legume
[12].meanIrrTarget	448.8
[12].meanIrrSim	432.2
[12].meanDrainTarget	47.68
[12].meanDrainSim	0
[12].resid	4.981
[12].inflow	9519
[12].relres	0.0005233
[12].Esim	6.612
[12].Etarget	7.687
[12].Tsim	469.3
[12].Ttarget	301.2
[12].kt	2.074
[13].soil	S06_Loam
[13].method	M01_Overhead
[13].crop	vegetable
[13].meanIrrTarget	921.7
[13].meanIrrSim	906.5
[13].meanDrainTarget	149.9
[13].meanDrainSim	130.2
[13].resid	5.387
[13].inflow	1.937e+04
[13].relres	0.0002781
[13].Esim	24.66
[13].Etarget	26.94
[13].Tsim	811.3
[13].Ttarget	113.5
[13].kt	7.364
[14].soil	S03_Sand
[14].method	M06_Drip20-30cm
[14].crop	legume
[14].meanIrrTarget	353.8
[14].meanIrrSim	390.2
[14].meanDrainTarget	56.93
[14].meanDrainSim	357.7
[14].resid	5.373
[14].inflow	8759

quantity	value
[14].relres	0.0006134

p3_e2_grid.json

Source: `Manuscripts/results/p3_e2_grid.json`; 1300 values, first 220 shown. Workbook sheet `R_p3_e2_grid`.

quantity	value
[0].soil	S03_Sand
[0].trig	-100
[0].method	M02_Surface
[0].irrT	284.5
[0].irrS	277.4
[0].drT	77.01
[0].drS	130.6
[0].ET	58.22
[0].ES	36.41
[0].TT	207.6
[0].TS	170
[0].kt	1
[0].relres	0.0002886
[1].soil	S03_Sand
[1].trig	-100
[1].method	M01_Overhead
[1].irrT	284.5
[1].irrS	277.4
[1].drT	77.01
[1].drS	130.6
[1].ET	58.22
[1].ES	36.41
[1].TT	207.6
[1].TS	170
[1].kt	1
[1].relres	0.0002886
[2].soil	S03_Sand
[2].trig	-200
[2].method	M02_Surface
[2].irrT	278.2
[2].irrS	285.1
[2].drT	76.52
[2].drS	27.95
[2].ET	58.2
[2].ES	34.19
[2].TT	202.3
[2].TS	289.2

quantity	value
[2].kt	1.843
[2].relres	0.0003502
[3].soil	S03_Sand
[3].trig	-200
[3].method	M01_Overhead
[3].irrT	278.2
[3].irrS	285.1
[3].drT	76.52
[3].drS	27.95
[3].ET	58.2
[3].ES	34.19
[3].TT	202.3
[3].TS	289.2
[3].kt	1.843
[3].relres	0.0003502
[4].soil	S03_Sand
[4].trig	-300
[4].method	M02_Surface
[4].irrT	278.2
[4].irrS	285.1
[4].drT	76.52
[4].drS	5.15
[4].ET	58.2
[4].ES	32.31
[4].TT	202.3
[4].TS	314.8
[4].kt	2.045
[4].relres	0.0003502
[5].soil	S03_Sand
[5].trig	-300
[5].method	M01_Overhead
[5].irrT	278.2
[5].irrS	285.1
[5].drT	76.52
[5].drS	5.15
[5].ET	58.2
[5].ES	32.31
[5].TT	202.3
[5].TS	314.8
[5].kt	2.045
[5].relres	0.0003502
[6].soil	S03_Sand
[6].trig	-500
[6].method	M02_Surface
[6].irrT	278.2
[6].irrS	0
[6].drT	76.52
[6].drS	2.198

quantity	value
[6].ET	58.2
[6].ES	20.58
[6].TT	202.3
[6].TS	44.29
[6].kt	1
[6].relres	0.001843
[7].soil	S03_Sand
[7].trig	-500
[7].method	M01_Overhead
[7].irrT	278.2
[7].irrS	0
[7].drT	76.52
[7].drS	2.198
[7].ET	58.2
[7].ES	20.58
[7].TT	202.3
[7].TS	44.29
[7].kt	1
[7].relres	0.001843
[8].soil	S03_Sand
[8].trig	-1000
[8].method	M02_Surface
[8].irrT	278.2
[8].irrS	0
[8].drT	76.52
[8].drS	2.198
[8].ET	58.2
[8].ES	20.58
[8].TT	202.3
[8].TS	44.29
[8].kt	1
[8].relres	0.001843
[9].soil	S03_Sand
[9].trig	-1000
[9].method	M01_Overhead
[9].irrT	278.2
[9].irrS	0
[9].drT	76.52
[9].drS	2.198
[9].ET	58.2
[9].ES	20.58
[9].TT	202.3
[9].TS	44.29
[9].kt	1
[9].relres	0.001843
[10].soil	S04_LoamySand
[10].trig	-100
[10].method	M02_Surface

quantity	value
[10].irrT	415.9
[10].irrS	411.2
[10].drT	75.78
[10].drS	232.6
[10].ET	55.73
[10].ES	42.95
[10].TT	271.6
[10].TS	237.7
[10].kt	1
[10].relres	0.0005469
[11].soil	S04_LoamySand
[11].trig	-100
[11].method	M01_Overhead
[11].irrT	415.9
[11].irrS	411.2
[11].drT	75.78
[11].drS	232.6
[11].ET	55.73
[11].ES	42.95
[11].TT	271.6
[11].TS	237.7
[11].kt	1
[11].relres	0.0005469
[12].soil	S04_LoamySand
[12].trig	-200
[12].method	M02_Surface
[12].irrT	309.8
[12].irrS	308.3
[12].drT	70.98
[12].drS	3.792
[12].ET	72.1
[12].ES	42.24
[12].TT	224.6
[12].TS	327.8
[12].kt	1.872
[12].relres	0.0003135
[13].soil	S04_LoamySand
[13].trig	-200
[13].method	M01_Overhead
[13].irrT	309.8
[13].irrS	308.3
[13].drT	70.98
[13].drS	3.792
[13].ET	72.1
[13].ES	42.24
[13].TT	224.6
[13].TS	327.8
[13].kt	1.872

quantity	value
[13].relres	0.0003135
[14].soil	S04_LoamySand
[14].trig	-300
[14].method	M02_Surface
[14].irrT	293
[14].irrS	291.6
[14].drT	70.14
[14].drS	0
[14].ET	73.95
[14].ES	41.58
[14].TT	215
[14].TS	319.3
[14].kt	1.93
[14].relres	0.0004296
[15].soil	S04_LoamySand
[15].trig	-300
[15].method	M01_Overhead
[15].irrT	293
[15].irrS	291.6
[15].drT	70.14
[15].drS	0
[15].ET	73.95
[15].ES	41.58
[15].TT	215
[15].TS	319.3
[15].kt	1.93
[15].relres	0.0004296
[16].soil	S04_LoamySand
[16].trig	-500
[16].method	M02_Surface
[16].irrT	293
[16].irrS	337
[16].drT	70.14
[16].drS	0
[16].ET	73.95
[16].ES	31.94
[16].TT	215
[16].TS	374.5
[16].kt	3.837

p3_bench.json

Source: `Manuscripts/results/p3_bench.json`; 4 values. Workbook sheet `R_p3_bench`.

quantity	value
engine1_model_ms	0.0212
sprinkler_ms	1.664
drip_ms	0.251
engine2_s	0.1434

p3_surface.json

Source: `Manuscripts/results/p3_surface.json`; 58 values. Workbook sheet `R_p3_surface`.

quantity	value
first.Treq	378.7
first.rows[0].Q	0.1
first.rows[0].tL	81.65
first.rows[0].r	0.4827
first.rows[0].Ea	0.6516
first.rows[1].Q	0.075
first.rows[1].tL	198.7
first.rows[1].r	0.3852
first.rows[1].Ea	0.6928
first.rows[2].Q	0.0625
first.rows[2].tL	442.3
first.rows[2].r	0.3195
first.rows[2].Ea	0.5846
first.rows[3].Q	0.05
first.rows[3].tL	2091
first.rows[3].r	0.2281
first.rows[3].Ea	0.243
first.rows[4].Q	0.0375
first.rows[4].tL	nan
first.rows[4].r	0.08246
first.rows[4].Ea	0
first.rows[5].Q	0.025
first.rows[5].tL	nan
first.rows[5].r	0.5
first.rows[5].Ea	0
first.rows[6].Q	0.0125
first.rows[6].tL	nan
first.rows[6].r	0.5
first.rows[6].Ea	0
later.Treq	868.5
later.rows[0].Q	0.1
later.rows[0].tL	26.16
later.rows[0].r	0.7569
later.rows[0].Ea	0.3353
later.rows[1].Q	0.075
later.rows[1].tL	36.05

quantity	value
later.rows[1].r	0.6942
later.rows[1].Ea	0.4422
later.rows[2].Q	0.0625
later.rows[2].tL	46.35
later.rows[2].r	0.6459
later.rows[2].Ea	0.5247
later.rows[3].Q	0.05
later.rows[3].tL	68.23
later.rows[3].r	0.5749
later.rows[3].Ea	0.6405
later.rows[4].Q	0.0375
later.rows[4].tL	143.6
later.rows[4].r	0.4532
later.rows[4].Ea	0.7904
later.rows[5].Q	0.025
later.rows[5].tL	nan
later.rows[5].r	0.1755
later.rows[5].Ea	0
later.rows[6].Q	0.0125
later.rows[6].tL	nan
later.rows[6].r	0.5
later.rows[6].Ea	0

p3_surface_shipped.json

Source: `Manuscripts/results/p3_surface_shipped.json`; 104 values. Workbook sheet `R_p3_surface_shipped`.

quantity	value
[0].irrigation	first
[0].Q	0.1
[0].A0	0.007773
[0].r	0.4827
[0].tL_book	80.2
[0].tL_reference	81.65
[0].tL_shipped	81.66
[0].tL_shipped_rAdv_r	81.66
[0].tL_v100	78.81
[0].err_reference_pct	1.814
[0].err_shipped_pct	1.815
[0].err_shipped_rAdv_r_pct	1.815
[0].err_v100_pct	-1.729
[1].irrigation	first
[1].Q	0.075
[1].A0	0.006298

quantity	value
[1].r	0.3852
[1].tL_book	198.6
[1].tL_reference	198.7
[1].tL_shipped	198.7
[1].tL_shipped_rAdv_r	198.7
[1].tL_v100	169.5
[1].err_reference_pct	0.03707
[1].err_shipped_pct	0.03792
[1].err_shipped_rAdv_r_pct	0.03782
[1].err_v100_pct	-14.67
[2].irrigation	first
[2].Q	0.0625
[2].A0	0.005511
[2].r	0.3195
[2].tL_book	441.6
[2].tL_reference	442.3
[2].tL_shipped	442.3
[2].tL_shipped_rAdv_r	442.3
[2].tL_v100	313.7
[2].err_reference_pct	0.1522
[2].err_shipped_pct	0.153
[2].err_shipped_rAdv_r_pct	0.1528
[2].err_v100_pct	-28.96
[3].irrigation	later
[3].Q	0.1
[3].A0	0.007773
[3].r	0.7569
[3].tL_book	26.2
[3].tL_reference	26.16
[3].tL_shipped	26.16
[3].tL_shipped_rAdv_r	26.16
[3].tL_v100	27.07
[3].err_reference_pct	-0.1565
[3].err_shipped_pct	-0.1566
[3].err_shipped_rAdv_r_pct	-0.1566
[3].err_v100_pct	3.338
[4].irrigation	later
[4].Q	0.075
[4].A0	0.006298
[4].r	0.6942
[4].tL_book	36
[4].tL_reference	36.05
[4].tL_shipped	36.05
[4].tL_shipped_rAdv_r	36.05
[4].tL_v100	37.42
[4].err_reference_pct	0.1366
[4].err_shipped_pct	0.1365
[4].err_shipped_rAdv_r_pct	0.1365

quantity	value
[4].err_v100_pct	3.957
[5].irrigation	later
[5].Q	0.0625
[5].A0	0.005511
[5].r	0.6459
[5].tL_book	46.3
[5].tL_reference	46.35
[5].tL_shipped	46.35
[5].tL_shipped_rAdv_r	46.35
[5].tL_v100	47.95
[5].err_reference_pct	0.1122
[5].err_shipped_pct	0.1119
[5].err_shipped_rAdv_r_pct	0.1119
[5].err_v100_pct	3.574
[6].irrigation	later
[6].Q	0.05
[6].A0	0.004681
[6].r	0.5749
[6].tL_book	68.2
[6].tL_reference	68.23
[6].tL_shipped	68.23
[6].tL_shipped_rAdv_r	68.23
[6].tL_v100	69.16
[6].err_reference_pct	0.04479
[6].err_shipped_pct	0.04405
[6].err_shipped_rAdv_r_pct	0.04414
[6].err_v100_pct	1.404
[7].irrigation	later
[7].Q	0.0375
[7].A0	0.003793
[7].r	0.4532
[7].tL_book	143.5
[7].tL_reference	143.6
[7].tL_shipped	143.6
[7].tL_shipped_rAdv_r	143.6
[7].tL_v100	129.4
[7].err_reference_pct	0.0976
[7].err_shipped_pct	0.09911
[7].err_shipped_rAdv_r_pct	0.0988
[7].err_v100_pct	-9.814

p3_border.json

Source: `Manuscripts/results/p3_border.json`; 10 values. Workbook sheet `R_p3_border`.

quantity	value
book.Treq_first	146.4
book.Treq_later	162.8
book.Qmax	1.88
book.tL_first_Q014_L360	158
book.A0	0.0303
computed.Treq_first	146.4
computed.Treq_later	162.2
computed.Qmax	1.883
computed.tL_first_Q014_L360	158.7
computed.A0	0.03034

p3_sprinkler.json

Source: `Manuscripts/results/p3_sprinkler.json`; 123 values. Workbook sheet `R_p3_sprinkler`.

quantity	value
0.675.umax	8
0.675.qmax	1.44
0.675.tset	16
0.675.sets	1
0.675.N	15
0.675.Nm	12
0.675.Tint	10
0.675.opts[0].n	1
0.675.opts[0].Nd	12
0.675.opts[0].Qin	10.12
0.675.opts[0].okT	False
0.675.opts[0].okQ	True
0.675.opts[1].n	2
0.675.opts[1].Nd	6
0.675.opts[1].Qin	20.25
0.675.opts[1].okT	True
0.675.opts[1].okQ	True
0.675.opts[2].n	3
0.675.opts[2].Nd	4
0.675.opts[2].Qin	30.37
0.675.opts[2].okT	True
0.675.opts[2].okQ	False
0.675.opts[3].n	4
0.675.opts[3].Nd	3
0.675.opts[3].Qin	40.5
0.675.opts[3].okT	True
0.675.opts[3].okQ	False
0.675.pick	2
0.675.Rw_ok	1
0.675.chk[0]	Spacing along the lateral S ≤ k·Rw, True

quantity	value
0.675.chk[1]	Spacing between laterals Sm ≤ 1.5 Rw, True
0.675.chk[2]	Wind below 16 km/h, True
0.675.chk[3]	Application rate r ≤ maximum allowed, True
0.675.chk[4]	Rate below the soil’s saturated conductivity, True
0.675.chk[5]	Christiansen uniformity CU ≥ 85 /%, False
0.675.chk[6]	Lateral head loss ≤ 20 /% of hs, True
0.675.chk[7]	Mainline gradient J < 0.05, True
0.675.chk[8]	Whole field done within the interval T, True
0.675.chk[9]	Flow within the farm supply, True
1.35.umax	8
1.35.qmax	1.44
1.35.tset	8
1.35.sets	2
1.35.N	15
1.35.Nm	12
1.35.Tint	10
1.35.opts[0].n	1
1.35.opts[0].Nd	6
1.35.opts[0].Qin	20.25
1.35.opts[0].okT	True
1.35.opts[0].okQ	True
1.35.opts[1].n	2
1.35.opts[1].Nd	3
1.35.opts[1].Qin	40.5
1.35.opts[1].okT	True
1.35.opts[1].okQ	False
1.35.opts[2].n	3
1.35.opts[2].Nd	2
1.35.opts[2].Qin	60.75
1.35.opts[2].okT	True
1.35.opts[2].okQ	False
1.35.opts[3].n	4
1.35.opts[3].Nd	2
1.35.opts[3].Qin	81
1.35.opts[3].okT	True
1.35.opts[3].okQ	False
1.35.pick	1
1.35.Rw_ok	1
1.35.chk[0]	Spacing along the lateral S ≤ k·Rw, True
1.35.chk[1]	Spacing between laterals Sm ≤ 1.5 Rw, True
1.35.chk[2]	Wind below 16 km/h, True
1.35.chk[3]	Application rate r ≤ maximum allowed, True
1.35.chk[4]	Rate below the soil’s saturated conductivity, True
1.35.chk[5]	Christiansen uniformity CU ≥ 85 /%, False
1.35.chk[6]	Lateral head loss ≤ 20 /% of hs, True
1.35.chk[7]	Mainline gradient J < 0.05, True
1.35.chk[8]	Whole field done within the interval T, True
1.35.chk[9]	Flow within the farm supply, True

quantity	value
2.03.umax	8
2.03.qmax	1.44
2.03.tset	5.32
2.03.sets	3
2.03.N	15
2.03.Nm	12
2.03.Tint	10
2.03.opts[0].n	1
2.03.opts[0].Nd	4
2.03.opts[0].Qin	30.45
2.03.opts[0].okT	True
2.03.opts[0].okQ	False
2.03.opts[1].n	2
2.03.opts[1].Nd	2
2.03.opts[1].Qin	60.9
2.03.opts[1].okT	True
2.03.opts[1].okQ	False
2.03.opts[2].n	3
2.03.opts[2].Nd	2
2.03.opts[2].Qin	91.35
2.03.opts[2].okT	True
2.03.opts[2].okQ	False
2.03.opts[3].n	4
2.03.opts[3].Nd	1
2.03.opts[3].Qin	121.8
2.03.opts[3].okT	True
2.03.opts[3].okQ	False
2.03.pick	1
2.03.Rw_ok	1
2.03.chk[0]	Spacing along the lateral S ≤ k·Rw, True
2.03.chk[1]	Spacing between laterals Sm ≤ 1.5 Rw, True
2.03.chk[2]	Wind below 16 km/h, True
2.03.chk[3]	Application rate r ≤ maximum allowed, False
2.03.chk[4]	Rate below the soil’s saturated conductivity, True
2.03.chk[5]	Christiansen uniformity CU ≥ 85 /%, False
2.03.chk[6]	Lateral head loss ≤ 20 /% of hs, True
2.03.chk[7]	Mainline gradient J < 0.05, True
2.03.chk[8]	Whole field done within the interval T, True
2.03.chk[9]	Flow within the farm supply, False
nozzle.q	22.38
nozzle.R	14.49
nozzle.h	24
nozzle_check.q	22.38
nozzle_check.R	14.49
nozzle_check.h	24

p3_drip_swafl.json

Source: `Manuscripts/results/p3_drip_swafl.json`; 960 values, first 220 shown. Workbook sheet `R_p3_drip_swafl`.

quantity	value
[0].kind	online
[0].q	3.773
[0].Se	1.296
[0].LL	138
[0].H0	19.12
[0].D	13
[0].x	0.5
[0].Hmin	0
[0].cv	0.07
[0].ne	1
[0].hf	3.44
[0].Hend	15.68
[0].Eu	0.8865
[0].qa	3.512
[0].qmin	3.417
[0].N	106
[1].kind	inline
[1].q	2.627
[1].Se	0.6332
[1].LL	48.5
[1].H0	8.825
[1].D	25
[1].x	0.5
[1].Hmin	0
[1].cv	0.04
[1].ne	1
[1].hf	0.01682
[1].Hend	8.809
[1].Eu	0.949
[1].qa	2.625
[1].qmin	2.624
[1].N	77
[2].kind	tape
[2].q	1.384
[2].Se	0.4125
[2].LL	112.3
[2].H0	14.57
[2].D	20
[2].x	0.6
[2].Hmin	0
[2].cv	0.08
[2].ne	2

quantity	value
[2].hf	0.3499
[2].Hend	14.22
[2].Eu	0.9246
[2].qa	1.369
[2].qmin	1.364
[2].N	272
[3].kind	online
[3].q	5.066
[3].Se	1.136
[3].LL	106.9
[3].H0	13.69
[3].D	25
[3].x	0.5
[3].Hmin	0
[3].cv	0.07
[3].ne	1
[3].hf	0.1776
[3].Hend	13.51
[3].Eu	0.9095
[3].qa	5.042
[3].qmin	5.033
[3].N	94
[4].kind	inline
[4].q	2.035
[4].Se	0.7778
[4].LL	54.71
[4].H0	16.99
[4].D	25
[4].x	0.5
[4].Hmin	0
[4].cv	0.04
[4].ne	1
[4].hf	0.009837
[4].Hend	16.98
[4].Eu	0.9491
[4].qa	2.035
[4].qmin	2.034
[4].N	70
[5].kind	tape
[5].q	1.015
[5].Se	0.2715
[5].LL	197.1
[5].H0	9.465
[5].D	16
[5].x	0.6
[5].Hmin	0
[5].cv	0.08
[5].ne	2

quantity	value
[5].hf	3.295
[5].Hend	6.169
[5].Eu	0.8636
[5].qa	0.8437
[5].qmin	0.785
[5].N	726
[6].kind	online
[6].q	4.295
[6].Se	0.3296
[6].LL	58.24
[6].H0	15.46
[6].D	25
[6].x	0.5
[6].Hmin	0
[6].cv	0.07
[6].ne	1
[6].hf	0.2288
[6].Hend	15.23
[6].Eu	0.9093
[6].qa	4.272
[6].qmin	4.263
[6].N	177
[7].kind	inline
[7].q	1.701
[7].Se	0.9267
[7].LL	188.2
[7].H0	11.19
[7].D	20
[7].x	0.5
[7].Hmin	0
[7].cv	0.04
[7].ne	1
[7].hf	0.4948
[7].Hend	10.69
[7].Eu	0.9435
[7].qa	1.673
[7].qmin	1.663
[7].N	203
[8].kind	tape
[8].q	1.012
[8].Se	0.3251
[8].LL	122
[8].H0	16.54
[8].D	16
[8].x	0.6
[8].Hmin	0
[8].cv	0.08
[8].ne	2

quantity	value
[8].hf	1.095
[8].Hend	15.45
[8].Eu	0.9181
[8].qa	0.9815
[8].qmin	0.9709
[8].N	375
[9].kind	online
[9].q	3.58
[9].Se	1.249
[9].LL	140.1
[9].H0	12.55
[9].D	25
[9].x	0.5
[9].Hmin	0
[9].cv	0.07
[9].ne	1
[9].hf	0.1687
[9].Hend	12.38
[9].Eu	0.9095
[9].qa	3.562
[9].qmin	3.556
[9].N	112
[10].kind	inline
[10].q	2.181
[10].Se	0.4138
[10].LL	80.59
[10].H0	19.05
[10].D	20
[10].x	0.5
[10].Hmin	0
[10].cv	0.04
[10].ne	1
[10].hf	0.3194
[10].Hend	18.73
[10].Eu	0.9471
[10].qa	2.168
[10].qmin	2.163
[10].N	195
[11].kind	tape
[11].q	1.096
[11].Se	0.27
[11].LL	109.4
[11].H0	10.14
[11].D	13
[11].x	0.6
[11].Hmin	0
[11].cv	0.08
[11].ne	2

quantity	value
[11].hf	2.779
[11].Hend	7.364
[11].Eu	0.8794
[11].qa	0.9542
[11].qmin	0.9041
[11].N	405
[12].kind	online
[12].q	4.86
[12].Se	1.112
[12].LL	63.15
[12].H0	14.52
[12].D	20
[12].x	0.5
[12].Hmin	0
[12].cv	0.07
[12].ne	1
[12].hf	0.1162
[12].Hend	14.4
[12].Eu	0.9101
[12].qa	4.846
[12].qmin	4.84
[12].N	57
[13].kind	inline
[13].q	2.659
[13].Se	0.6456
[13].LL	105.1
[13].H0	15.66
[13].D	25
[13].x	0.5
[13].Hmin	0
[13].cv	0.04
[13].ne	1
[13].hf	0.1466
[13].Hend	15.51

p3_drip_bands2.json

Source: `Manuscripts/results/p3_drip_bands2.json`; 6 values. Workbook sheet `R_p3_drip_bands2`.

quantity	value
new -1-0.2	244, 5.341e-09, 3.21e-07, 2.644e-08
new 0.2-0.6	80, 1.138e-07, 2.549e-07, 9.205e-08
new 0.6-2	76, 4.764e-16, 0.3092, 0.0797
old -1-0.2	244, 1.025e-08, 0.001344, 4.567e-05
old 0.2-0.6	80, 0.01911, 0.8116, 0.1326

quantity	value
old 0.6-2	76, 0.6731, 0.8912, 0.4438

p3_ws61.csv

Source: `Manuscripts/results/p3_ws61.csv`; 16 rows. Workbook sheet `R_p3_ws61`.

irr	quantity	book	swafl	err_pct
first	Treq	378.7	378.7	0.0007214
first	tL Q=0.1	80.2	81.65	1.814
first	Ea Q=0.1	65.4	65.16	-0.3631
first	tL Q=0.075	198.6	198.7	0.03707
first	Ea Q=0.075	69.3	69.28	-0.03564
first	tL Q=0.0625	441.6	442.3	0.1522
first	Ea Q=0.0625	58.5	58.46	-0.05991
later	Treq	868.5	868.5	0.002155
later	tL Q=0.1	26.2	26.16	-0.1565
later	tL Q=0.075	36	36.05	0.1366
later	Ea Q=0.075	44.2	44.22	0.04527
later	tL Q=0.0625	46.3	46.35	0.1122
later	tL Q=0.05	68.2	68.23	0.04479
later	Ea Q=0.05	64.1	64.05	-0.07599
later	tL Q=0.0375	143.5	143.6	0.0976
later	Ea Q=0.0375	79.1	79.04	-0.07714

fao_out.json

Source: `Manuscripts/results/fao_out.json`; 98 values. Workbook sheet `R_fao_out`.

quantity	value
[0].id	first
[0].treq	388.5
[0].treqRef	388.5
[0].qm	0.02258
[0].qmRef	0.0226
[0].rows[0].q	0.036
[0].rows[0].ref	6.36
[0].rows[0].adv	6.357
[0].rows[0].err	-0.0539
[0].rows[0].adv_v100	6.391
[0].rows[0].err_v100	0.4827
[0].rows[1].q	0.0545
[0].rows[1].ref	3.11
[0].rows[1].adv	3.119

quantity	value
[0].rows[1].err	0.2913
[0].rows[1].adv_v100	3.188
[0].rows[1].err_v100	2.519
[0].rows[2].q	0.072
[0].rows[2].ref	2.14
[0].rows[2].adv	2.135
[0].rows[2].err	-0.2347
[0].rows[2].adv_v100	2.182
[0].rows[2].err_v100	1.961
[0].rows[3].q	0.09
[0].rows[3].ref	1.64
[0].rows[3].adv	1.641
[0].rows[3].err	0.0496
[0].rows[3].adv_v100	1.673
[0].rows[3].err_v100	2.036
[1].id	later
[1].treq	678.9
[1].treqRef	678.9
[1].qm	0.02258
[1].qmRef	0.0226
[1].rows[0].q	0.018
[1].rows[0].ref	15.55
[1].rows[0].adv	15.56
[1].rows[0].err	0.03386
[1].rows[0].adv_v100	15.57
[1].rows[0].err_v100	0.1467
[1].rows[1].q	0.036
[1].rows[1].ref	5.03
[1].rows[1].adv	5.028
[1].rows[1].err	-0.04831
[1].rows[1].adv_v100	5.176
[1].rows[1].err_v100	2.904
[1].rows[2].q	0.0545
[1].rows[2].ref	3.15
[1].rows[2].adv	3.154
[1].rows[2].err	0.1409
[1].rows[2].adv_v100	3.228
[1].rows[2].err_v100	2.465
[2].id	first100
[2].treq	388.5
[2].treqRef	388.5
[2].qm	0.03193
[2].qmRef	0.032
[2].rows[0].q	0.018
[2].rows[0].ref	5.27
[2].rows[0].adv	5.268
[2].rows[0].err	-0.04728
[2].rows[0].adv_v100	5.188

quantity	value
[2].rows[0].err_v100	-1.548
[2].rows[1].q	0.0269
[2].rows[1].ref	2.35
[2].rows[1].adv	2.348
[2].rows[1].err	-0.09153
[2].rows[1].adv_v100	2.371
[2].rows[1].err_v100	0.9027
[2].rows[2].q	0.036
[2].rows[2].ref	1.44
[2].rows[2].adv	1.442
[2].rows[2].err	0.1328
[2].rows[2].adv_v100	1.462
[2].rows[2].err_v100	1.554
[2].rows[3].q	0.045
[2].rows[3].ref	1.03
[2].rows[3].adv	1.035
[2].rows[3].err	0.4668
[2].rows[3].adv_v100	1.05
[2].rows[3].err_v100	1.92
[3].id	later100
[3].treq	678.9
[3].treqRef	678.9
[3].qm	0.03193
[3].qmRef	0.032
[3].rows[0].q	0.009
[3].rows[0].ref	12.89
[3].rows[0].adv	12.89
[3].rows[0].err	0.03702
[3].rows[0].adv_v100	12.61
[3].rows[0].err_v100	-2.198
[3].rows[1].q	0.018
[3].rows[1].ref	3.45
[3].rows[1].adv	3.446
[3].rows[1].err	-0.1056
[3].rows[1].adv_v100	3.52
[3].rows[1].err_v100	2.034

spr_out.json

Source: `Manuscripts/results/spr_out.json`; 432 values, first 220 shown. Workbook sheet `R_spr_out`.

quantity	value
new.nozzle.q	22.38
new.nozzle.R	14.49
new.nozzle.h	24
new.D	28.98

quantity	value
new.cu[0].u	0
new.cu[0].ratio	0.4
new.cu[0].S	11.59
new.cu[0].CU	0.974
new.cu[0].AR	9.992
new.cu[0].Ea	0.9873
new.cu[1].u	0
new.cu[1].ratio	0.5
new.cu[1].S	14.49
new.cu[1].CU	0.9435
new.cu[1].AR	6.392
new.cu[1].Ea	0.9873
new.cu[2].u	0
new.cu[2].ratio	0.6
new.cu[2].S	17.39
new.cu[2].CU	0.9289
new.cu[2].AR	4.445
new.cu[2].Ea	0.9873
new.cu[3].u	0
new.cu[3].ratio	0.65
new.cu[3].S	18.84
new.cu[3].CU	0.8573
new.cu[3].AR	3.785
new.cu[3].Ea	0.9873
new.cu[4].u	0
new.cu[4].ratio	0.7
new.cu[4].S	20.29
new.cu[4].CU	0.7868
new.cu[4].AR	3.261
new.cu[4].Ea	0.9873
new.cu[5].u	0
new.cu[5].ratio	0.8
new.cu[5].S	23.18
new.cu[5].CU	0.6247
new.cu[5].AR	2.5
new.cu[5].Ea	0.9873
new.cu[6].u	0
new.cu[6].ratio	1
new.cu[6].S	28.98
new.cu[6].CU	0.2207
new.cu[6].AR	1.6
new.cu[6].Ea	0.9873
new.cu[7].u	0
new.cu[7].ratio	1.2
new.cu[7].S	34.78
new.cu[7].CU	-0.08095
new.cu[7].AR	1.109
new.cu[7].Ea	0.9873

quantity	value
new.cu[8].u	3
new.cu[8].ratio	0.4
new.cu[8].S	11.59
new.cu[8].CU	0.9744
new.cu[8].AR	9.993
new.cu[8].Ea	0.9677
new.cu[9].u	3
new.cu[9].ratio	0.5
new.cu[9].S	14.49
new.cu[9].CU	0.9416
new.cu[9].AR	6.394
new.cu[9].Ea	0.9677
new.cu[10].u	3
new.cu[10].ratio	0.6
new.cu[10].S	17.39
new.cu[10].CU	0.9281
new.cu[10].AR	4.443
new.cu[10].Ea	0.9677
new.cu[11].u	3
new.cu[11].ratio	0.65
new.cu[11].S	18.84
new.cu[11].CU	0.8581
new.cu[11].AR	3.785
new.cu[11].Ea	0.9677
new.cu[12].u	3
new.cu[12].ratio	0.7
new.cu[12].S	20.29
new.cu[12].CU	0.7877
new.cu[12].AR	3.262
new.cu[12].Ea	0.9677
new.cu[13].u	3
new.cu[13].ratio	0.8
new.cu[13].S	23.18
new.cu[13].CU	0.6262
new.cu[13].AR	2.5
new.cu[13].Ea	0.9677
new.cu[14].u	3
new.cu[14].ratio	1
new.cu[14].S	28.98
new.cu[14].CU	0.2226
new.cu[14].AR	1.6
new.cu[14].Ea	0.9677
new.cu[15].u	3
new.cu[15].ratio	1.2
new.cu[15].S	34.78
new.cu[15].CU	-0.07702
new.cu[15].AR	1.11
new.cu[15].Ea	0.9677

quantity	value
new.cutri[0].ratio	0.5
new.cutri[0].CU	0.9463
new.cutri[1].ratio	0.6
new.cutri[1].CU	0.937
new.cutri[2].ratio	0.7
new.cutri[2].CU	0.9279
new.cutri[3].ratio	0.8
new.cutri[3].CU	0.7518
new.q[0].d	2.4
new.q[0].P	200
new.q[0].q	5.157
new.q[0].qth	5.157
new.q[0].R	9.443
new.q[0].Rkb	9.443
new.q[1].d	2.4
new.q[1].P	300
new.q[1].q	6.316
new.q[1].qth	6.316
new.q[1].R	11.57
new.q[1].Rkb	11.57
new.q[2].d	2.4
new.q[2].P	400
new.q[2].q	7.293
new.q[2].qth	7.293
new.q[2].R	13.35
new.q[2].Rkb	13.35
new.q[3].d	3.2
new.q[3].P	200
new.q[3].q	9.168
new.q[3].qth	9.168
new.q[3].R	10.9
new.q[3].Rkb	10.9
new.q[4].d	3.2
new.q[4].P	300
new.q[4].q	11.23
new.q[4].qth	11.23
new.q[4].R	13.35
new.q[4].Rkb	13.35
new.q[5].d	3.2
new.q[5].P	400
new.q[5].q	12.97
new.q[5].qth	12.97
new.q[5].R	15.42
new.q[5].Rkb	15.42
new.q[6].d	4
new.q[6].P	200
new.q[6].q	14.33
new.q[6].qth	14.33

quantity	value
new.q[6].R	12.19
new.q[6].Rkb	12.19
new.q[7].d	4
new.q[7].P	300
new.q[7].q	17.55
new.q[7].qth	17.55
new.q[7].R	14.93
new.q[7].Rkb	14.93
new.q[8].d	4
new.q[8].P	400
new.q[8].q	20.26
new.q[8].qth	20.26
new.q[8].R	17.24
new.q[8].Rkb	17.24
new.q[9].d	4.8
new.q[9].P	200
new.q[9].q	20.63
new.q[9].qth	20.63
new.q[9].R	13.35
new.q[9].Rkb	13.35
new.q[10].d	4.8
new.q[10].P	300
new.q[10].q	25.27
new.q[10].qth	25.27
new.q[10].R	16.36
new.q[10].Rkb	16.36
new.q[11].d	4.8
new.q[11].P	400
new.q[11].q	29.17
new.q[11].qth	29.17
new.q[11].R	18.89
new.q[11].Rkb	18.89
new.q[12].d	6
new.q[12].P	200
new.q[12].q	32.23
new.q[12].qth	32.23
new.q[12].R	14.93
new.q[12].Rkb	14.93
new.q[13].d	6
new.q[13].P	300
new.q[13].q	39.48
new.q[13].qth	39.48
new.q[13].R	18.29
new.q[13].Rkb	18.29
new.q[14].d	6
new.q[14].P	400
new.q[14].q	45.58
new.q[14].qth	45.58

quantity	value
new.q[14].R	21.12
new.q[14].Rkb	21.12
new.q[15].d	8
new.q[15].P	200
new.q[15].q	57.3
new.q[15].qth	57.3
new.q[15].R	17.24
new.q[15].Rkb	17.24
new.q[16].d	8
new.q[16].P	300
new.q[16].q	70.18
new.q[16].qth	70.18
new.q[16].R	21.12
new.q[16].Rkb	21.12
new.q[17].d	8
new.q[17].P	400
new.q[17].q	81.04
new.q[17].qth	81.04
new.q[17].R	24.38
new.q[17].Rkb	24.38
old.nozzle.q	22.38
old.nozzle.R	12.34
old.nozzle.h	24
old.D	24.69

drip_out.json

Source: `Manuscripts/results/drip_out.json`; 14470 values, first 220 shown. Workbook sheet `R_drip_out`.

quantity	value
new.rows[0].kind	online
new.rows[0].q	3.265
new.rows[0].Se	0.421
new.rows[0].LL	124.9
new.rows[0].H0	7.351
new.rows[0].D	25
new.rows[0].x	0.5
new.rows[0].Hmin	0
new.rows[0].cv	0.07
new.rows[0].ne	1
new.rows[0].N	297
new.rows[0].hf	0.7154
new.rows[0].Hend	6.635
new.rows[0].Eu	0.8987

quantity	value
new.rows[0].qa	3.145
new.rows[0].qmin	3.102
new.rows[0].conv	True
new.rows[0].iters	9
new.rows[0].hfFrac	0.09732
new.rows[0].warn	False
new.rows[1].kind	inline
new.rows[1].q	2.724
new.rows[1].Se	0.2681
new.rows[1].LL	223.8
new.rows[1].H0	7.866
new.rows[1].D	13
new.rows[1].x	0.5
new.rows[1].Hmin	0
new.rows[1].cv	0.04
new.rows[1].ne	1
new.rows[1].N	835
new.rows[1].hf	7.366
new.rows[1].Hend	0.5
new.rows[1].Eu	0.7249
new.rows[1].qa	0.8993
new.rows[1].qmin	0.6868
new.rows[1].conv	True
new.rows[1].iters	25
new.rows[1].hfFrac	0.9364
new.rows[1].warn	True
new.rows[2].kind	tape
new.rows[2].q	1.309
new.rows[2].Se	0.4164
new.rows[2].LL	127
new.rows[2].H0	12.39
new.rows[2].D	20
new.rows[2].x	0.6
new.rows[2].Hmin	0
new.rows[2].cv	0.08
new.rows[2].ne	2
new.rows[2].N	305
new.rows[2].hf	0.4366
new.rows[2].Hend	11.96
new.rows[2].Eu	0.9229
new.rows[2].qa	1.288
new.rows[2].qmin	1.281
new.rows[2].conv	True
new.rows[2].iters	5
new.rows[2].hfFrac	0.03523
new.rows[2].warn	False
new.rows[3].kind	pc
new.rows[3].q	2.355

quantity	value
new.rows[3].Se	0.5139
new.rows[3].LL	159.3
new.rows[3].H0	18.03
new.rows[3].D	16
new.rows[3].x	0.05
new.rows[3].Hmin	8
new.rows[3].cv	0.03
new.rows[3].ne	1
new.rows[3].N	310
new.rows[3].hf	5.049
new.rows[3].Hend	12.98
new.rows[3].Eu	0.9591
new.rows[3].qa	2.336
new.rows[3].qmin	2.329
new.rows[3].conv	True
new.rows[3].iters	4
new.rows[3].hfFrac	0.28
new.rows[3].warn	True
new.rows[4].kind	online
new.rows[4].q	2.857
new.rows[4].Se	0.7863
new.rows[4].LL	238.7
new.rows[4].H0	6.648
new.rows[4].D	13
new.rows[4].x	0.5
new.rows[4].Hmin	0
new.rows[4].cv	0.07
new.rows[4].ne	1
new.rows[4].N	304
new.rows[4].hf	6.148
new.rows[4].Hend	0.5
new.rows[4].Eu	0.6143
new.rows[4].qa	1.393
new.rows[4].qmin	0.9395
new.rows[4].conv	False
new.rows[4].iters	500
new.rows[4].hfFrac	0.9248
new.rows[4].warn	True
new.rows[5].kind	inline
new.rows[5].q	1.426
new.rows[5].Se	0.6609
new.rows[5].LL	43.52
new.rows[5].H0	12.51
new.rows[5].D	32
new.rows[5].x	0.5
new.rows[5].Hmin	0
new.rows[5].cv	0.04
new.rows[5].ne	1

quantity	value
new.rows[5].N	66
new.rows[5].hf	0.0011
new.rows[5].Hend	12.51
new.rows[5].Eu	0.9492
new.rows[5].qa	1.426
new.rows[5].qmin	1.426
new.rows[5].conv	True
new.rows[5].iters	4
new.rows[5].hfFrac	8.798e-05
new.rows[5].warn	False
new.rows[6].kind	tape
new.rows[6].q	1.326
new.rows[6].Se	0.7828
new.rows[6].LL	291.2
new.rows[6].H0	6.849
new.rows[6].D	16
new.rows[6].x	0.6
new.rows[6].Hmin	0
new.rows[6].cv	0.08
new.rows[6].ne	2
new.rows[6].N	372
new.rows[6].hf	2.78
new.rows[6].Hend	4.069
new.rows[6].Eu	0.8466
new.rows[6].qa	1.064
new.rows[6].qmin	0.9702
new.rows[6].conv	True
new.rows[6].iters	15
new.rows[6].hfFrac	0.406
new.rows[6].warn	True
new.rows[7].kind	pc
new.rows[7].q	2.596
new.rows[7].Se	0.2255
new.rows[7].LL	87.88
new.rows[7].H0	21.47
new.rows[7].D	13
new.rows[7].x	0.05
new.rows[7].Hmin	8
new.rows[7].cv	0.03
new.rows[7].ne	1
new.rows[7].N	390
new.rows[7].hf	13.8
new.rows[7].Hend	7.669
new.rows[7].Eu	0.9533
new.rows[7].qa	2.554
new.rows[7].qmin	2.531
new.rows[7].conv	True
new.rows[7].iters	22

quantity	value
new.rows[7].hfFrac	0.6428
new.rows[7].warn	True
new.rows[8].kind	online
new.rows[8].q	3.418
new.rows[8].Se	0.897
new.rows[8].LL	232.7
new.rows[8].H0	22.88
new.rows[8].D	25
new.rows[8].x	0.5
new.rows[8].Hmin	0
new.rows[8].cv	0.07
new.rows[8].ne	1
new.rows[8].N	259
new.rows[8].hf	1.169
new.rows[8].Hend	21.71
new.rows[8].Eu	0.9048
new.rows[8].qa	3.353
new.rows[8].qmin	3.329
new.rows[8].conv	True
new.rows[8].iters	5
new.rows[8].hfFrac	0.05109
new.rows[8].warn	False
new.rows[9].kind	inline
new.rows[9].q	1.478
new.rows[9].Se	0.5697
new.rows[9].LL	190.2
new.rows[9].H0	16.27
new.rows[9].D	16
new.rows[9].x	0.5
new.rows[9].Hmin	0
new.rows[9].cv	0.04
new.rows[9].ne	1
new.rows[9].N	334
new.rows[9].hf	2.59
new.rows[9].Hend	13.68
new.rows[9].Eu	0.9271
new.rows[9].qa	1.388
new.rows[9].qmin	1.356
new.rows[9].conv	True
new.rows[9].iters	12
new.rows[9].hfFrac	0.1591
new.rows[9].warn	False
new.rows[10].kind	tape
new.rows[10].q	0.8285
new.rows[10].Se	0.694
new.rows[10].LL	153
new.rows[10].H0	6.85
new.rows[10].D	25

quantity	value
new.rows[10].x	0.6
new.rows[10].Hmin	0
new.rows[10].cv	0.08
new.rows[10].ne	2
new.rows[10].N	220
new.rows[10].hf	0.04271
new.rows[10].Hend	6.807
new.rows[10].Eu	0.9272
new.rows[10].qa	0.8262
new.rows[10].qmin	0.8254
new.rows[10].conv	True
new.rows[10].iters	4
new.rows[10].hfFrac	0.006235
new.rows[10].warn	False

e12.json

Source: `Manuscripts/results/e12.json`; 12000 values, first 220 shown. Workbook sheet `R_e12`.

quantity	value
[0].trig	-100
[0].z	100
[0].P	0.5
[0].ET_mm	674.7
[0].irrigation_mm	438.2
[0].drainage_1m_mm	3.799
[0].yield_Mg_ha	4.935
[0].transpiration_mm	244.4
[0].soil_evap_mm	97.75
[0].N_leached_kg_ha	43.62
[0].treatment	z100cm_h100_P0.5
[0].soil	S01_CoarseLoam
[1].trig	-100
[1].z	100
[1].P	0.6
[1].ET_mm	712.6
[1].irrigation_mm	436.6
[1].drainage_1m_mm	8.117
[1].yield_Mg_ha	6.598
[1].transpiration_mm	263.4
[1].soil_evap_mm	108.6
[1].N_leached_kg_ha	10.62
[1].treatment	z100cm_h100_P0.6
[1].soil	S01_CoarseLoam
[2].trig	-100
[2].z	100

quantity	value
[2].P	0.7
[2].ET_mm	715.4
[2].irrigation_mm	408.6
[2].drainage_1m_mm	22.13
[2].yield_Mg_ha	6.625
[2].transpiration_mm	256.6
[2].soil_evap_mm	116.3
[2].N_leached_kg_ha	14.91
[2].treatment	z100cm_h100_P0.7
[2].soil	S01_CoarseLoam
[3].trig	-100
[3].z	100
[3].P	0.8
[3].ET_mm	704.6
[3].irrigation_mm	358.6
[3].drainage_1m_mm	31.85
[3].yield_Mg_ha	6.355
[3].transpiration_mm	242.8
[3].soil_evap_mm	127.5
[3].N_leached_kg_ha	23.82
[3].treatment	z100cm_h100_P0.8
[3].soil	S01_CoarseLoam
[4].trig	-100
[4].z	100
[4].P	0.9
[4].ET_mm	705.5
[4].irrigation_mm	324.7
[4].drainage_1m_mm	42.4
[4].yield_Mg_ha	6.397
[4].transpiration_mm	240.8
[4].soil_evap_mm	134.5
[4].N_leached_kg_ha	25.07
[4].treatment	z100cm_h100_P0.9
[4].soil	S01_CoarseLoam
[5].trig	-100
[5].z	100
[5].P	1
[5].ET_mm	699.6
[5].irrigation_mm	289.1
[5].drainage_1m_mm	59.81
[5].yield_Mg_ha	6.263
[5].transpiration_mm	232.8
[5].soil_evap_mm	140.7
[5].N_leached_kg_ha	30.91
[5].treatment	z100cm_h100_P1
[5].soil	S01_CoarseLoam
[6].trig	-100
[6].z	100

quantity	value
[6].P	1.1
[6].ET_mm	696.8
[6].irrigation_mm	256.2
[6].drainage_1m_mm	75.08
[6].yield_Mg_ha	6.356
[6].transpiration_mm	232.8
[6].soil_evap_mm	145.1
[6].N_leached_kg_ha	27.9
[6].treatment	z100cm_h100_P1.1
[6].soil	S01_CoarseLoam
[7].trig	-100
[7].z	100
[7].P	1.2
[7].ET_mm	699.9
[7].irrigation_mm	235.8
[7].drainage_1m_mm	96.75
[7].yield_Mg_ha	6.511
[7].transpiration_mm	235
[7].soil_evap_mm	148.6
[7].N_leached_kg_ha	23.25
[7].treatment	z100cm_h100_P1.2
[7].soil	S01_CoarseLoam
[8].trig	-100
[8].z	100
[8].P	1.3
[8].ET_mm	698.8
[8].irrigation_mm	214.7
[8].drainage_1m_mm	121.1
[8].yield_Mg_ha	6.605
[8].transpiration_mm	235.1
[8].soil_evap_mm	152.7
[8].N_leached_kg_ha	22.41
[8].treatment	z100cm_h100_P1.3
[8].soil	S01_CoarseLoam
[9].trig	-100
[9].z	100
[9].P	1.4
[9].ET_mm	696.9
[9].irrigation_mm	192
[9].drainage_1m_mm	143.5
[9].yield_Mg_ha	6.62
[9].transpiration_mm	232.6
[9].soil_evap_mm	159.8
[9].N_leached_kg_ha	22.29
[9].treatment	z100cm_h100_P1.4
[9].soil	S01_CoarseLoam
[10].trig	-1200
[10].z	100

quantity	value
[10].P	0.5
[10].ET_mm	663.3
[10].irrigation_mm	426.5
[10].drainage_1m_mm	3.744
[10].yield_Mg_ha	4.54
[10].transpiration_mm	236.1
[10].soil_evap_mm	98.74
[10].N_leached_kg_ha	55.36
[10].treatment	z100cm_h1200_P0.5
[10].soil	S01_CoarseLoam
[11].trig	-1200
[11].z	100
[11].P	0.6
[11].ET_mm	677.2
[11].irrigation_mm	397
[11].drainage_1m_mm	6.238
[11].yield_Mg_ha	5.187
[11].transpiration_mm	239.1
[11].soil_evap_mm	109
[11].N_leached_kg_ha	47.29
[11].treatment	z100cm_h1200_P0.6
[11].soil	S01_CoarseLoam
[12].trig	-1200
[12].z	100
[12].P	0.7
[12].ET_mm	672.7
[12].irrigation_mm	349.1
[12].drainage_1m_mm	10.19
[12].yield_Mg_ha	5.43
[12].transpiration_mm	233
[12].soil_evap_mm	118.4
[12].N_leached_kg_ha	45.91
[12].treatment	z100cm_h1200_P0.7
[12].soil	S01_CoarseLoam
[13].trig	-1200
[13].z	100
[13].P	0.8
[13].ET_mm	671.1
[13].irrigation_mm	304
[13].drainage_1m_mm	14.8
[13].yield_Mg_ha	5.555
[13].transpiration_mm	227.4
[13].soil_evap_mm	129.6
[13].N_leached_kg_ha	50.08
[13].treatment	z100cm_h1200_P0.8
[13].soil	S01_CoarseLoam
[14].trig	-1200
[14].z	100

quantity	value
[14].P	0.9
[14].ET_mm	669
[14].irrigation_mm	257
[14].drainage_1m_mm	18.59
[14].yield_Mg_ha	5.826
[14].transpiration_mm	230.7
[14].soil_evap_mm	137.9
[14].N_leached_kg_ha	41.62
[14].treatment	z100cm_h1200_P0.9
[14].soil	S01_CoarseLoam
[15].trig	-1200
[15].z	100
[15].P	1
[15].ET_mm	673
[15].irrigation_mm	217.7
[15].drainage_1m_mm	21.76
[15].yield_Mg_ha	6.011
[15].transpiration_mm	231.8
[15].soil_evap_mm	149.1
[15].N_leached_kg_ha	37.34
[15].treatment	z100cm_h1200_P1
[15].soil	S01_CoarseLoam
[16].trig	-1200
[16].z	100
[16].P	1.1
[16].ET_mm	662.5
[16].irrigation_mm	166.8
[16].drainage_1m_mm	30.3
[16].yield_Mg_ha	6.184
[16].transpiration_mm	232.1
[16].soil_evap_mm	157.6
[16].N_leached_kg_ha	34.88
[16].treatment	z100cm_h1200_P1.1
[16].soil	S01_CoarseLoam
[17].trig	-1200
[17].z	100
[17].P	1.2
[17].ET_mm	658.8
[17].irrigation_mm	122.5
[17].drainage_1m_mm	35.24
[17].yield_Mg_ha	6.292
[17].transpiration_mm	231.7
[17].soil_evap_mm	169.2
[17].N_leached_kg_ha	32.42
[17].treatment	z100cm_h1200_P1.2
[17].soil	S01_CoarseLoam
[18].trig	-1200
[18].z	100

quantity	value
[18].P	1.3
[18].ET_mm	644.9

e12_pred.json

Source: `Manuscripts/results/e12_pred.json`; 11000 values, first 220 shown. Workbook sheet `R_e12_pred`.

quantity	value
[0].evap	78.21
[0].transp	280.6
[0].et	818.9
[0].irr	667.3
[0].yield	1.619
[0].drain	50.44
[0].nl	33.99
[0].rain	232
[0].other	460.1
[0].store	30
[0].ie	0.2427
[1].evap	93.39
[1].transp	287.3
[1].et	827
[1].irr	629.8
[1].yield	2.614
[1].drain	51.24
[1].nl	27.94
[1].rain	278.4
[1].other	446.4
[1].store	30
[1].ie	0.4151
[2].evap	106.5
[2].transp	301.2
[2].et	840.3
[2].irr	597.2
[2].yield	3.975
[2].drain	51.7
[2].nl	5.26
[2].rain	324.8
[2].other	432.6
[2].store	30
[2].ie	0.6655
[3].evap	118
[3].transp	305.3
[3].et	832.6

quantity	value
[3].irr	545.2
[3].yield	5.087
[3].drain	53.74
[3].nl	0
[3].rain	371.2
[3].other	409.4
[3].store	30
[3].ie	0.9331
[4].evap	134.5
[4].transp	270.9
[4].et	780.5
[4].irr	452.5
[4].yield	6.289
[4].drain	59.66
[4].nl	3.96
[4].rain	417.6
[4].other	375.1
[4].store	30
[4].ie	1.39
[5].evap	144.7
[5].transp	270.2
[5].et	780.7
[5].irr	393.3
[5].yield	6.633
[5].drain	68.84
[5].nl	8.32
[5].rain	464
[5].other	365.8
[5].store	7.743
[5].ie	1.686
[6].evap	154.3
[6].transp	267.2
[6].et	782.3
[6].irr	344.5
[6].yield	6.633
[6].drain	84.62
[6].nl	14.29
[6].rain	510.4
[6].other	360.8
[6].store	-12
[6].ie	1.925
[7].evap	159.1
[7].transp	261.5
[7].et	764.6
[7].irr	297.8
[7].yield	6.633
[7].drain	105
[7].nl	12.99

quantity	value
[7].rain	556.8
[7].other	344
[7].store	-15
[7].ie	2.227
[8].evap	162.1
[8].transp	256.8
[8].et	751
[8].irr	261.5
[8].yield	6.633
[8].drain	128.7
[8].nl	10.94
[8].rain	603.2
[8].other	332.2
[8].store	-15
[8].ie	2.536
[9].evap	163.6
[9].transp	256
[9].et	745.4
[9].irr	235.7
[9].yield	6.633
[9].drain	154.9
[9].nl	6.95
[9].rain	649.6
[9].other	325.9
[9].store	-15
[9].ie	2.814
[10].evap	63.48
[10].transp	209.7
[10].et	596.9
[10].irr	361.5
[10].yield	1.426
[10].drain	11.54
[10].nl	50.76
[10].rain	232
[10].other	323.7
[10].store	-15
[10].ie	0.3944
[11].evap	78.24
[11].transp	221.5
[11].et	622.2
[11].irr	341.1
[11].yield	2.301
[11].drain	12.34
[11].nl	44.71
[11].rain	278.4
[11].other	322.4
[11].store	-15
[11].ie	0.6746

quantity	value
[12].evap	91.82
[12].transp	239
[12].et	650.5
[12].irr	323.5
[12].yield	3.499
[12].drain	12.8
[12].nl	22.03
[12].rain	324.8
[12].other	319.6
[12].store	-15
[12].ie	1.082
[13].evap	105.2
[13].transp	250.6
[13].et	666.7
[13].irr	295.3
[13].yield	4.479
[13].drain	14.84
[13].nl	8.33
[13].rain	371.2
[13].other	310.9
[13].store	-15
[13].ie	1.517
[14].evap	126.1
[14].transp	233.7
[14].et	657
[14].irr	245.1
[14].yield	5.537
[14].drain	20.76
[14].nl	20.73
[14].rain	417.6
[14].other	297.1
[14].store	-15
[14].ie	2.259
[15].evap	136.7
[15].transp	234.9
[15].et	662.1
[15].irr	213
[15].yield	6.345
[15].drain	29.94
[15].nl	25.09
[15].rain	464
[15].other	290.5
[15].store	-15
[15].ie	2.979
[16].evap	146.4
[16].transp	233.3
[16].et	666.3
[16].irr	186.6

quantity	value
[16].yield	6.633
[16].drain	45.72
[16].nl	31.06
[16].rain	510.4
[16].other	286.6
[16].store	-15
[16].ie	3.554
[17].evap	154.6
[17].transp	233.9
[17].et	667
[17].irr	161.3
[17].yield	6.519
[17].drain	66.09
[17].nl	29.76
[17].rain	556.8
[17].other	278.6
[17].store	-15
[17].ie	4.042
[18].evap	161
[18].transp	234.8
[18].et	670
[18].irr	141.7
[18].yield	6.541
[18].drain	89.81
[18].nl	27.71
[18].rain	603.2
[18].other	274.2
[18].store	-15
[18].ie	4.618
[19].evap	165.2
[19].transp	238.1
[19].et	676.3
[19].irr	127.7
[19].yield	6.607
[19].drain	116
[19].nl	23.72
[19].rain	649.6
[19].other	273
[19].store	-15
[19].ie	5.175

Reproducing every number

Sheet `EQ_P2` holds Parts 1, 2, 3, 6 and 7 as formulas; the verification scripts of the article are listed in its last section.

S14. Where every table and figure comes from

Item	How it is made	Script(s)	Input and output
Table 1	Table typed from the result file (4 of its numbers found there)	Manuscripts/reproducibility/audit_numbers.py; Manuscripts/reproducibility/p3_js/e1_heldout.js; Manuscripts/reproducibility/analysis/p3_figs.py	Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/results/e12_pred.json
Figure 1	Figure drawn by the script from the input data	Manuscripts/reproducibility/make_a4.py; Manuscripts/reproducibility/analysis/p3_figs.py; Manuscripts/reproducibility/analysis/p3_fig01_workflow_v9.py	Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/figures/p3/fig01_architecture_A4.pdf
Table 2	Table typed from the result file (10 of its numbers found there)	Manuscripts/reproducibility/audit_numbers.py; Manuscripts/reproducibility/p3_js/e1_heldout.js; Manuscripts/reproducibility/analysis/p3_figs.py	Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/results/e12_pred.json
Table 3	Table typed from the result file (5 of its numbers found there)	Manuscripts/reproducibility/p3_js/run_drip.js; Manuscripts/reproducibility/analysis/p3_lit_figs.py; Manuscripts/reproducibility/analysis/make_supplement_data.py	Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/results/drip_out.json
Table 4	Table typed in the manuscript from the result files of the script	(see the reproducibility appendix of the paper)	Data/daisy_ensemble.csv.gz (10 190 runs)
Table 5	Descriptive table (definitions, design, parameters or tests); no computation	-	
Table 6	Descriptive table (definitions, design, parameters or tests); no computation	-	
Table 7	Table typed from the result file (11 of its numbers found there)	Manuscripts/reproducibility/audit_numbers.py; Manuscripts/reproducibility/p3_js/e1_heldout.js; Manuscripts/reproducibility/analysis/p3_figs.py	Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/results/e12_pred.json
Table 8	Table typed from the result file (39 of its numbers found there)	Manuscripts/reproducibility/p3_js/run_drip.js; Manuscripts/reproducibility/analysis/p3_lit_figs.py; Manuscripts/reproducibility/analysis/make_supplement_data.py; swa_vars.js; p3_swa_vars.js; p3_swa_fig.py	Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/results/drip_out.json
Table 9	Table typed from the result file (15 of its numbers found there)	Manuscripts/reproducibility/audit_numbers.py; Manuscripts/reproducibility/p3_js/e1_heldout.js; Manuscripts/reproducibility/analysis/p3_figs.py; swa_vars.js; p3_swa_vars.js; p3_swa_fig.py	Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/results/e12_pred.json

Item	How it is made	Script(s)	Input and output
Table 10	Table typed from the result file (36 of its numbers found there)	Manuscripts/reproducibility/p3_js/swa_vars.js; Manuscripts/reproducibility/analysis/p3_swa_fig.py; Manuscripts/reproducibility/analysis/make_supplement_data.py; p3_swa_vars.js	Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/results/p3_swa_vars.json
Figure 2	Figure drawn by the script from the input data	Manuscripts/reproducibility/analysis/p3_swa_fig.py; swa_vars.js; p3_swa_vars.js	Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/figures/p3/fig04_swa_state.pdf
Figure 3	Screen capture of SWAFL Desktop 1.1.0 (not regenerated)	-	Manuscripts/figures/p3/fig_views_A.jpg
Figure 4	Screen capture of SWAFL Desktop 1.1.0 (not regenerated)	-	Manuscripts/figures/p3/fig_views_B.jpg
Figure 5	Screen capture of SWAFL Desktop 1.1.0 (not regenerated)	-	Manuscripts/figures/p3/fig_views_C.jpg
Table 11	Table typed from the result file (11 of its numbers found there)	Manuscripts/reproducibility/audit_numbers.py; Manuscripts/reproducibility/p3_js/e1_heldout.js; Manuscripts/reproducibility/analysis/p3_figs.py; Results/08_Scripts/novel.py	Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/results/e12_pred.json
Figure 6	Figure drawn by the script from the input data	Manuscripts/reproducibility/analysis/p3_sens_fig.py; e1_sens.js; l_sens.js; p3_e1_sens.js; p3_l_sens.js	Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/figures/p3/fig08_sensitivity.pdf
Table 12	Table typed from the result file (22 of its numbers found there)	Manuscripts/reproducibility/audit_numbers.py; Manuscripts/reproducibility/analysis/p3_figs.py; Manuscripts/reproducibility/analysis/make_supplement_data.py; lib/wsmodel.js; v_surface.js; p3_surface.js	Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/results/p3_ws61.csv
Figure 7	Figure drawn by the script from the input data	Manuscripts/reproducibility/analysis/p3_figs.py	Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/figures/p3/fig03_ws_verification.pdf
Table 13	Table typed from the result file (11 of its numbers found there)	Manuscripts/reproducibility/run_all.sh; Manuscripts/reproducibility/make_e12.py; Manuscripts/reproducibility/audit_numbers.py; v_border.js; p3_border.js; v_sprinkler.js; run_spr.js; p3_sprinkler.js; spr_out.js	Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/results/e12.json
Table 14	Table typed from the result file (15 of its numbers found there)	Manuscripts/reproducibility/run_all.sh; Manuscripts/reproducibility/make_e12.py; Manuscripts/reproducibility/audit_numbers.py; e1_heldout.js; e12.js; e12_pred.js	Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/results/e12.json
Figure 8	Figure drawn by the script from the input data	Manuscripts/reproducibility/analysis/p3_figs.py; make_e12.py; e1_heldout.js; e12.js; e12_pred.js	Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/figures/p3/fig05_engine1_heldout.pdf

Item	How it is made	Script(s)	Input and output
Figure 9	Figure drawn by the script from the input data	Manuscripts/reproducibility/analysis/p3_figs.py; e2_closure.js; e2_grid.js; p3_e2_closure.js; p3_e2_grid.js	Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/figures/p3/fig06_engine2.pdf
Figure 10	Figure drawn by the script from the input data	Manuscripts/reproducibility/analysis/p3_figs.py; Manuscripts/reproducibility/analysis/p3_lit_figs.py; v_drip.js; run_drip.js; p3_drip_swafl.js; drip_out.js	Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/figures/p3/fig07_drip.pdf
Figure 11	Figure drawn by the script from the input data	Manuscripts/reproducibility/analysis/p3_lit_figs.py; run_fao.js; fao_out.js; v_sprinkler.js; run_spr.js; p3_sprinkler.js; spr_out.js	Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/figures/p3/fig09_literature.pdf
Table 15	Table typed from the result file (4 of its numbers found there)	Manuscripts/reproducibility/p3_js/run_drip.js; Manuscripts/reproducibility/analysis/p3_lit_figs.py; Manuscripts/reproducibility/analysis/make_supplement_data.py; v_drip.js; p3_drip_swafl.js; drip_out.js	Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/results/drip_out.json
Figure 12	Screen capture of SWAFL Desktop 1.1.0 (not regenerated)	-	Manuscripts/figures/shots_jpg/wb_plot_views.png
Table 16	Table typed from the result file (4 of its numbers found there)	Manuscripts/reproducibility/audit_numbers.py; Manuscripts/reproducibility/p3_js/e1_heldout.js; Manuscripts/reproducibility/analysis/p3_figs.py	Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/results/e12_pred.json
Figure 13	Screen capture of SWAFL Desktop 1.1.0 (not regenerated)	-	Manuscripts/figures/shots_jpg/layer_field.jpg; Manuscripts/figures/shots_jpg/layer_spac.jpg
Figure 14	Screen capture of SWAFL Desktop 1.1.0 (not regenerated)	-	Manuscripts/figures/shots_jpg/layer_v1.jpg; Manuscripts/figures/shots_jpg/layer_v2.jpg
Figure 15	Screen capture of SWAFL Desktop 1.1.0 (not regenerated)	-	Manuscripts/figures/shots_jpg/layer_n1.jpg
Figure 16	Screen capture of SWAFL Desktop 1.1.0 (not regenerated)	-	Manuscripts/figures/shots_jpg/layer_v3.jpg; Manuscripts/figures/shots_jpg/layer_v7.jpg
Figure 17	Screen capture of SWAFL Desktop 1.1.0 (not regenerated)	-	Manuscripts/figures/shots_jpg/layer_r1.jpg
Figure 18	Screen capture of SWAFL Desktop 1.1.0 (not regenerated)	-	Manuscripts/figures/shots_jpg/layer_e1.jpg; Manuscripts/figures/shots_jpg/layer_v4.jpg
Figure 19	Screen capture of SWAFL Desktop 1.1.0 (not regenerated)	-	Manuscripts/figures/shots_jpg/layer_v6.jpg; Manuscripts/figures/shots_jpg/layer_n2.jpg
Figure 20	Screen capture of SWAFL Desktop 1.1.0 (not regenerated)	-	Manuscripts/figures/shots_jpg/layer_n3.jpg; Manuscripts/figures/shots_jpg/layer_v5.jpg

Item	How it is made	Script(s)	Input and output
Figure 21	Screen capture of SWAFL Desktop 1.1.0 (not regenerated)	-	Manuscripts/figures/shots_jpg/layer_sid.jpg; Manuscripts/figures/shots_jpg/sid_sec_long.jpg
Figure 22	Screen capture of SWAFL Desktop 1.1.0 (not regenerated)	-	Manuscripts/figures/shots_jpg/sid_sec_sched.jpg; Manuscripts/figures/shots_jpg/sid_sec_table.jpg
Figure 23	Screen capture of SWAFL Desktop 1.1.0 (not regenerated)	-	Manuscripts/figures/shots_jpg/layer_drip.jpg; Manuscripts/figures/shots_jpg/drip_sec_layout.jpg
Figure 24	Screen capture of SWAFL Desktop 1.1.0 (not regenerated)	-	Manuscripts/figures/shots_jpg/drip_sec_lateral.jpg; Manuscripts/figures/shots_jpg/drip_sec_result.jpg
Figure 25	Screen capture of SWAFL Desktop 1.1.0 (not regenerated)	-	Manuscripts/figures/shots_jpg/drip_sec_table.jpg
Figure 26	Screen capture of SWAFL Desktop 1.1.0 (not regenerated)	-	Manuscripts/figures/shots_jpg/layer_spr.jpg; Manuscripts/figures/shots_jpg/spr_sec_checks.jpg
Figure 27	Screen capture of SWAFL Desktop 1.1.0 (not regenerated)	-	Manuscripts/figures/shots_jpg/spr_sec_intake.jpg; Manuscripts/figures/shots_jpg/dash_design.jpg
Figure 28	Screen capture of SWAFL Desktop 1.1.0 (not regenerated)	-	Manuscripts/figures/shots_jpg/dash_results.jpg; Manuscripts/figures/shots_jpg/dash_charts.jpg
Figure 29	Screen capture of SWAFL Desktop 1.1.0 (not regenerated)	-	Manuscripts/figures/shots_jpg/dash_messages.jpg
Figure 30	Screen capture of SWAFL Desktop 1.1.0 (not regenerated)	-	Manuscripts/figures/shots_jpg/real_field_balance_charts.jpg; Manuscripts/figures/shots_jpg/real_crop_water.jpg
Figure 31	Screen capture of SWAFL Desktop 1.1.0 (not regenerated)	-	Manuscripts/figures/shots_jpg/real_tillage_flux.jpg

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